Figure atlas
three experiments, no interpretation

Every graph, with what it plots — for reading together

Lior Baruch · Reichman University · 24 August 2026

Exp1 and Exp2 are PTO only. GRPO appears from the Exp3 section onward.

One figure per slide, and a strip that says what it is

- **No conclusions on any slide.** Each strip states the x axis, the y axis, what the series are, and how the numbers were produced. It does not say what the figure shows. That is what we are here to work out.
- **Sign conventions are stated wherever a difference is plotted,** because they differ between families — and three rubrics are lower-is-better (MICI, and the MI-inconsistency channels).
- **Every Exp3 figure names its grader.** The two are never averaged: one of them WAS the training reward and the other is held out, and the level offset between them is model-dependent.
- **Exp1 and Exp2 figures did not exist before this deck.** Neither experiment's EDA writes figures to disk. These were generated from the raw per-conversation files; the generator is meetings/build/make_exp1_exp2_figs.py.

TWO ARTIFACT DEFECTS, SO NOBODY IS MISLED

The GRPO K=5 arm's last scored iteration is 6, and every figure here plots it to 6. But 33 rendered caption files in the EDA still say it is censored at iteration 5. The figures are current; that prose is stale. Descriptions in this deck were written from the images and tables.

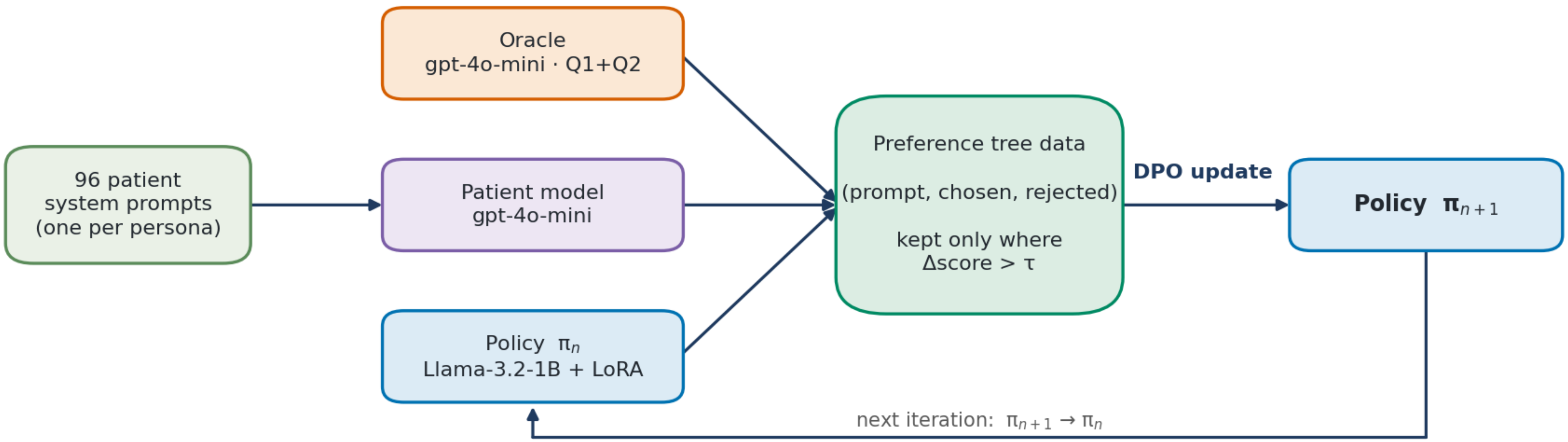
The reward-faithfulness tables pool iterations 1–6 for GRPO K=5 under a column labelled 1–5. Those two slides carry the note inline.

METHOD

How the two methods build training data

Hand-drawn schematics, not measurements. Included so the axes later in the deck have something to attach to.

PTO — preference tree + DPO



Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle

The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.

WHAT IT IS

A schematic of the PTO loop, not data.

READS

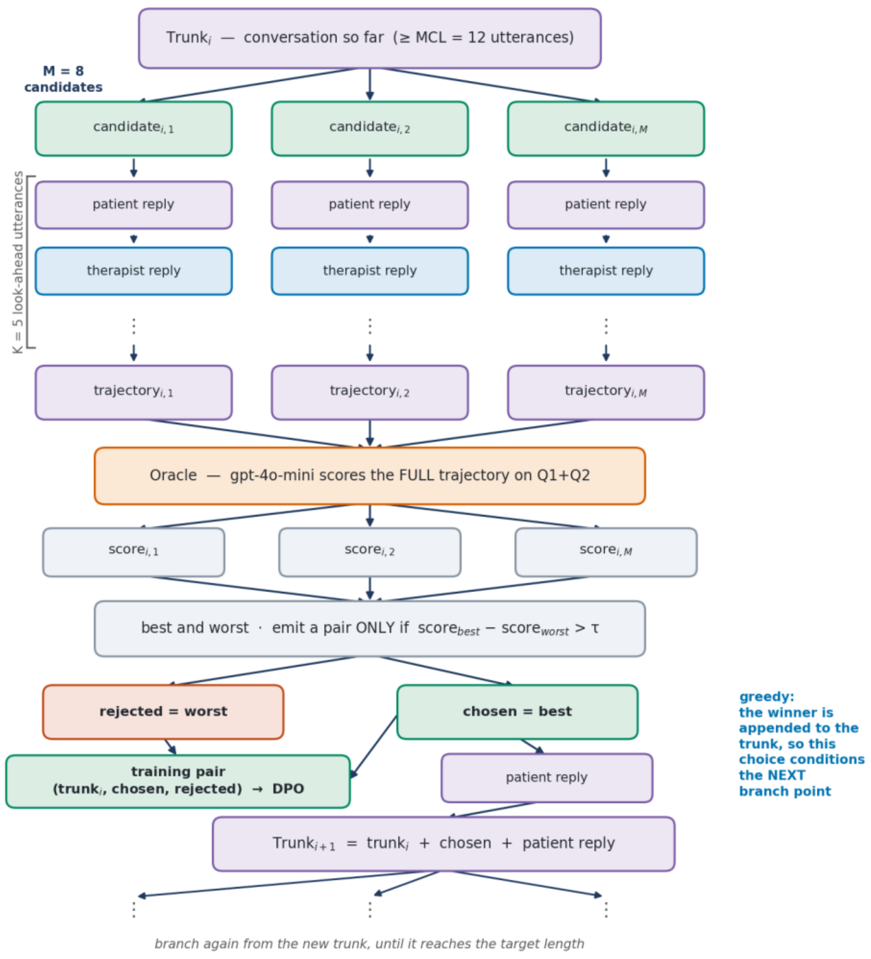
Left to right: branch M candidates at a therapist turn, simulate K look-ahead turns, score each with the oracle, keep the best/worst pair.

IN EXP3

$M = 8$, $K \in \{0, 5\}$, $\tau = 0.1$, MCL = 12.

source: `schematics/pto_framework.png`

PTO — the preference tree in detail



M candidates are generated, all M are scored, and exactly two of them survive into the training set.

WHAT IT IS

Schematic of one trunk being grown.

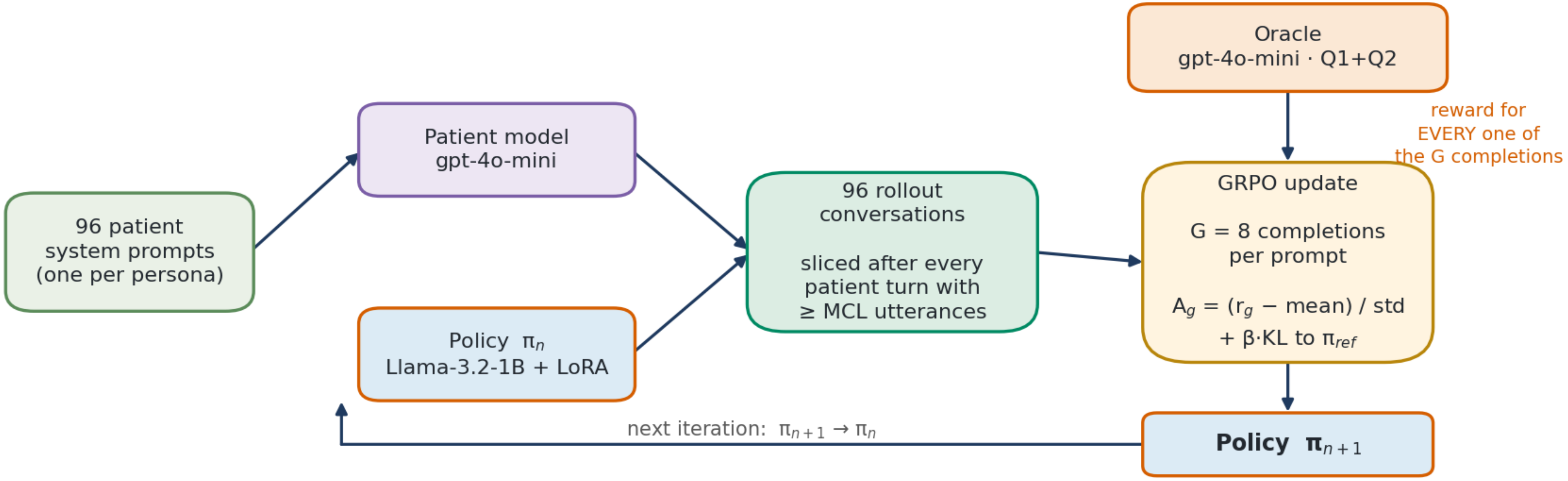
READS

At each branch point the best-of-M completion is appended to the trunk, so the chosen branch feeds the next branch point.

NOTE

This is `greedy` mode, the Exp3 default. The alternative branches a pre-recorded conversation with no feedback.

GRPO — group-relative rollout



Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.

As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

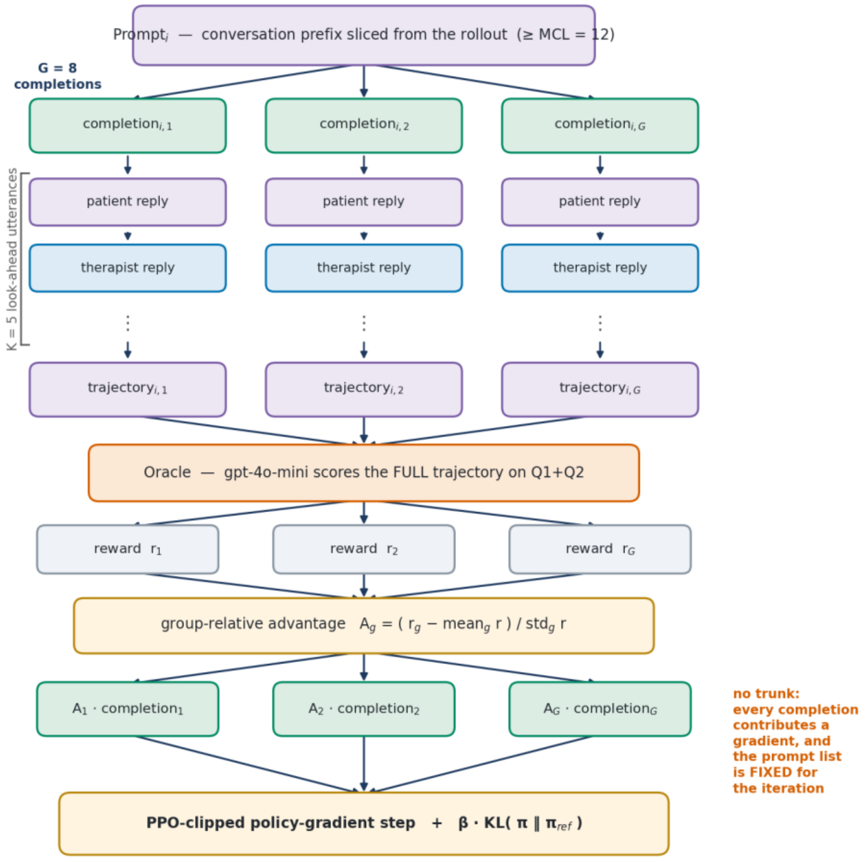
WHAT IT IS
A schematic of the GRPO loop, not data.

READS
Slice the rollout after every patient turn; sample G completions per prompt; score all of them; use the group-relative advantage.

IN EXP3
G = 8, K ∈ {0, 5}, MCL = 12. The oracle sits inside the training loop.

source: `schematics/grpo_framework.png`

GRPO — the sampled group



The next prompt is the next slice on the list — it is not conditioned on which completion won here.

G completions are generated, all G are scored, and all G carry a gradient. Rows match the PTO figure so the two can be compared.

WHAT IT IS

Schematic of one prompt's group of G completions.

READS

All G are kept and weighted by their advantage; none is discarded.

CONTRAST

PTO keeps two of M (best and worst) and only if their margin clears τ.

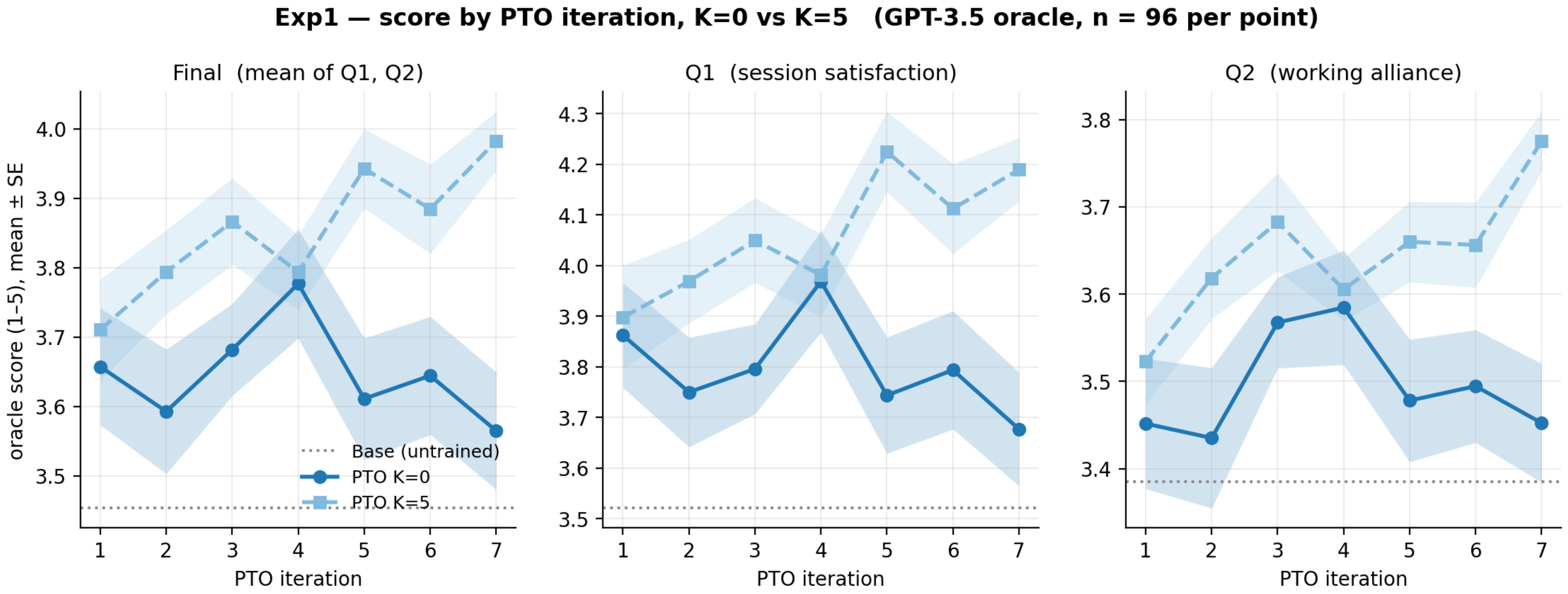
source: `schematics/grpo_group_rollout.png`

EXP1

Exp1 — ICLR 2025

Llama-2-7B therapist. GPT-3.5 as patient, oracle and training reward. PTO at $K \in \{0, 5\}$, 7 iterations, 96 personas. PTO only — there is no GRPO in this experiment.

Score by iteration, K=0 vs K=5



X AXIS
PTO iteration, 1–7.

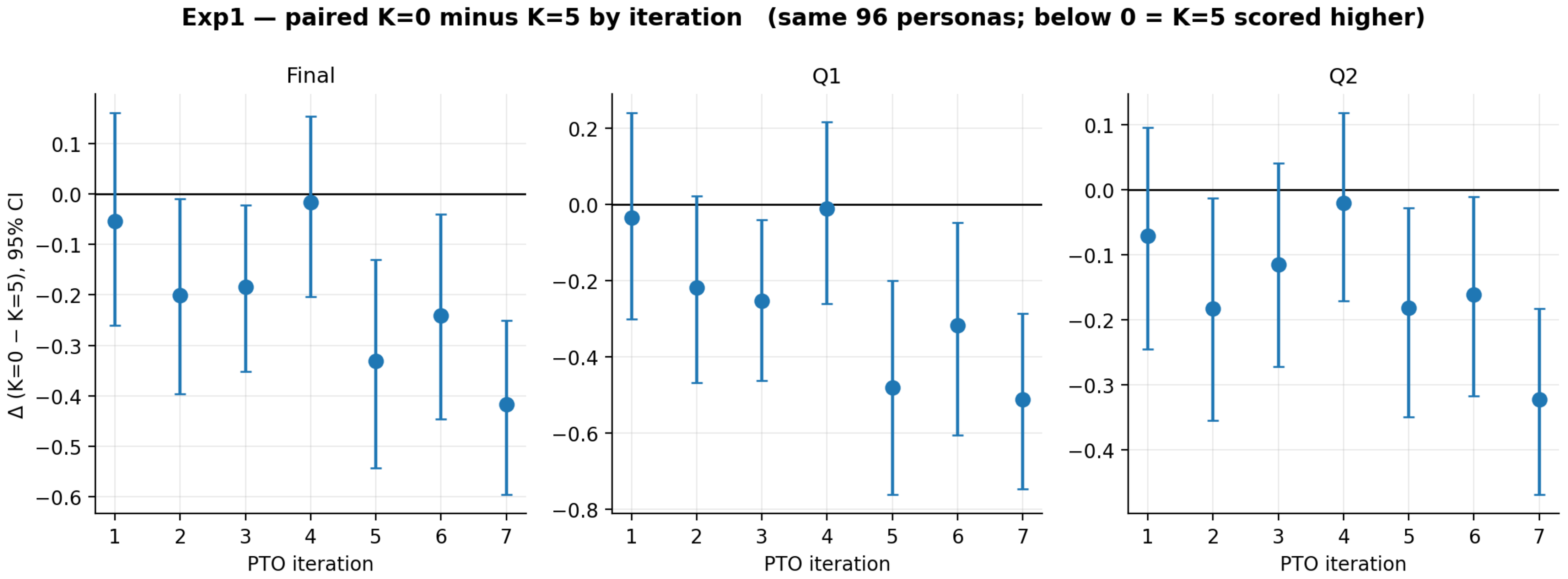
Y AXIS
Oracle score, 1–5. Band = ±1 SE of the mean over 96 conversations.

SERIES
K=0 solid circle, K=5 dashed square. Dotted grey = the untrained baseline. Panels: Final = mean(Q1, Q2), then Q1 and Q2 separately.

DATA
The 15 models of the paper's Table 1, recomputed from the shipped per-conversation score files. Grader: GPT-3.5.

source: `_figs/exp1/exp1_trajectory.png` ← `Exp1_ICLR2025/data/conversations_eval/**`

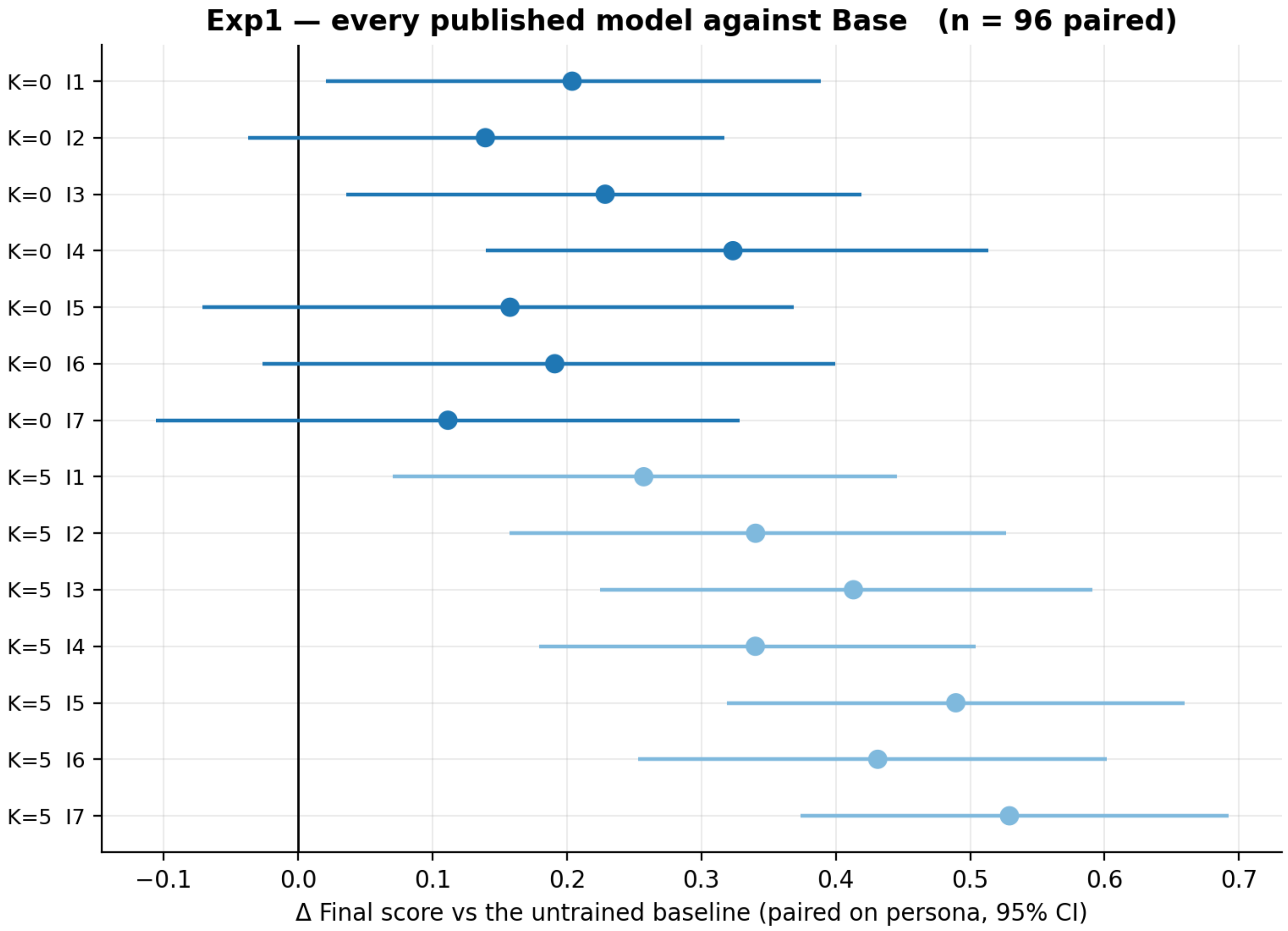
Paired difference between the two look-ahead arms



X AXIS	Y AXIS	SERIES	DATA
PTO iteration, 1–7.	Mean of (K=0 score − K=5 score) over the 96 shared personas. Positive = K=0 scored higher. Whiskers = 95% percentile bootstrap CI, 5,000 resamples.	One point per iteration; the three panels are Final, Q1, Q2.	Same conversation index = same patient permutation in both arms, so the pairing is exact. n = 96.

source: _figs/exp1/exp1_paired_delta.png

Every published model against the untrained baseline



X AXIS

Δ Final score vs Base, paired on persona. 95% bootstrap CI.

Y AXIS

The 14 trained models, K=5 above K=0, iteration 7 at the top of each block.

SERIES

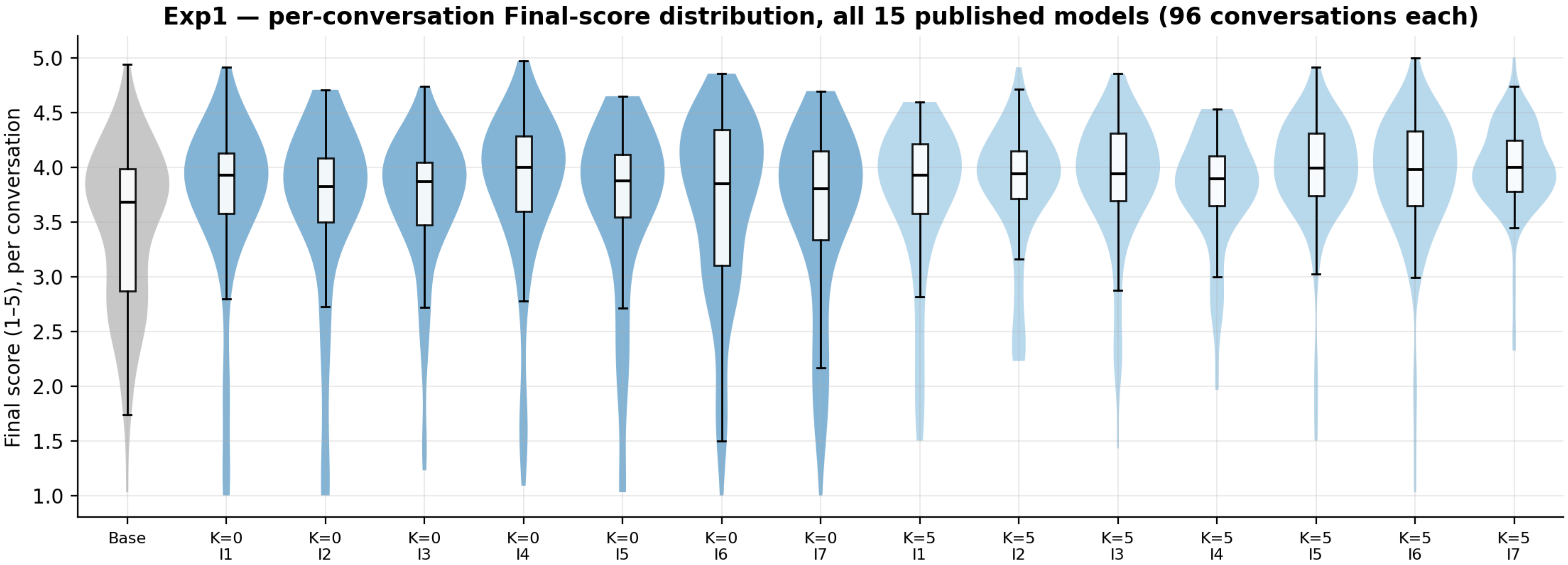
Colour = look-ahead arm.

DATA

n = 96 paired per row. Uncorrected — no multiplicity adjustment is applied in this figure.

source: `_figs/exp1/exp1_vs_base.png`

The spread behind each mean



X AXIS
The 15 models: Base, then K=0 iterations 1–7, then K=5 iterations 1–7.

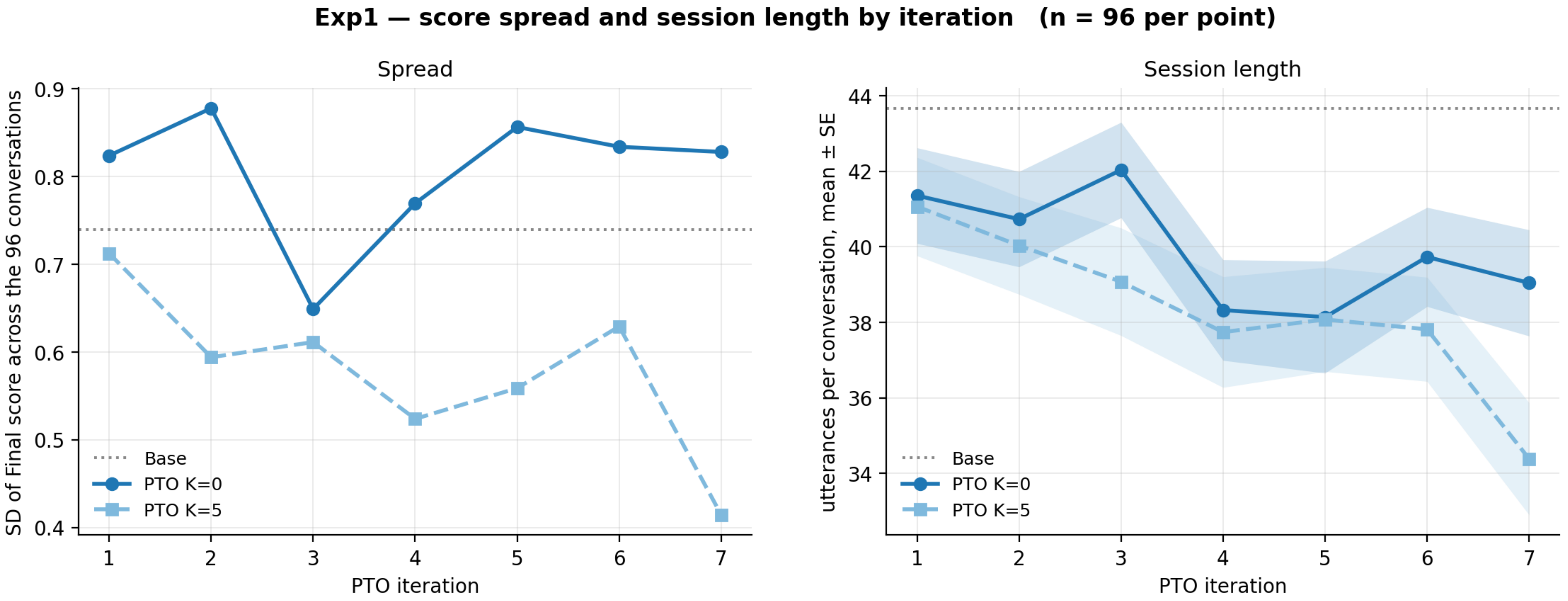
Y AXIS
Final score of a single conversation, 1–5.

SERIES
Violin = full distribution, box = median and quartiles, outliers hidden.

DATA
96 conversations per model, 1,440 in total.

source: `_figs/exp1/exp1_distributions.png`

Score spread and session length by iteration



LEFT PANEL

Y = standard deviation of the Final score across the 96 conversations (ddof = 1). One point per model.

RIGHT PANEL

Y = utterances per conversation, therapist and patient combined, mean ± SE. Counted as rows of the stored transcript.

SERIES

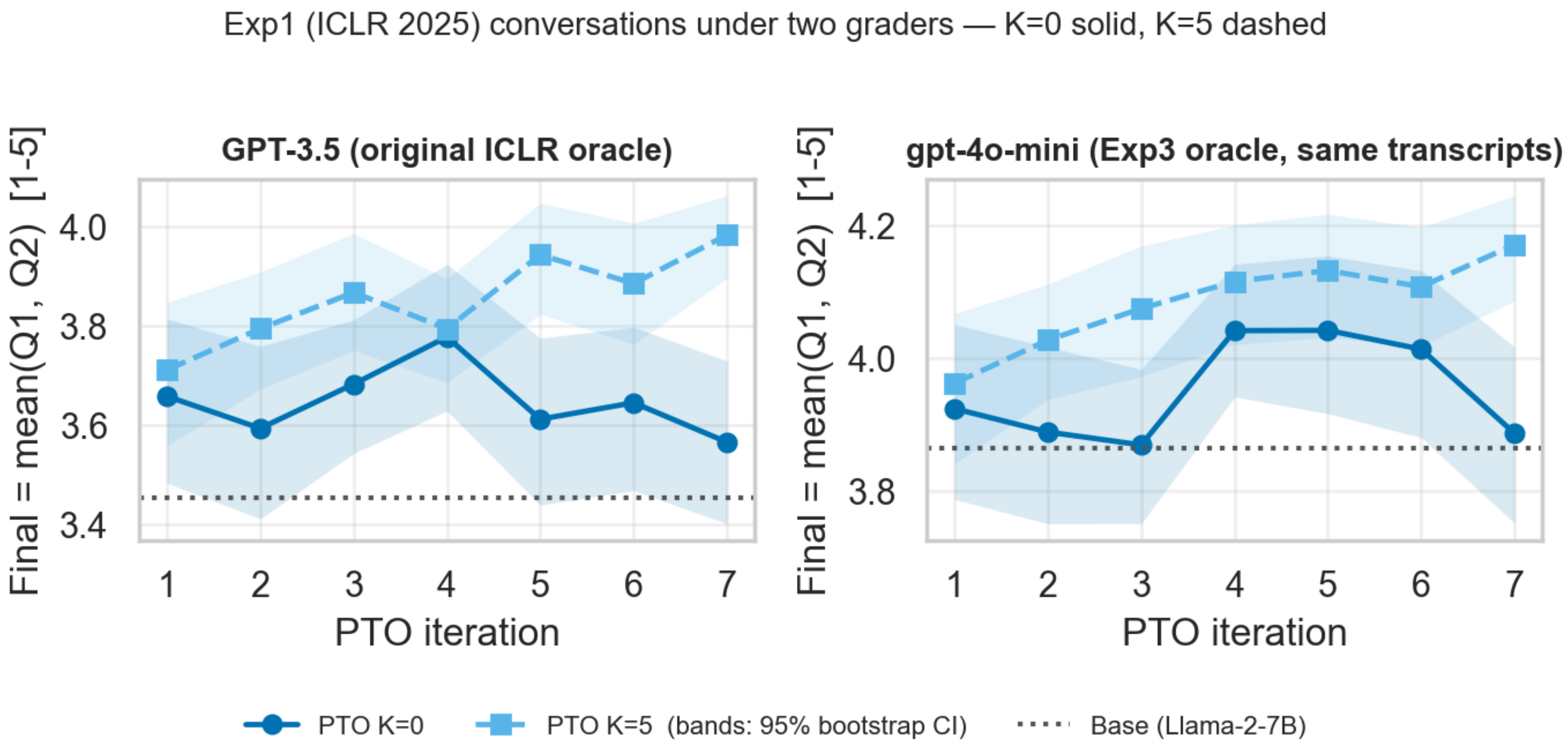
K=0 solid circle, K=5 dashed square, dotted grey = Base.

NOTE

The paper underlines the lowest SD per column in its Table 1 but runs no test on variance.

source: `_figs/exp1/exp1_sd_length.png`

The same transcripts under two different graders



X AXIS
PTO iteration, 1–7. Both arms complete 7 — no censoring here.

Y AXIS
Final = mean(Q1, Q2) over the 96 conversations. Band = 95% percentile bootstrap CI. Note the two panels have SEPARATE y scales.

SERIES
Left: the original GPT-3.5 oracle, i.e. the paper's Table 1. Right: the same stored transcripts re-scored by gpt-4o-mini on the Exp3 rubric.

DATA
1,440 conversations re-scored. The scales differ by roughly 0.19–0.43 points and are never averaged.

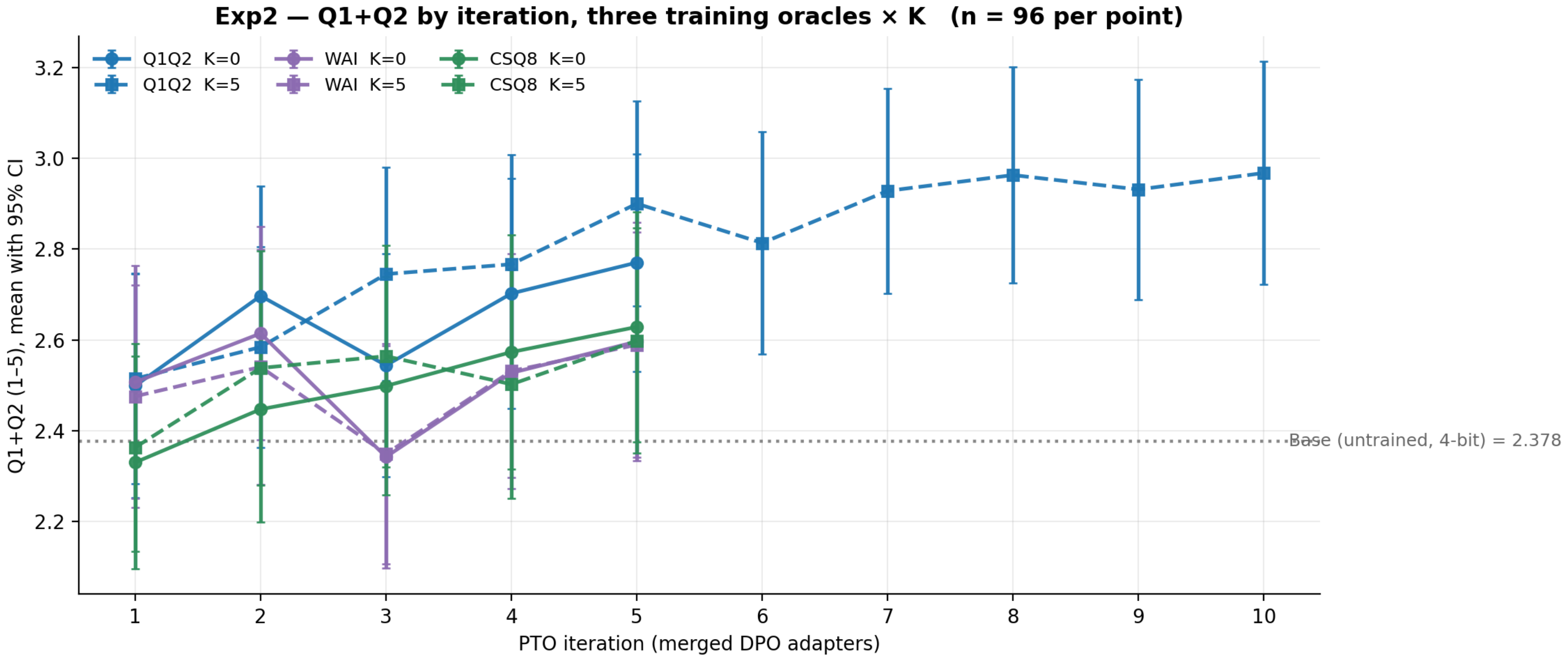
source: `lookahead/replication/figures/crossgen.png`

EXP2

Exp2 — the regime change

Llama-3.2-1B therapist in 4-bit NF4. gpt-4o-mini as patient and oracle, on a JSON schema. Less cooperative patients. PTO across three training instruments × K ∈ {0, 5}. PTO only.

Q1+Q2 by iteration, three training oracles × K



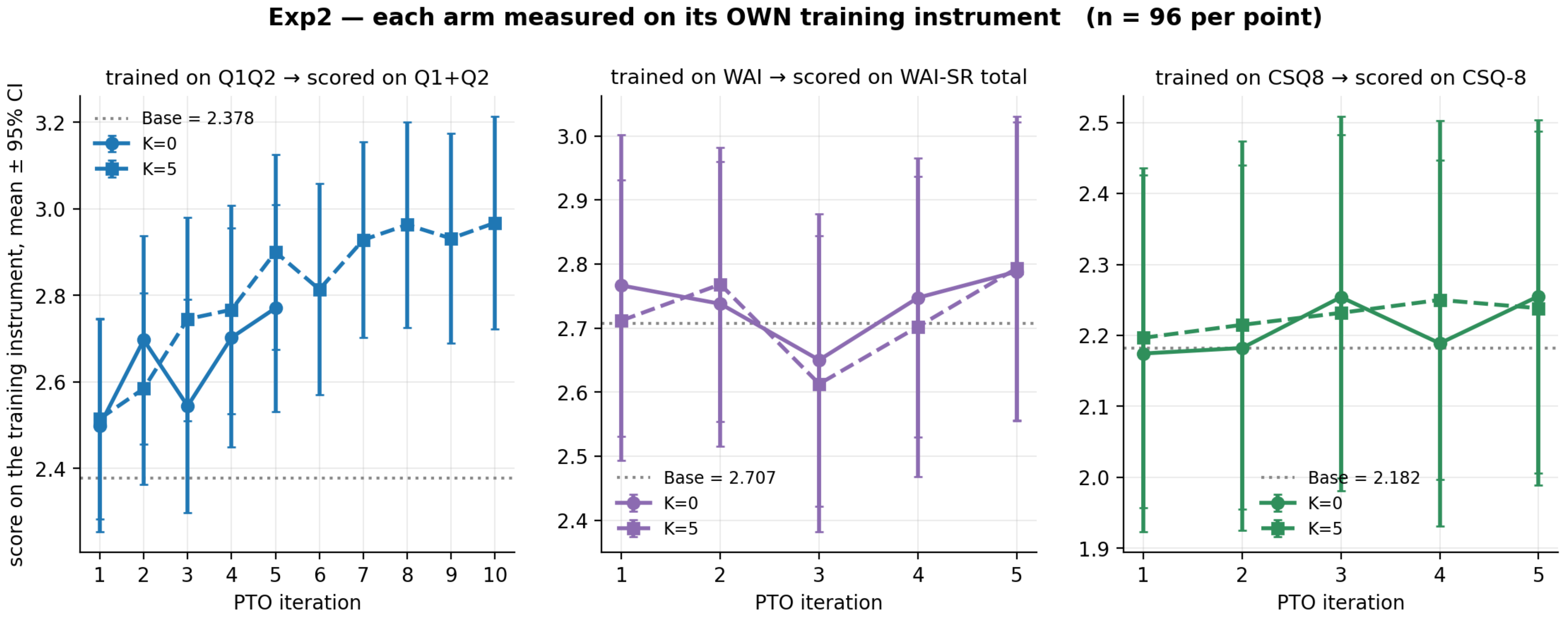
X AXIS
PTO iteration — the number of merged DPO adapters, 1–10.

Y AXIS
Q1+Q2 (mean of Q1 and Q2 per conversation), 1–5, with 95% CI.

SERIES
Colour = which instrument the arm was TRAINED on (Q1+Q2, WAI-SR, CSQ-8). Solid circle = K=0, dashed square = K=5. Dotted = untrained base.

DATA
n = 96 per point. Only the Q1+Q2 K=5 arm ran past iteration 5.

Each arm scored on the instrument it was trained on



X AXIS
PTO iteration.

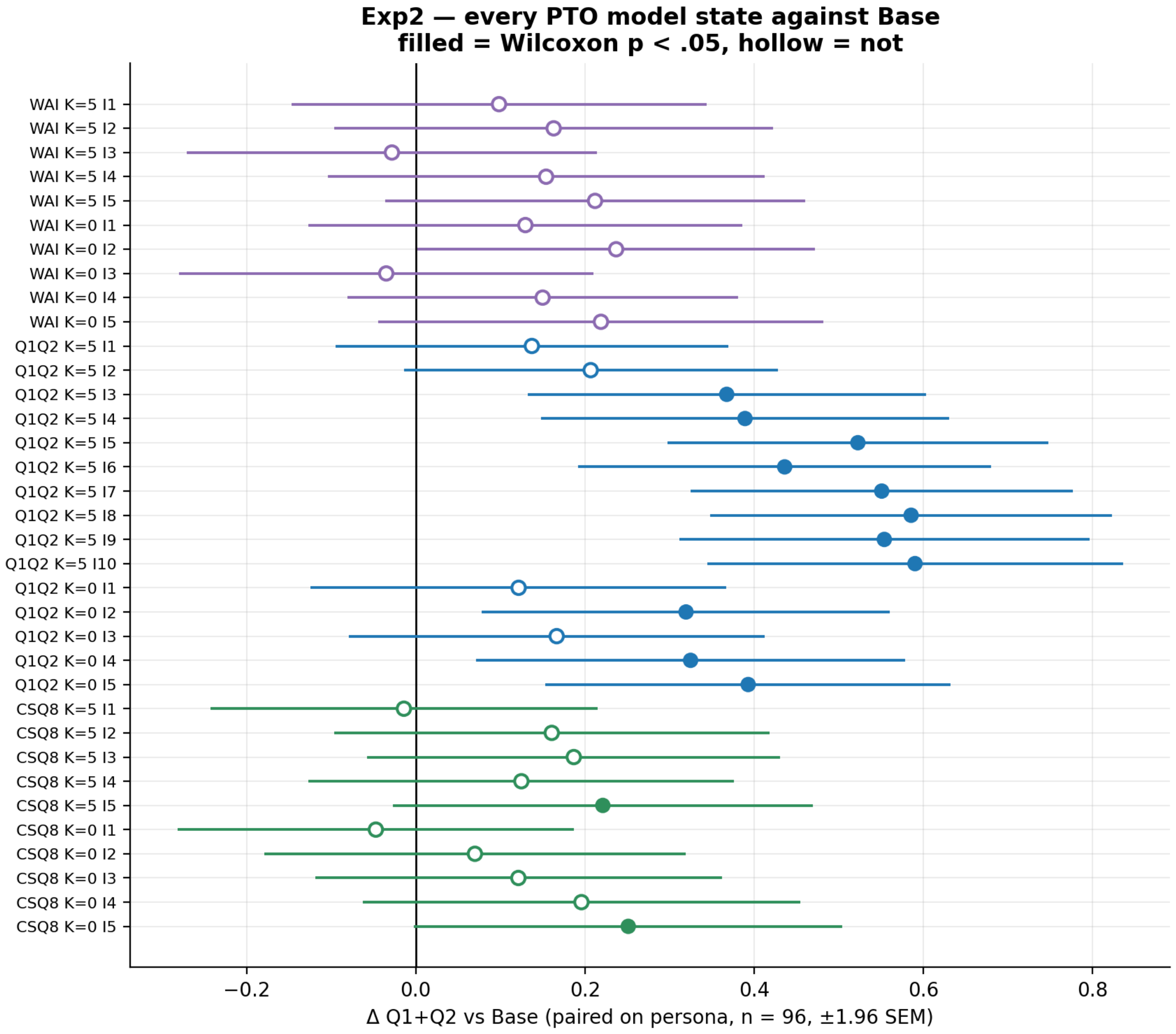
Y AXIS
Score on that arm's OWN training instrument — note each panel has its own scale. Mean with 95% CI.

SERIES
K=0 solid circle, K=5 dashed square. Dotted = the base model on that same instrument.

DATA
n = 96 per point. This is the like-for-like test of whether the optimisation moved the thing it optimised.

source: _figs/exp2/exp2_own_instrument.png

Every model state against the base model



X AXIS

Δ Q1+Q2 vs Base, paired on persona, ±1.96 SEM.

Y AXIS

The 35 trained model states, grouped by training oracle then K.

SERIES

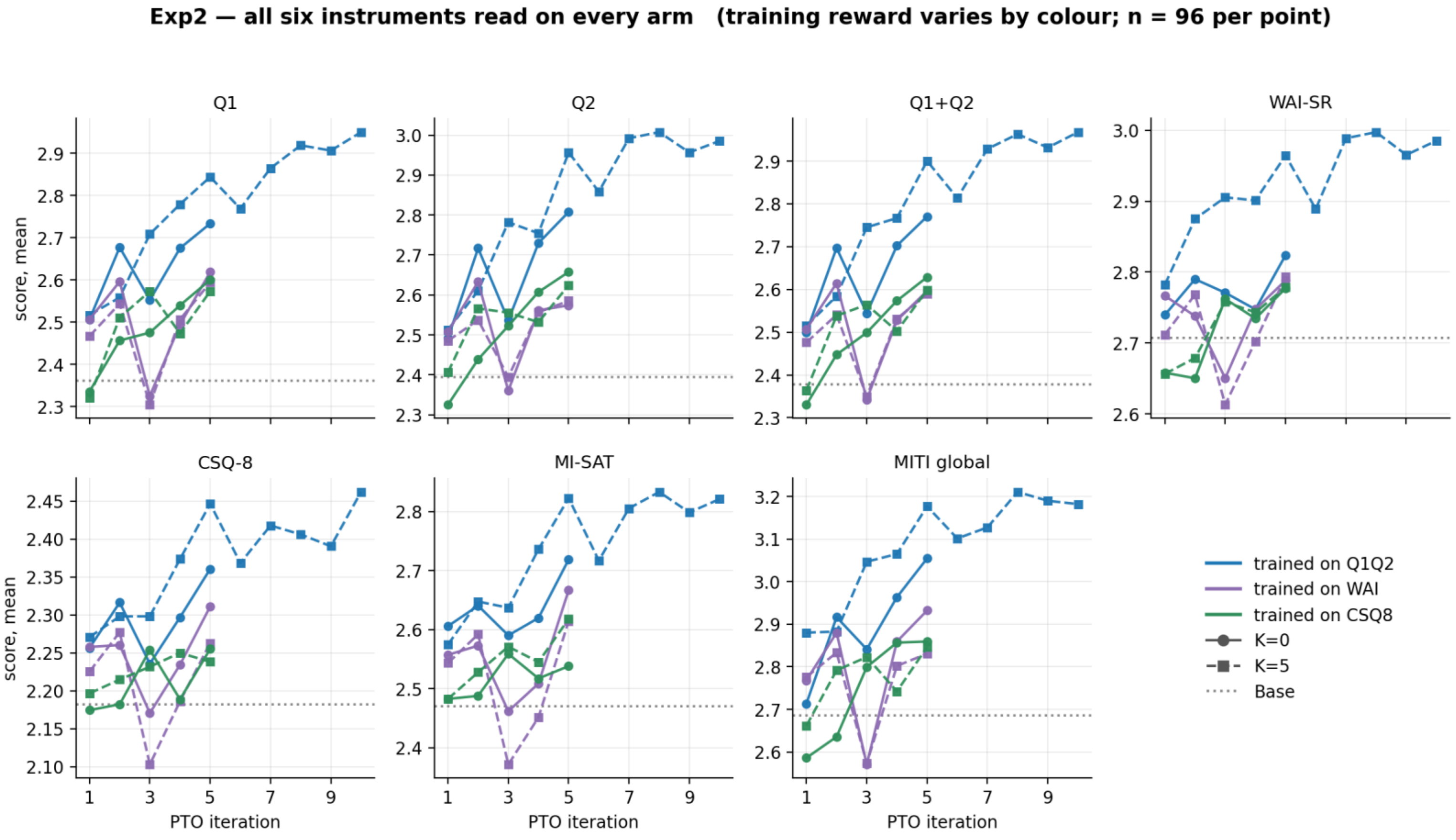
Filled marker = Wilcoxon signed-rank p < .05; hollow = not. Colour = training oracle.

DATA

n = 96 paired per row, uncorrected for multiplicity.

source: `_figs/exp2/exp2_vs_base_forest.png`

All six instruments read on every arm



X AXIS
PTO iteration, in every panel.

Y AXIS
Score on the panel's instrument — each panel has its own scale. Dotted = base on that instrument.

SERIES
Colour = training oracle, line style = K. Every arm is read on every instrument, including ones it never optimised.

DATA
n = 96 per point. Seven instruments shown; MITI global is the 1–5 clinician-style global rating.

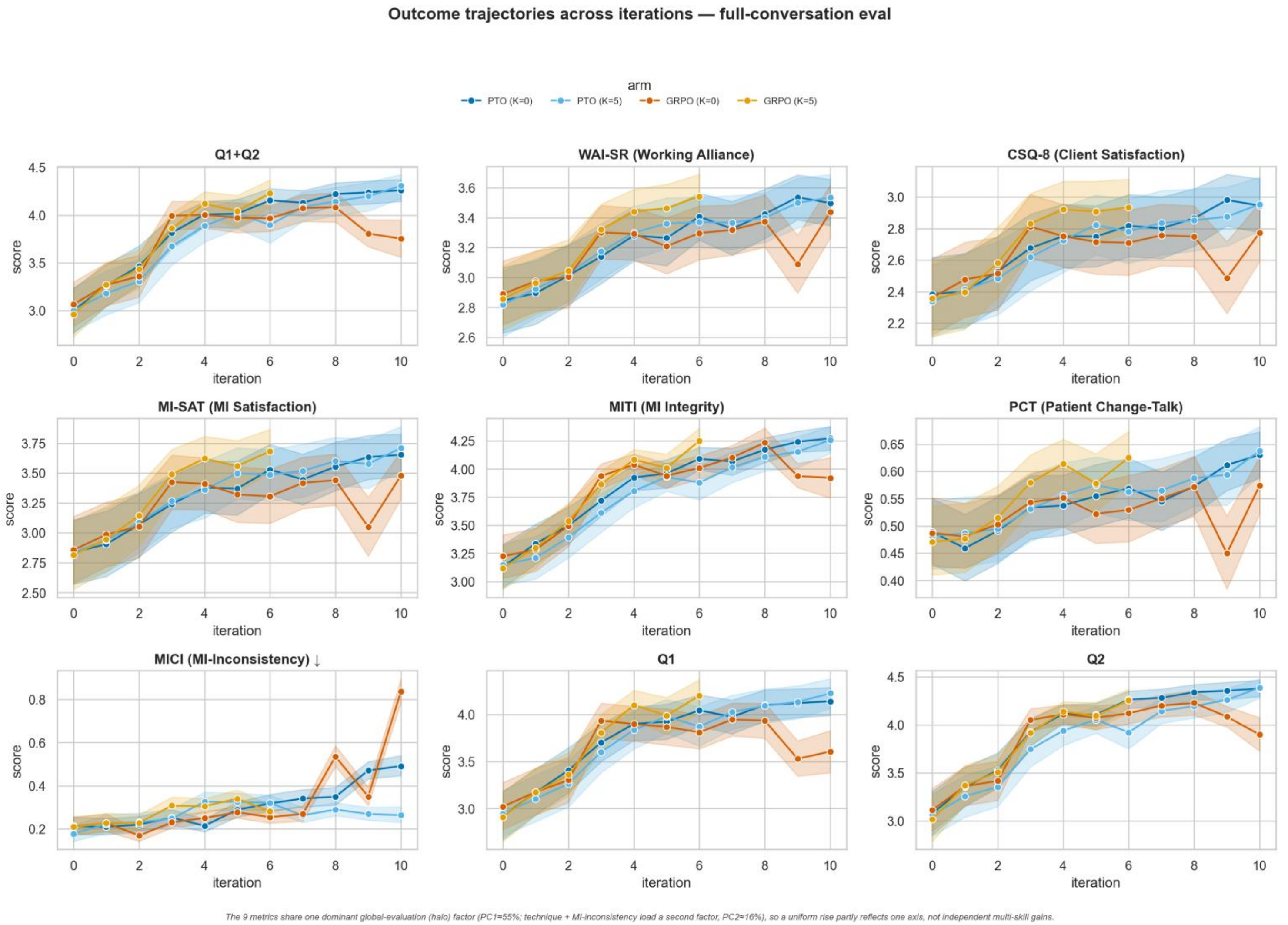
source: `_figs/exp2/exp2_instrument_grid.png`

EXP3 · OUTCOMES

Exp3 — four arms, two graders

Llama-3.2-1B in bf16. gpt-4o-mini patient + training oracle, Claude Haiku 4.5 held out. PTO and GRPO at $K \in \{0, 5\}$, matched MCL, $M = G = 8$, temperature and oracle. 40 model states × 8 instruments × 96 personas, per grader.

All nine rubrics by iteration — training oracle



X AXIS
Training iteration, 0–10, in every panel.

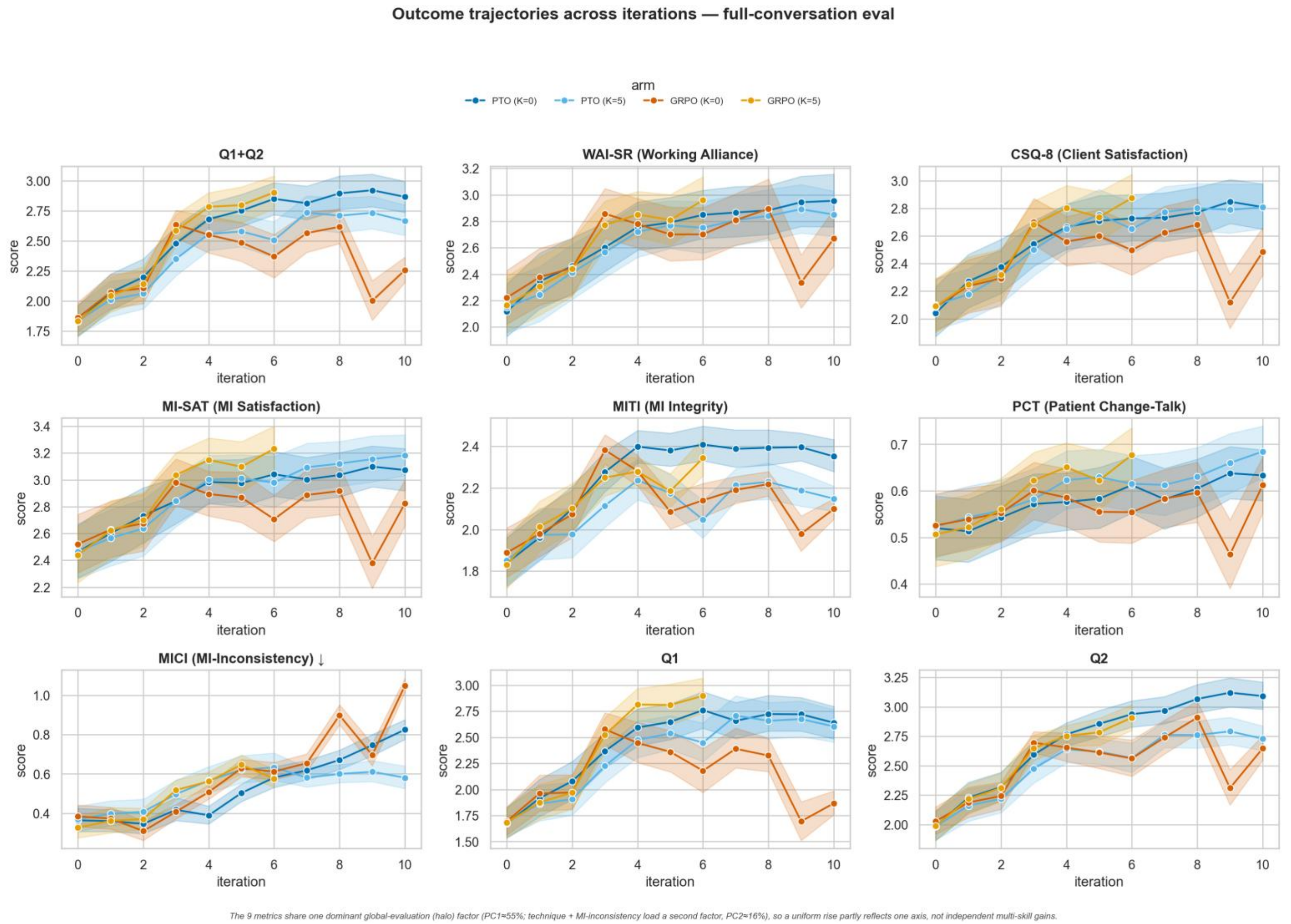
Y AXIS
The panel's rubric, on its own scale. MICI and PCT are 0–1 rates; the rest are 1–5.

SERIES
Four arms: PTO K=0, PTO K=5, GRPO K=0, GRPO K=5. K=0 solid + circle, K=5 dashed + square. GRPO K=5's line ends at iteration 6, its last scored state.

DATA
Grader: gpt-4o-mini, which was also the training reward. 96 personas, one conversation each. Contrasts are paired on persona identity.

source: `arms/outcomes/figures/gpt-4o-mini/trajectories_all_metrics.png`

All nine rubrics by iteration — held-out judge



X AXIS

Training iteration, 0–10.

Y AXIS

Same rubrics, same arms as the previous slide — but note the y scales differ: this grader sits 1.2–1.7 points lower on the point rubrics.

SERIES

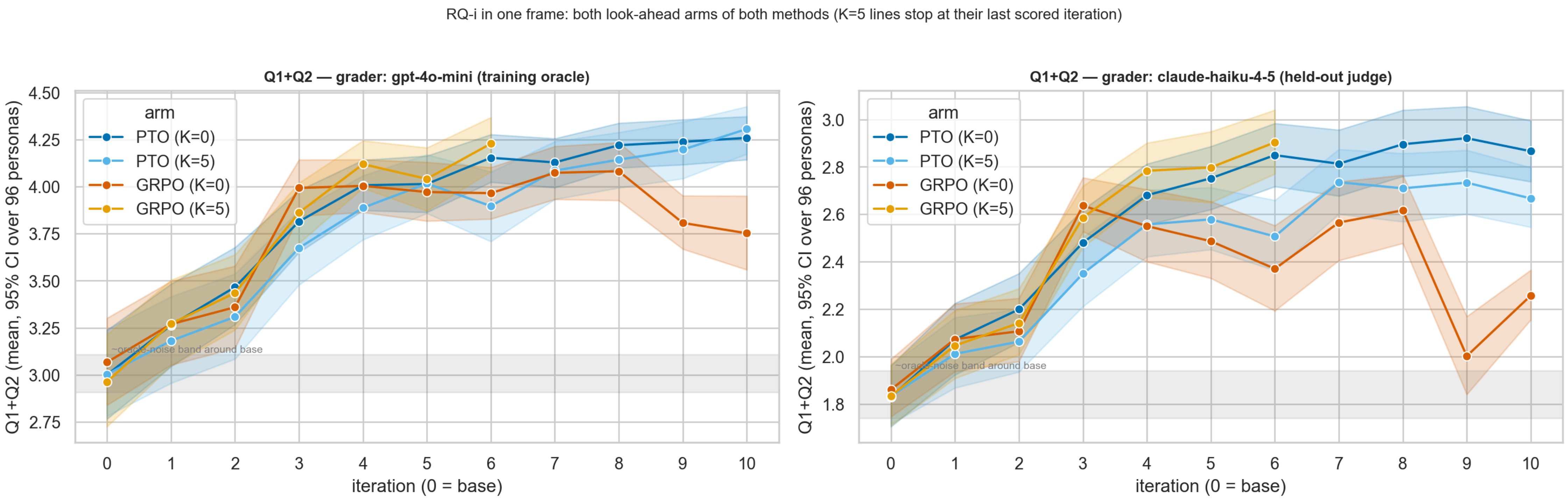
Four arms: PTO K=0, PTO K=5, GRPO K=0, GRPO K=5. K=0 solid + circle, K=5 dashed + square. GRPO K=5's line ends at iteration 6, its last scored state.

DATA

Grader: Claude Haiku 4.5, held out. Same 96 conversations per state as the previous slide — only the grader changes.

source: `arms/outcomes/figures/claude-haiku-4-5/trajectories_all_metrics.png`

The training-reward rubric alone, both graders



X AXIS

Training iteration, 0–10.

Y AXIS

Q1+Q2, mean with 95% CI. Left = gpt-4o-mini, right = Claude Haiku 4.5, on SEPARATE scales. Grey band = ±0.10 around the pooled base, the measured oracle repeatability.

SERIES

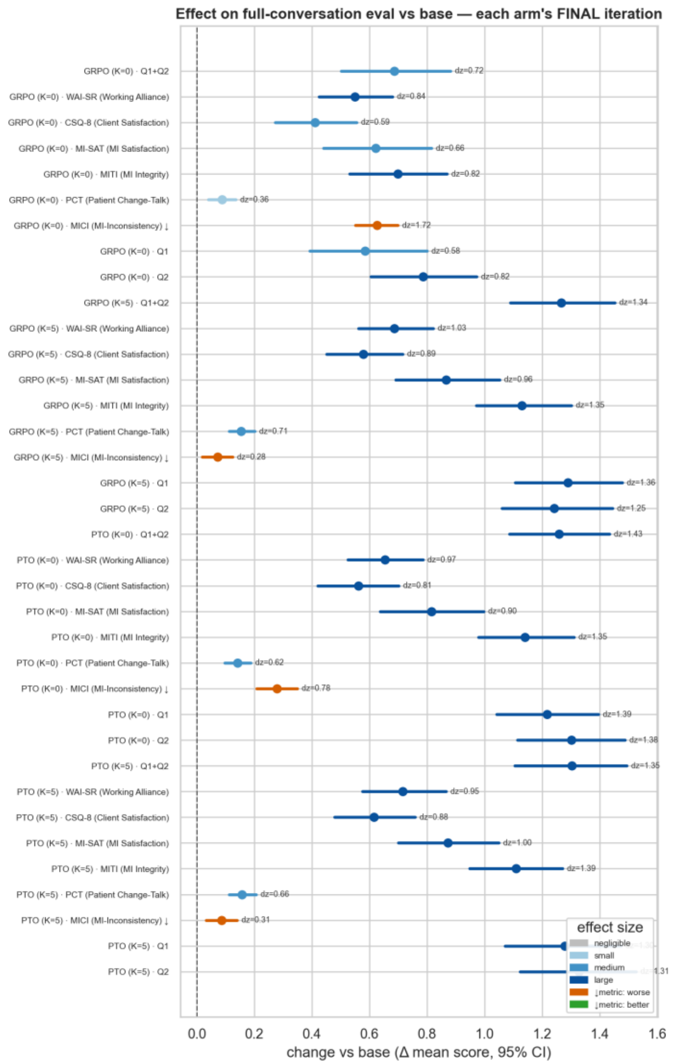
Four arms: PTO K=0, PTO K=5, GRPO K=0, GRPO K=5. K=0 solid + circle, K=5 dashed + square. GRPO K=5's line ends at iteration 6, its last scored state.

DATA

96 personas, one conversation each. Contrasts are paired on persona identity. Where a K=5 line stops, everything further right is that arm's K=0 sibling only.

source: `lookahead/reward/figures/k_trajectory_Q1Q2.png`

Each arm's endpoint against its own base — training oracle



X AXIS

Effect size (paired dz) of the arm's FINAL iteration against its own untrained base draw, with CI.

Y AXIS

Rubric × arm.

SERIES

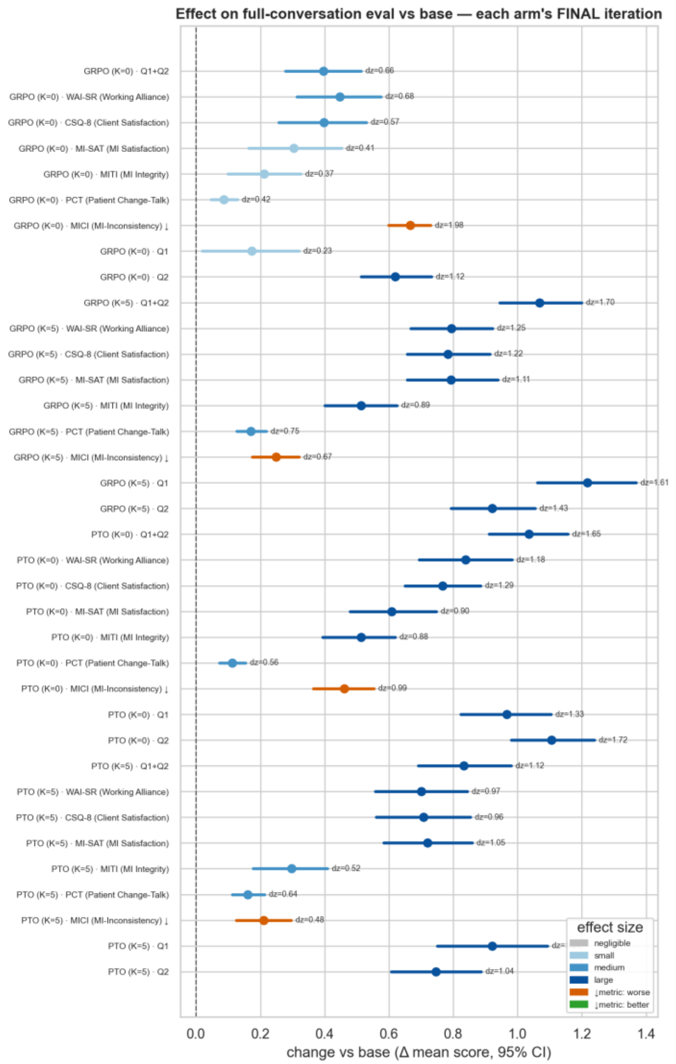
One row per (arm, rubric). Lower-is-better rubrics are flagged in the row label.

DATA

Grader: gpt-4o-mini. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

source: `arms/outcomes/figures/gpt-4o-mini/effect_vs_base_forest_final.png`

Each arm's endpoint against its own base — held-out judge



X AXIS
Paired dz vs own base at the final iteration, with CI.

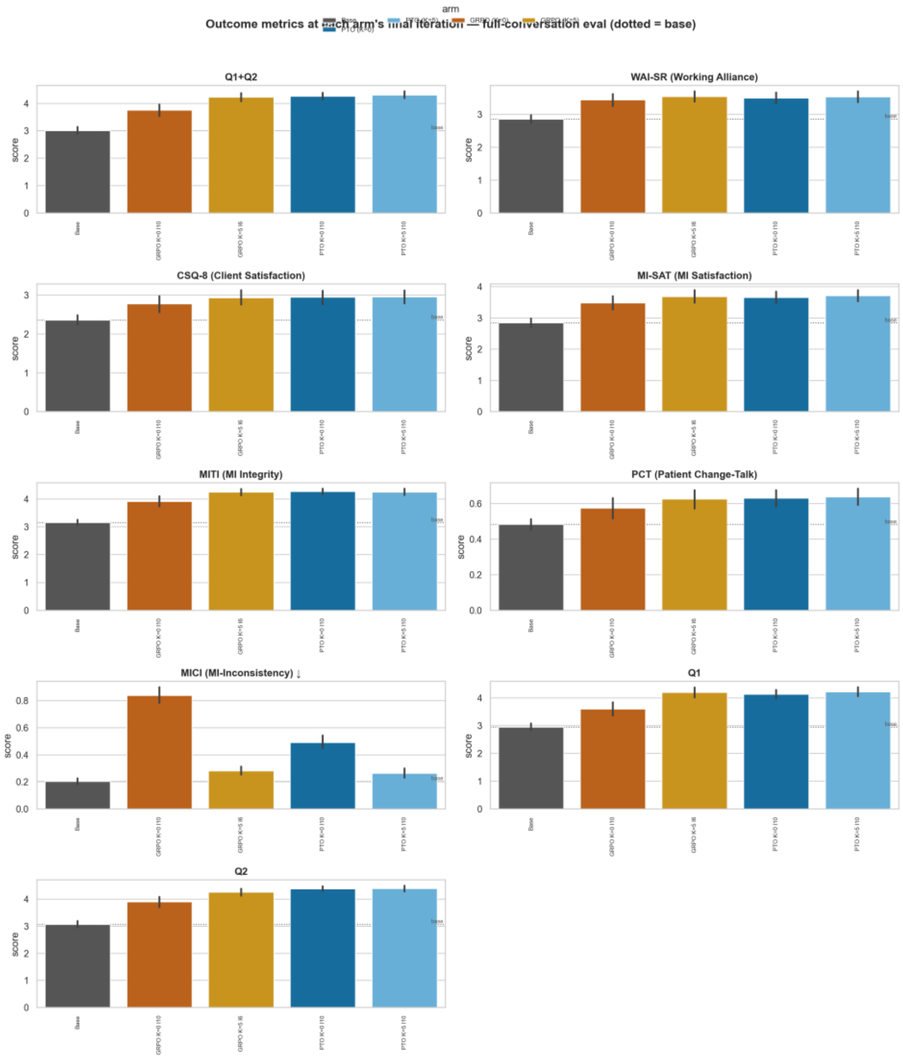
Y AXIS
Rubric × arm, same layout as the previous slide.

SERIES
One row per (arm, rubric).

DATA
Grader: Claude Haiku 4.5. Same conversations as the previous slide.

source: `arms/outcomes/figures/claude-haiku-4-5/effect_vs_base_forest_final.png`

Arm by arm across every rubric, at the final iteration



X AXIS
Model state.

Y AXIS
Score on each rubric, panel by panel.

SERIES
One bar or point per arm; base included.

DATA
Grader: gpt-4o-mini, final iteration of each arm. 96 personas, one conversation each. Contrasts are paired on persona identity.

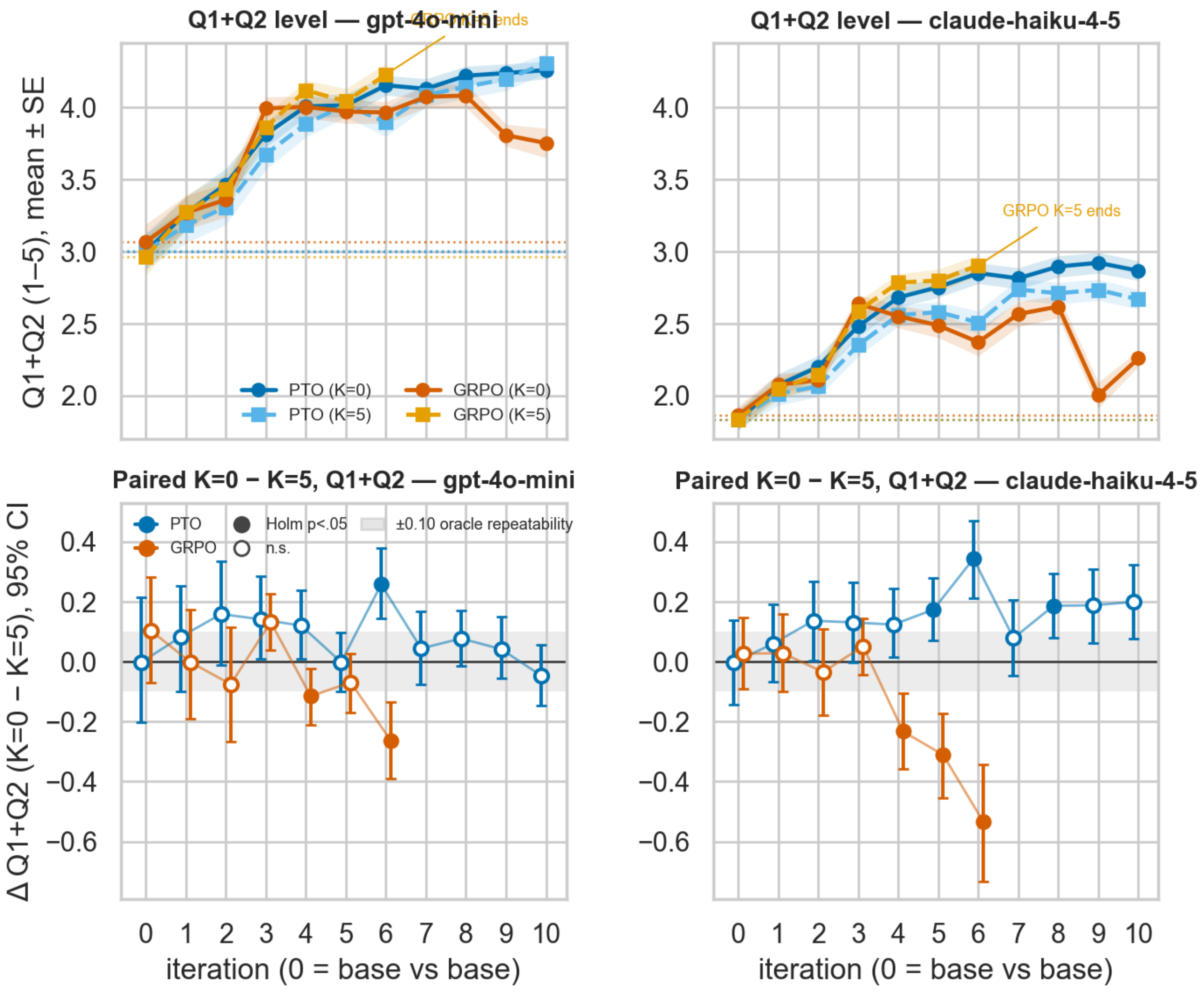
source: `arms/outcomes/figures/gpt-4o-mini/outcomes_by_model_final.png`

EXP3 · LOOK-AHEAD

Exp3 — the look-ahead contrast

$K = 0$ vs $K = 5$ within each optimizer, on the same 96 personas, under both graders. Sign convention throughout: positive = $K=0$ higher.

Level and paired difference in one frame



TOP ROW

Y = Q1+Q2 level, mean ± SE. Each arm's own base is the dotted line. Left = training oracle, right = held-out judge, shared row scale.

BOTTOM ROW

Y = paired (K=0 − K=5) on Q1+Q2, 95% CI. Positive = K=0 higher. Grey band = ±0.10 oracle repeatability. Filled = Holm p < .05 across iterations; hollow = not.

X AXIS

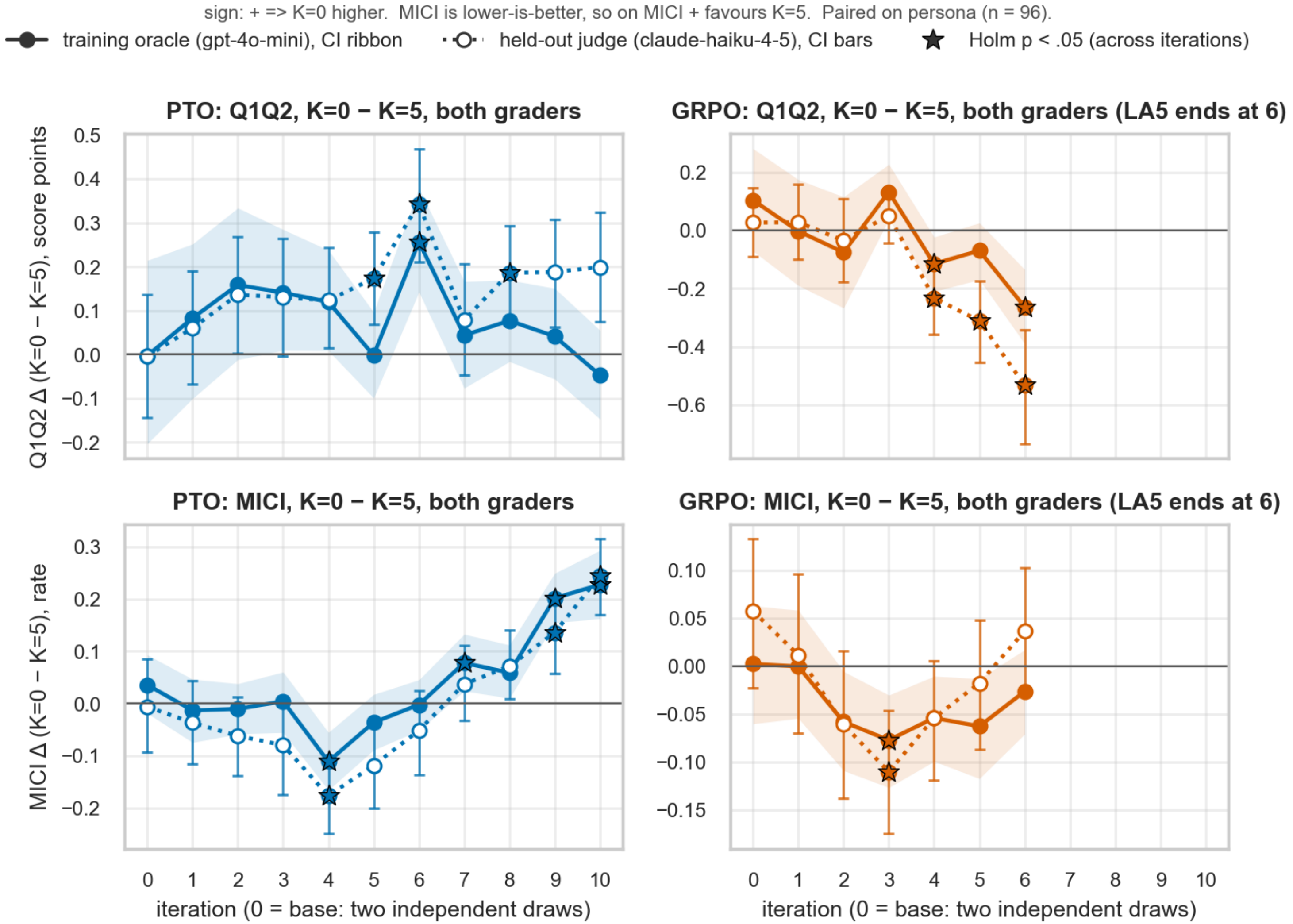
Training iteration. 0 = two independent draws of the same untrained base — the noise floor of this design.

DATA

Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

source: `lookahead/reward/figures/k_headline_q1q2.png`

The contrast on the reward rubric and the harm rubric



X AXIS
Training iteration.

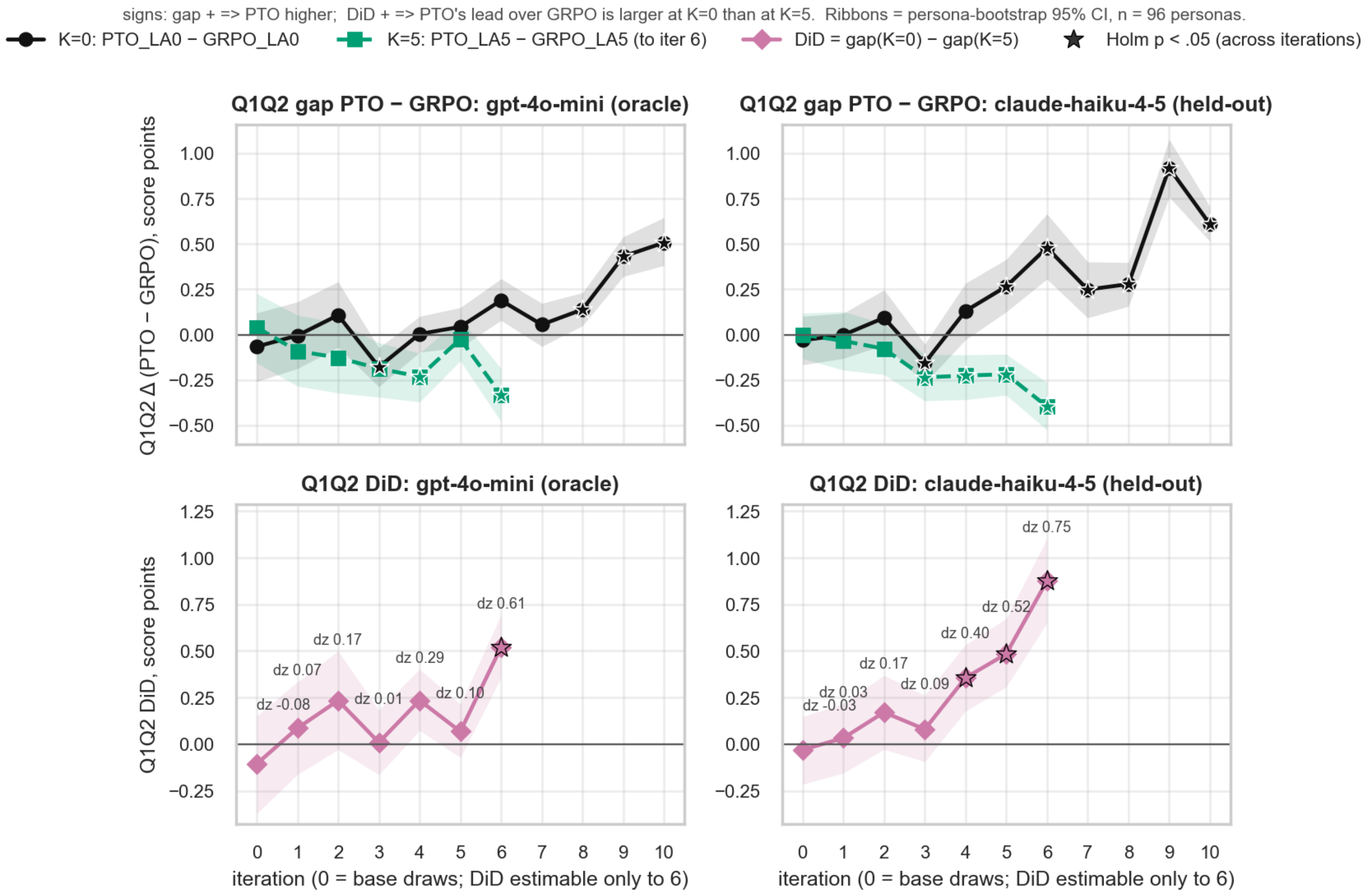
Y AXIS
Paired K=0 − K=5. Rows are Q1+Q2 and MICI; columns are PTO and GRPO.

SERIES
Solid line + filled circle + ribbon = training oracle; dotted line + open circle + bars = held-out judge. Star = Holm p < .05 under either grader.

SIGN
Positive = K=0 higher. MICI is lower-is-better, so on the MICI row a positive value means K=0 was more MI-inconsistent. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

source: `lookahead/reward/figures/k_contrast_both_judges.png`

The method gap at each K, and the difference between them



TOP ROW

Y = PTO minus GRPO on Q1+Q2 at a fixed K. Black solid = K=0, green dashed = K=5. Ribbons = persona-bootstrap 95% CI. Positive = PTO higher.

BOTTOM ROW

Y = difference-in-differences: the K=0 gap minus the K=5 gap, computed per persona. dz annotated at each iteration.

COLUMNS

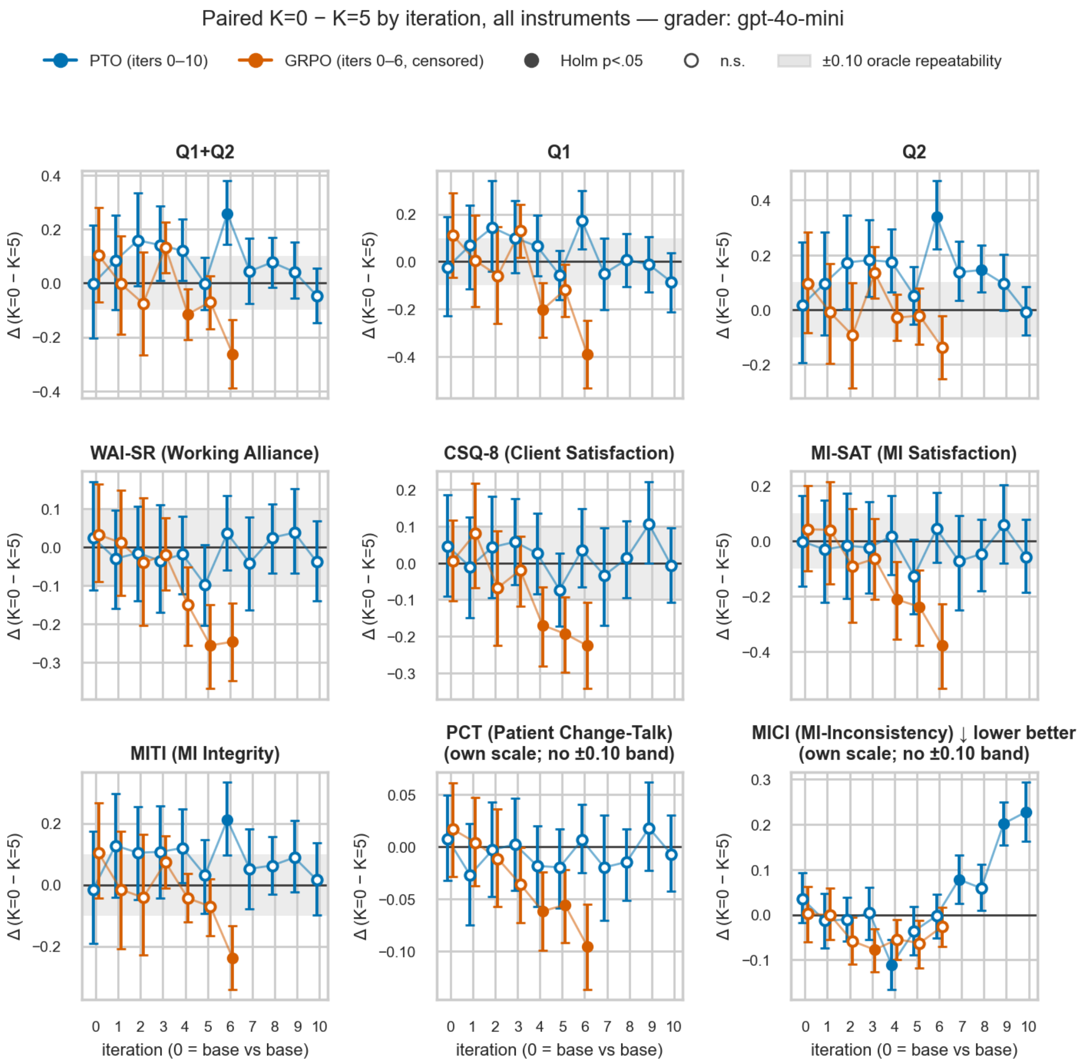
Left = training oracle, right = held-out judge. Row y-limits shared.

DATA

The DiD needs all four arms, so it is defined while GRPO K=5 runs. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

source: `lookahead/reward/figures/k_did.png`

The contrast on every rubric — training oracle



X AXIS

Training iteration, in each of the nine panels.

Y AXIS

Paired K=0 − K=5 on that panel's rubric, 95% CI whiskers.

SERIES

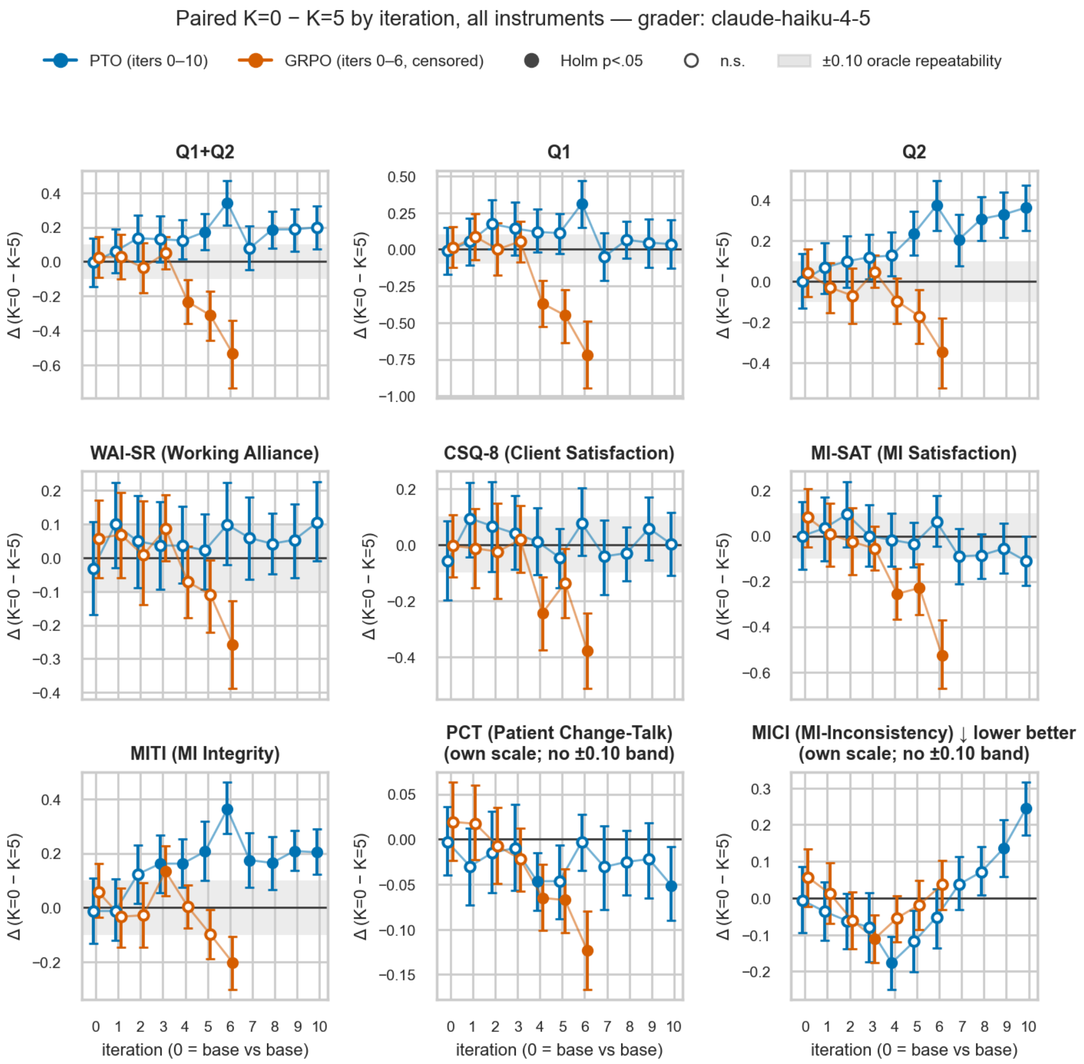
PTO points shifted left, GRPO shifted right. Filled = Holm p < .05 within (grader, method, rubric); hollow = not.

SCALES

Point rubrics carry the ±0.10 repeatability band; PCT and MICI are 0–1 rates and get a bare zero line. Lower-is-better rubrics are flagged.

source: `lookahead/reward/figures/k_delta_grid_gpt-4o-mini.png`

The contrast on every rubric — held-out judge



X AXIS
Training iteration.

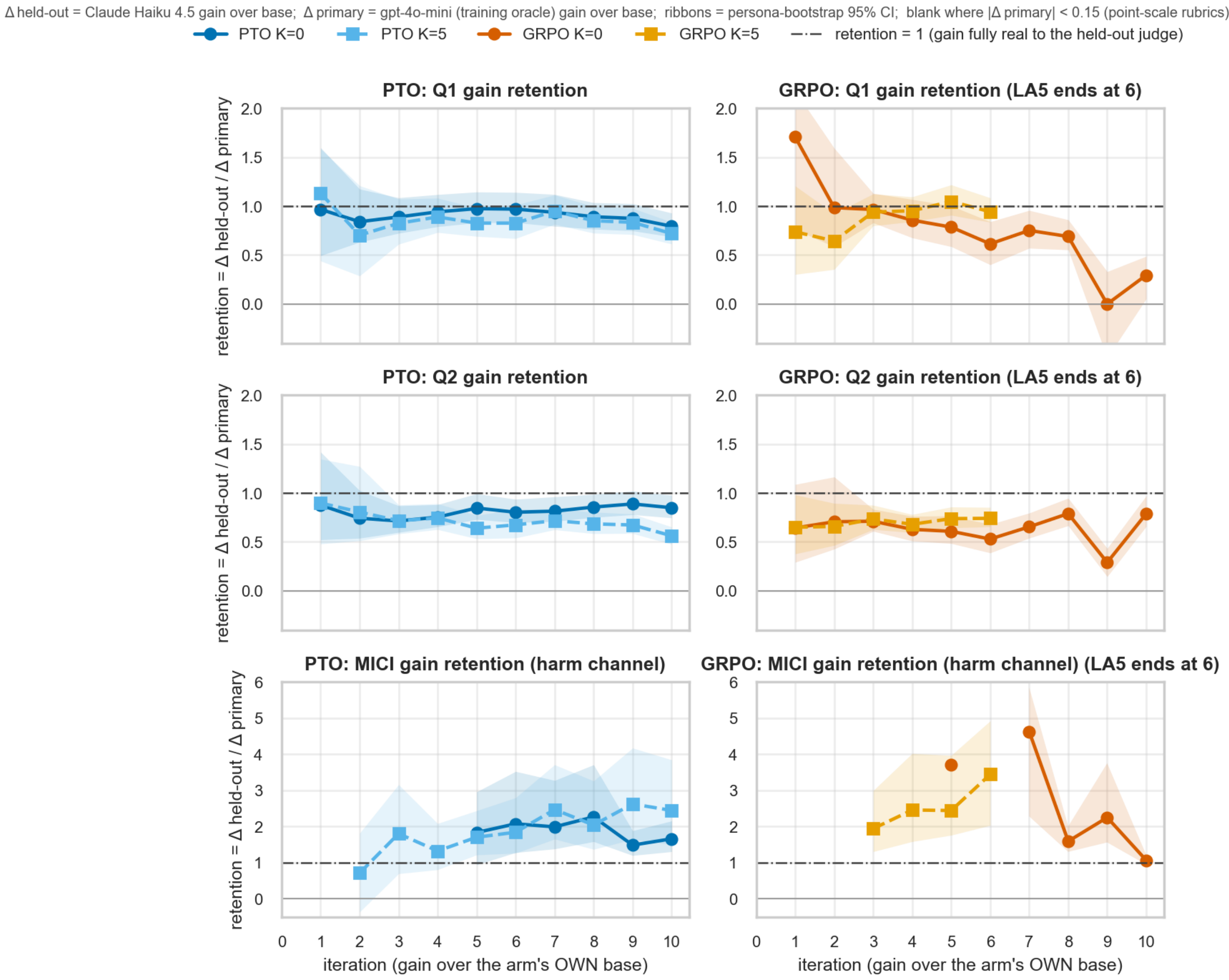
Y AXIS
Paired K=0 − K=5, same nine rubrics and same layout as the previous slide.

SERIES
PTO shifted left, GRPO shifted right. Filled = Holm $p < .05$.

DATA
Grader: Claude Haiku 4.5, on the same conversations. Persona-paired, $n = 96$. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

source: `lookahead/reward/figures/k_delta_grid_claude-haiku-4-5.png`

How much of each gain the held-out judge also sees



X AXIS
Training iteration.

Y AXIS
Retention = (gain measured by the held-out judge) ÷ (gain measured by the training oracle), each against that arm's own base. Dash-dot line at 1.0.

SERIES
Rows = Q1, Q2, MICI. Columns = PTO, GRPO. K=0 solid circle, K=5 dashed square, ribbons = persona-bootstrap 95% CI.

BLANKS
The curve is blanked where the training-oracle gain is under 0.15 points (0.05 on MICI), because the ratio is unstable there. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

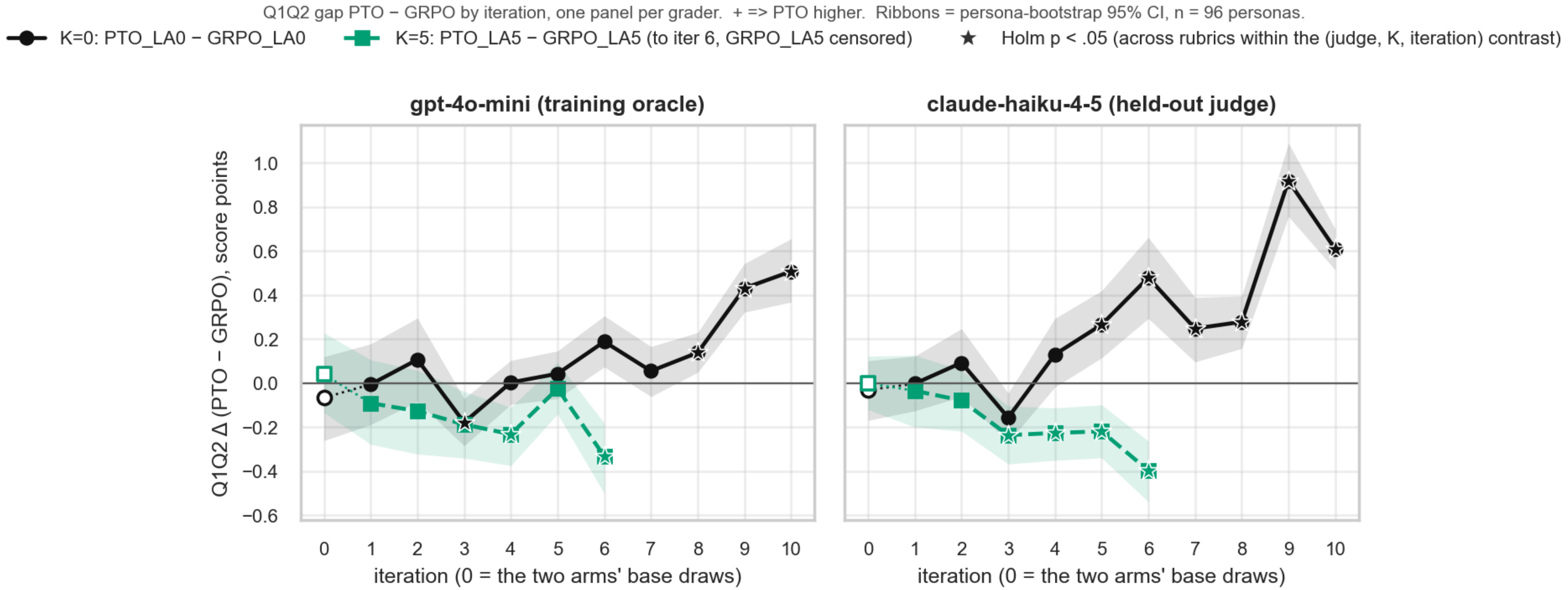
source: `lookahead/transfer/figures/k_retention.png`

EXP3 · METHOD

Exp3 — PTO against GRPO

The two optimizers at matched K and matched iteration, and then on the compute axis.

The method gap per iteration, at each K



X AXIS	Y AXIS	SERIES	NOTE
Training iteration.	PTO minus GRPO on Q1+Q2. Positive = PTO higher. Ribbons = persona-bootstrap 95% CI over the 96 shared personas.	Black solid circle = K=0, green dashed square = K=5. Stars = cleared Holm across rubrics within that grader / K / iteration.	Iteration 0 (hollow) is the two arms' independent base draws of the same untrained model — the noise floor. Matched on iteration, not on budget.

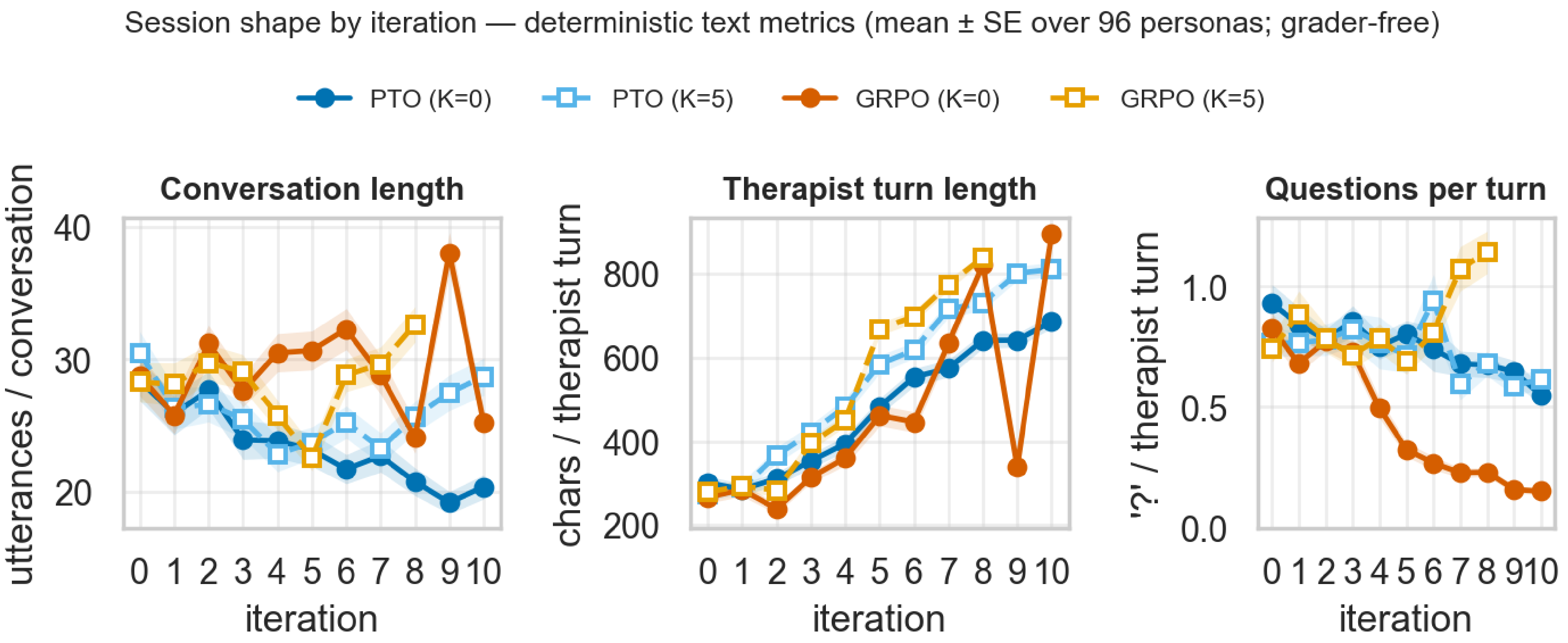
source: `method/contrast/figures/method_gap.png`

EXP3 · BEHAVIOUR

Exp3 — what the therapist actually does

Coded behaviour channels and deterministic text metrics, rather than rubric totals.

Session shape — no grader involved



X AXIS
Training iteration, 0–10.

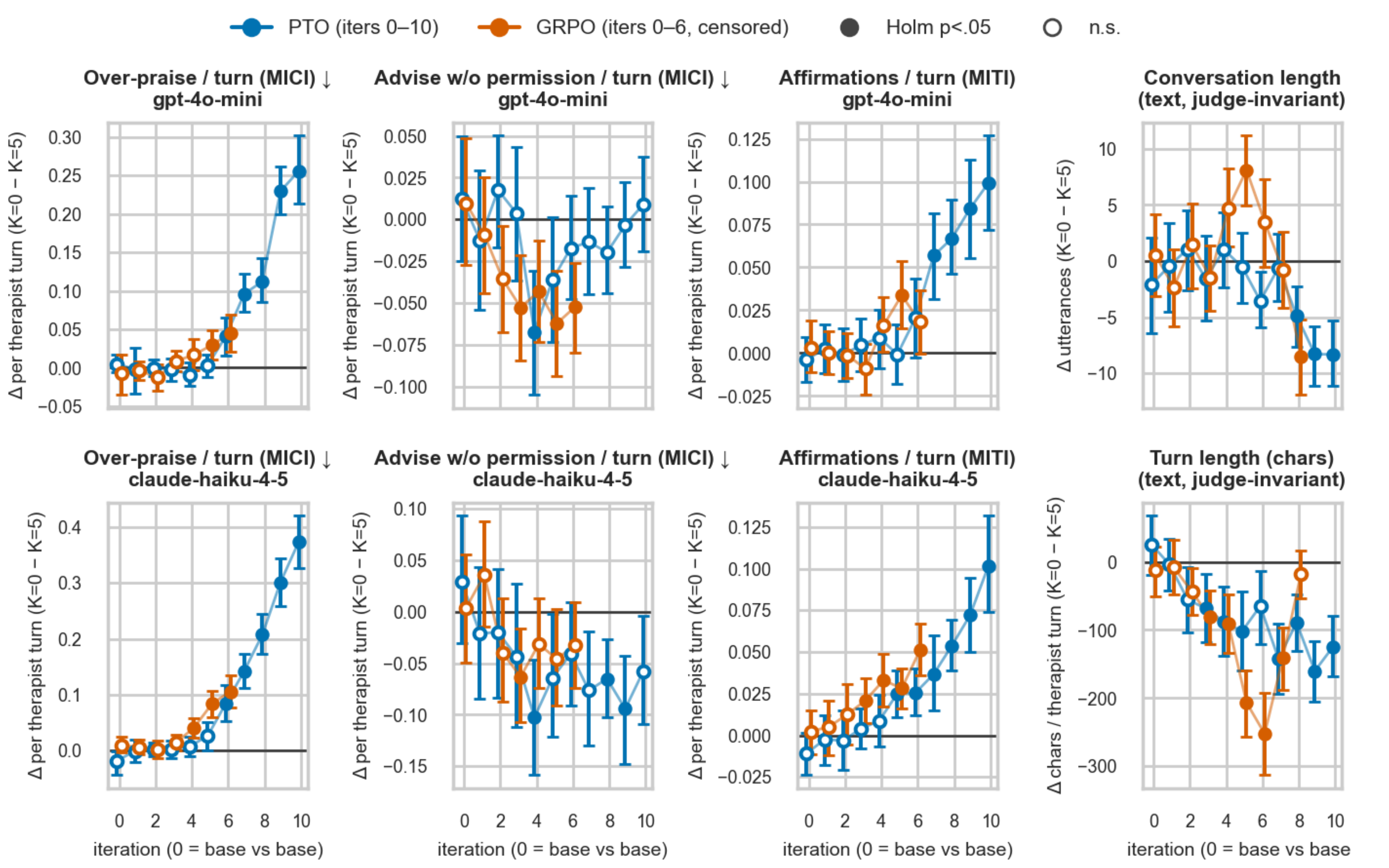
Y AXIS
Left: utterances per conversation. Middle: characters per therapist turn. Right: question marks per therapist turn. Mean ± SE.

SERIES
Four arms: PTO K=0, PTO K=5, GRPO K=0, GRPO K=5. K=0 solid + circle, K=5 dashed + square. GRPO K=5's line ends at iteration 6, its last scored state. PTO cool colours, GRPO warm.

DATA
These are computed directly from the transcript text — no LLM grades them, so they are identical under both graders. 96 personas, one conversation each. Contrasts are paired on persona identity.

source: `lookahead/behaviour/figures/session_shape.png`

The K contrast on coded behaviour channels



X AXIS
Training iteration.

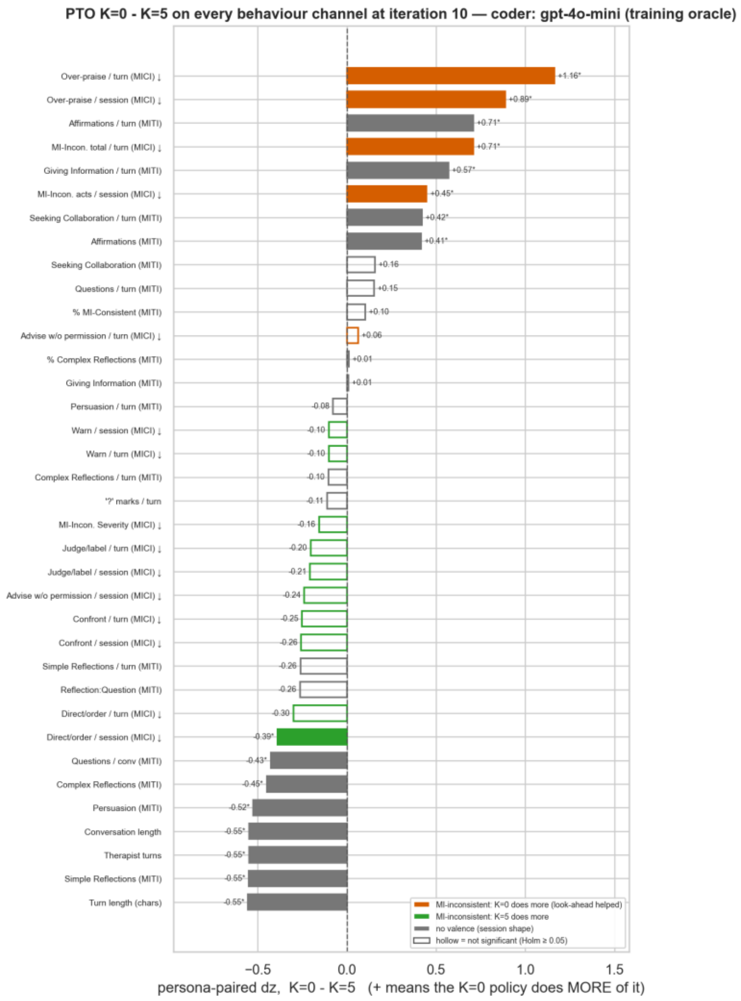
Y AXIS
Paired K=0 − K=5 on that column's channel, 95% CI.

COLUMNS
Over-praise per turn, advice-without-permission per turn, MITI affirmations per turn, then judge-invariant text channels.

SIGN
Positive = K=0 does MORE of it. The first two columns are MI-INCONSISTENT behaviours, so more is worse there. Rows = the two graders. Filled = Holm p < .05.

source: `lookahead/behaviour/figures/k_channels_grid.png`

Every channel at once, at PTO's last iteration — training oracle



Sign convention + = the K=0 policy does MORE of the channel. Colour is VALENCE, not direction: red = a MI-inconsistent behaviour K=0 does more of, green = one K=5 does more of, grey = unvalenced session shape. Red AND green bars both being large would mean the violations were swapped, not removed. Hollow = did not clear Holm within its (iteration, family) set.

X AXIS

Paired dz of K=0 − K=5 at iteration 10. Positive = the K=0 policy does more of it.

Y AXIS

One row per behaviour channel.

COLOUR

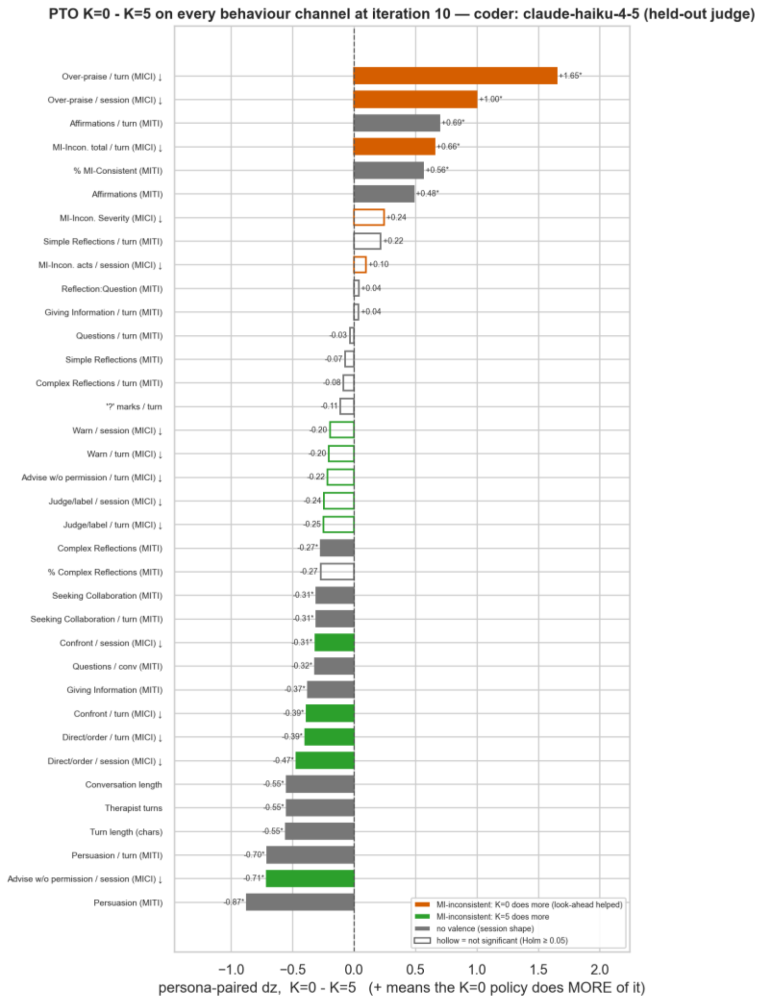
Red = MI-inconsistent and higher under K=0; green = MI-inconsistent and higher under K=5; grey = unvalenced session shape.

MARKERS

Hollow = did not clear Holm within its (iteration, family) set. Coder: gpt-4o-mini. n = 96.

source: `lookahead/behaviour/figures/k_channel_forest_gpt-4o-mini.png`

Every channel at once, at PTO's last iteration — held-out judge



Sign convention + = the K=0 policy does MORE of the channel. Colour is VALENCE, not direction: red = a MI-inconsistent behaviour K=0 does more of, green = one K=5 does more of, grey = unvalenced session shape. Red AND green bars both being large would mean the violations were swapped, not removed. Hollow = did not clear Holm within its (iteration, family) set.

X AXIS

Paired dz of K=0 − K=5 at iteration 10.

Y AXIS

Same channels, same layout as the previous slide.

COLOUR

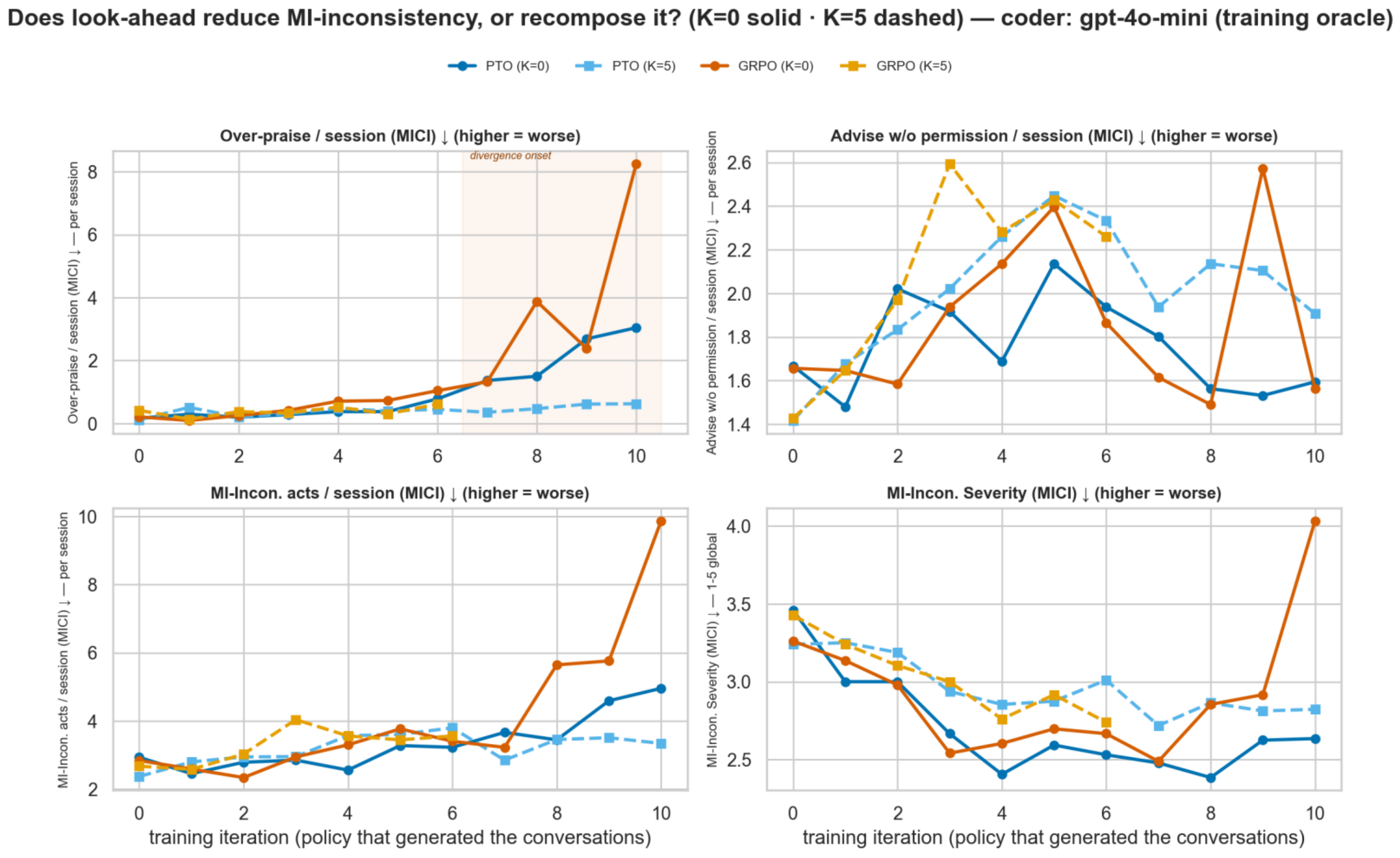
Same valence encoding. Hollow = did not clear Holm.

DATA

Coder: Claude Haiku 4.5, on the same 96 conversations.

source: `lookahead/behaviour/figures/k_channel_forest_claude-haiku-4-5.png`

MI-inconsistent acts, counted per session



X AXIS
Training iteration.

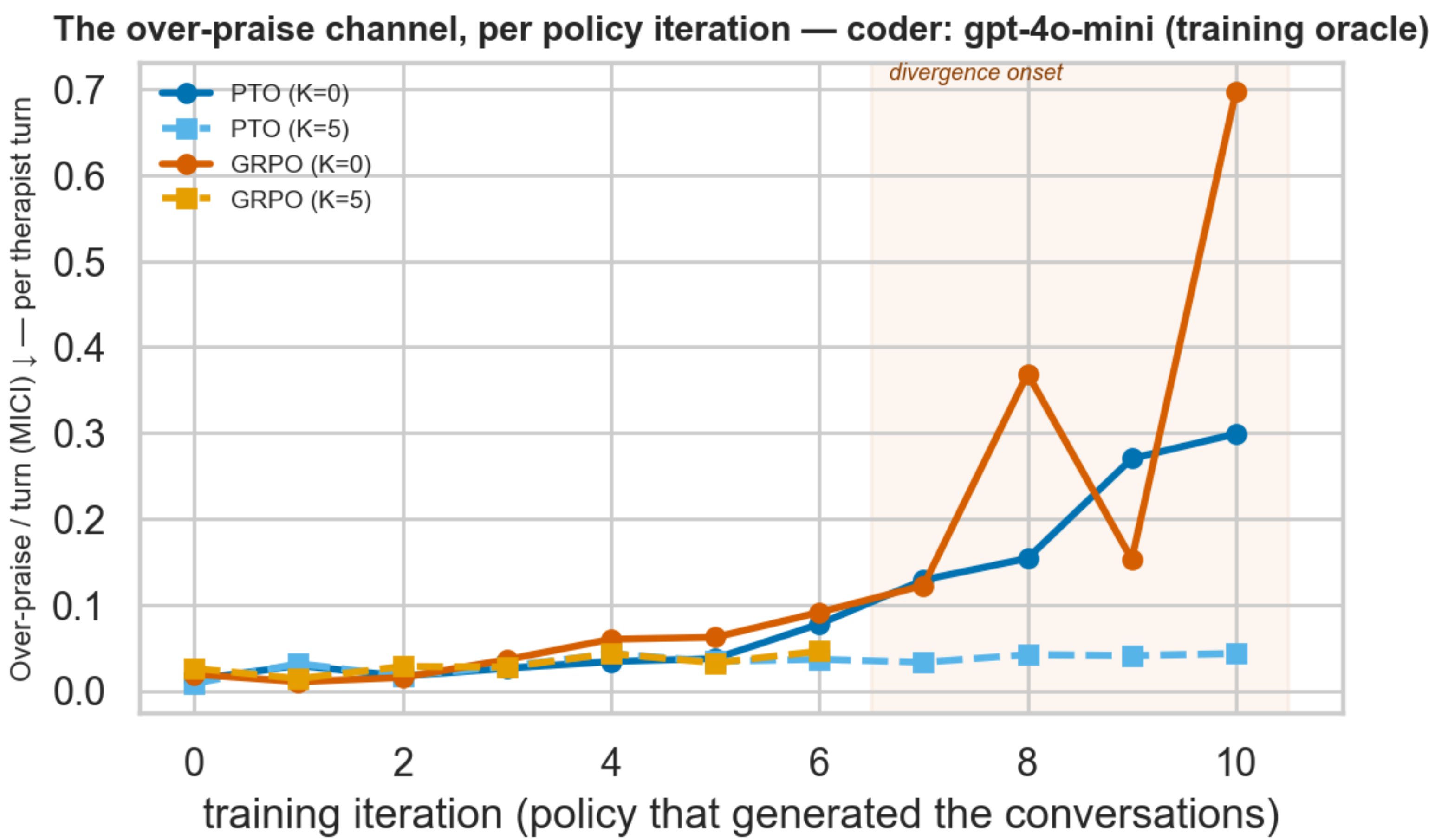
Y AXIS
Counts PER SESSION, so no denominator is involved. Panels: over-praise, advice without permission, the TOTAL number of MI-inconsistent acts, and the severity global.

SERIES
Arm means over 96 personas. Higher is worse on every panel.

WHY PER SESSION
The per-turn rate has a moving denominator — the arms differ in turn count. The total panel is the aggregate control. Coder: gpt-4o-mini.

source: `lookahead/behaviour/figures/k_mici_composition_grid_gpt-4o-mini.png`

The over-praise channel per therapist turn



X AXIS
Training iteration.

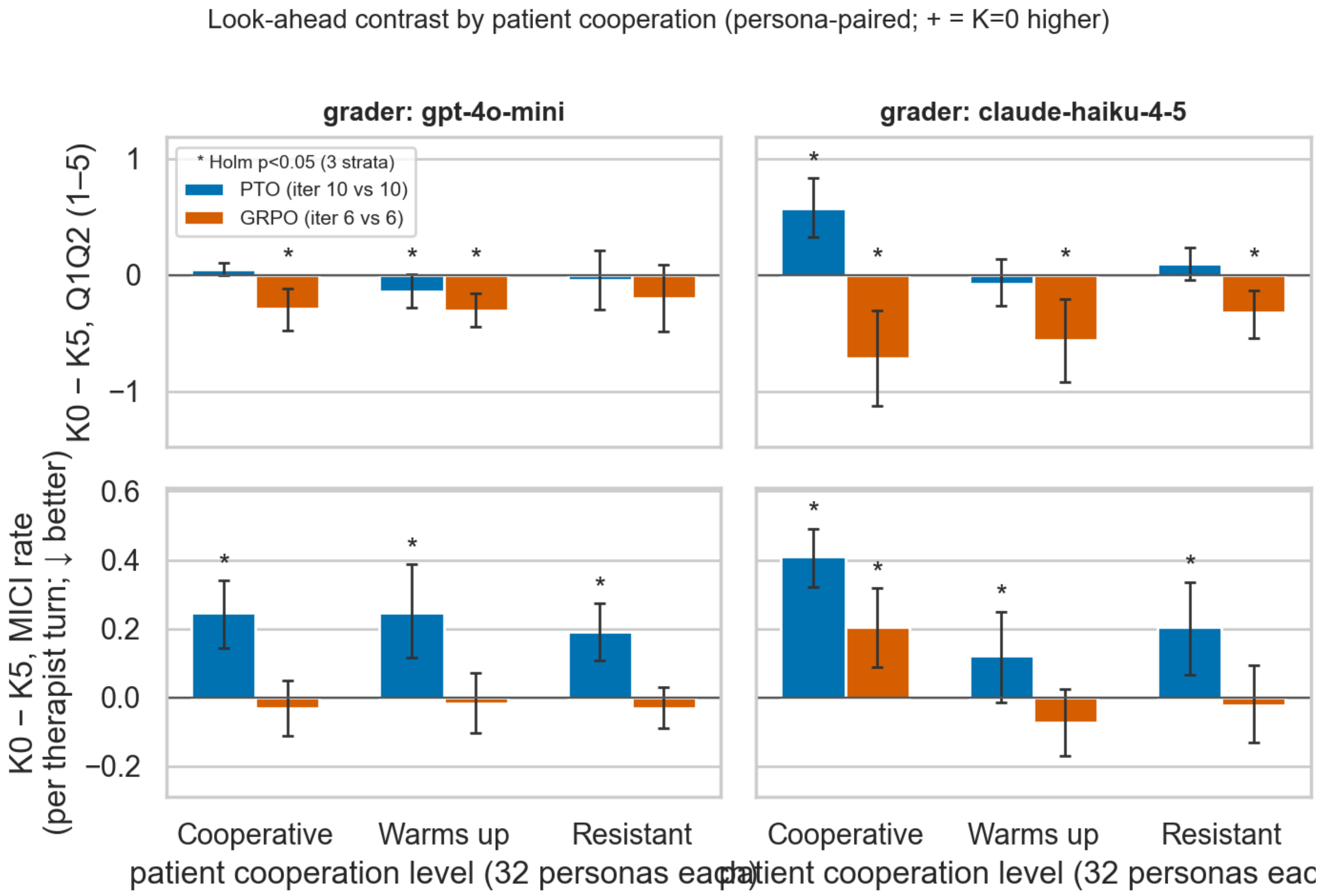
Y AXIS
Over-praise instances per therapist turn. Higher is worse.

SERIES
All four arms; K=0 solid, K=5 dashed. Shading marks the first iteration at which the PTO K contrast cleared Holm within its family.

CAVEAT
Per-turn rate — the denominator moves between arms. Read against the per-session panel two slides back. Coder: gpt-4o-mini.

source: `lookahead/behaviour/figures/k_overpraise_trajectory_gpt-4o-mini.png`

The contrast split by how cooperative the patient is



X AXIS

Patient stratum: Cooperative / Warms up / Resistant, 32 personas each.

Y AXIS

Paired K=0 − K=5 at each method's matched endpoint. Rows: Q1+Q2 (higher better) and MICI (lower better, so positive = K=0 worse).

SERIES

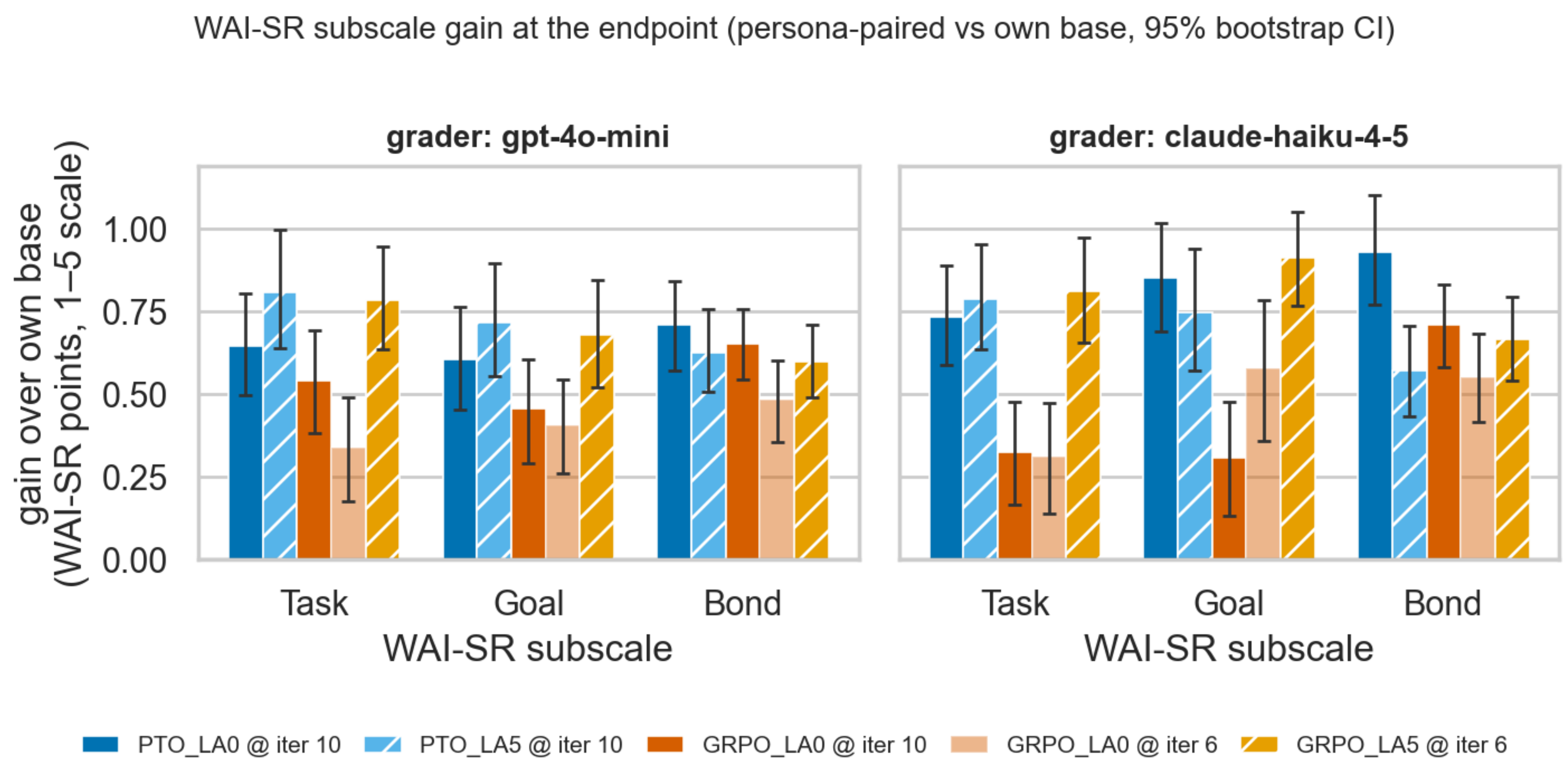
PTO and GRPO bars, coloured by the method. Whiskers = 95% bootstrap CI. Star = Holm p < .05 across the three strata.

COLUMNS

Left = training oracle, right = held-out judge.

source: `lookahead/behaviour/figures/hetero.png`

WAI-SR subscale gains at each arm's endpoint



X AXIS
WAI-SR subscale: Bond, Goal, Task.

Y AXIS
Gain over that arm's OWN base, paired on persona, 95% bootstrap CI.

SERIES
One bar per arm; K=5 bars hatched. Left panel = training oracle, right = held-out judge.

NOTE
This is an arm-vs-own-base figure, NOT a K contrast. n = 96 paired.

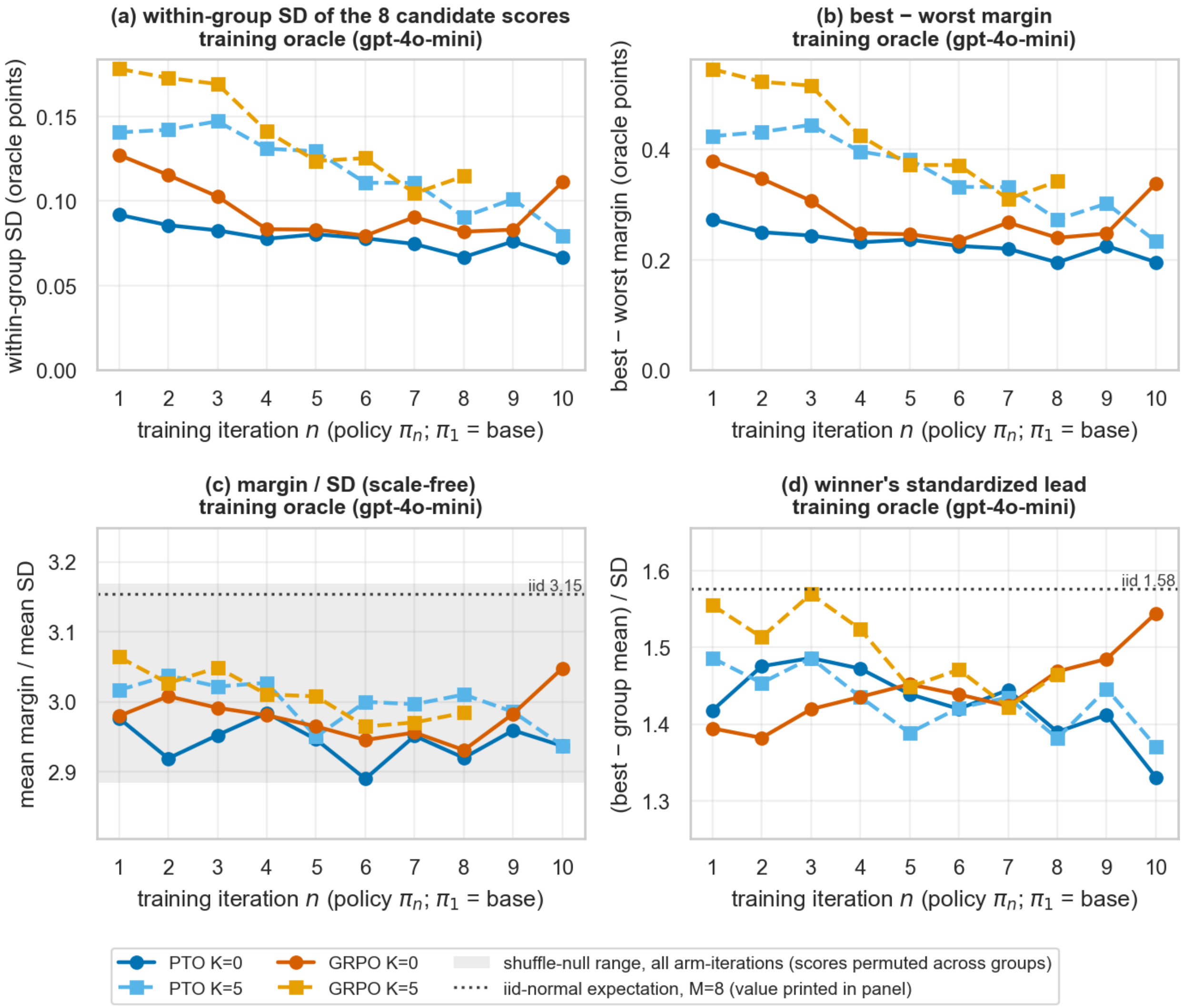
source: `lookahead/behaviour/figures/wai.png`

EXP3 · MECHANISM

Exp3 — inside the training signal

These read the trainer's own per-candidate records: what the oracle scored, what it selected, and how faithful a short-prefix score is to the finished conversation.

Dispersion of the reward within each sampled group



X AXIS

Training iteration.

Y AXIS

(a) within-group SD of the 8 candidate scores; (b) best-minus-worst margin; (c) margin ÷ SD; (d) the winner's standardized lead.

REFERENCES

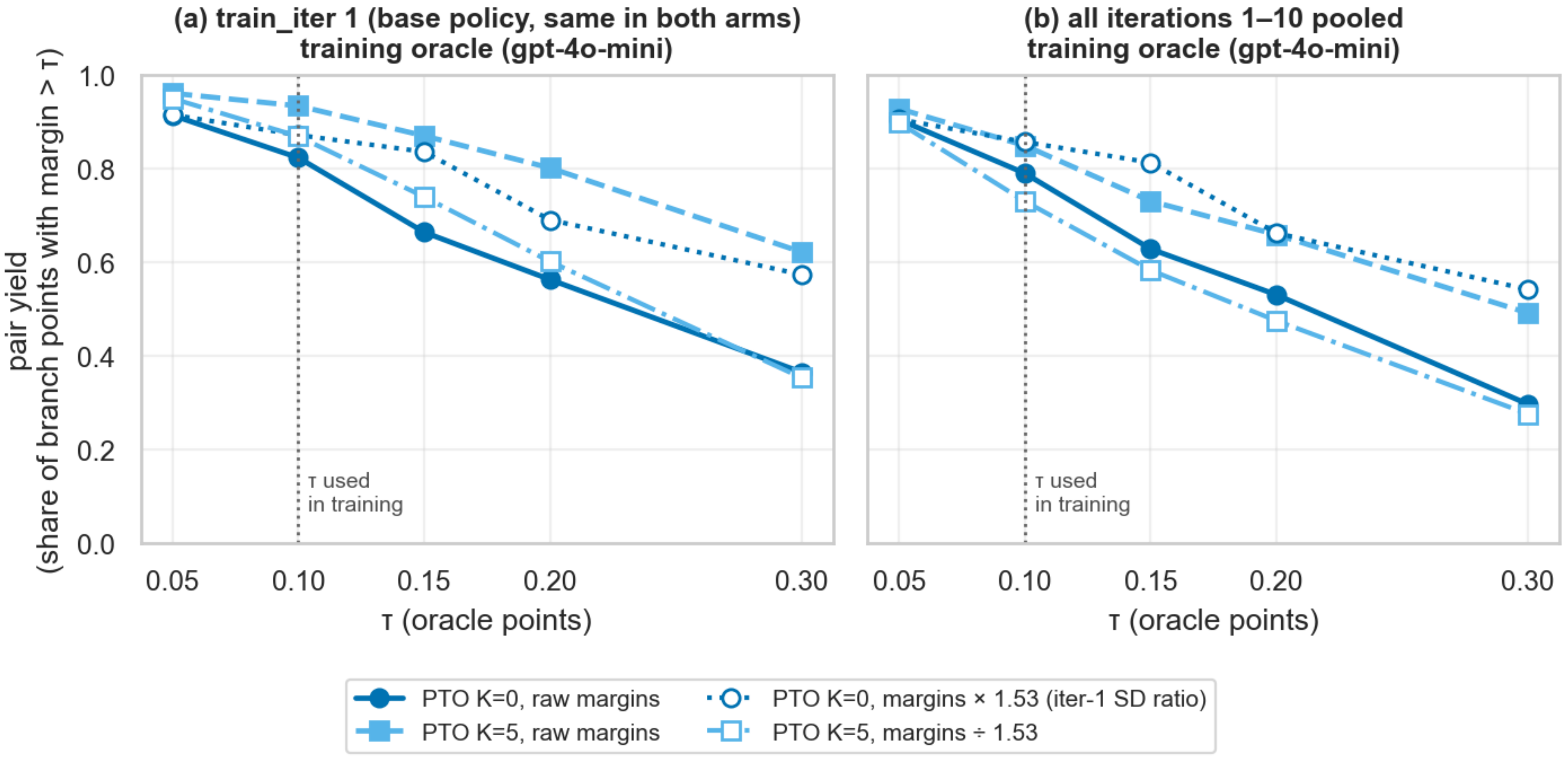
Panel (c) has the iid-normal expectation 3.15 dotted and a shuffle null shaded; panel (d) has 1.58 dotted.

UNIT

One GROUP = the 8 candidates sampled at one branch point, keyed (arm, iteration, conversation_id, branch_id). Grader: gpt-4o-mini.

source: `lookahead/mechanism/figures/dispersion.png`

How many preference pairs survive the τ filter



X AXIS
τ, the minimum best-worst margin required to emit a preference pair.

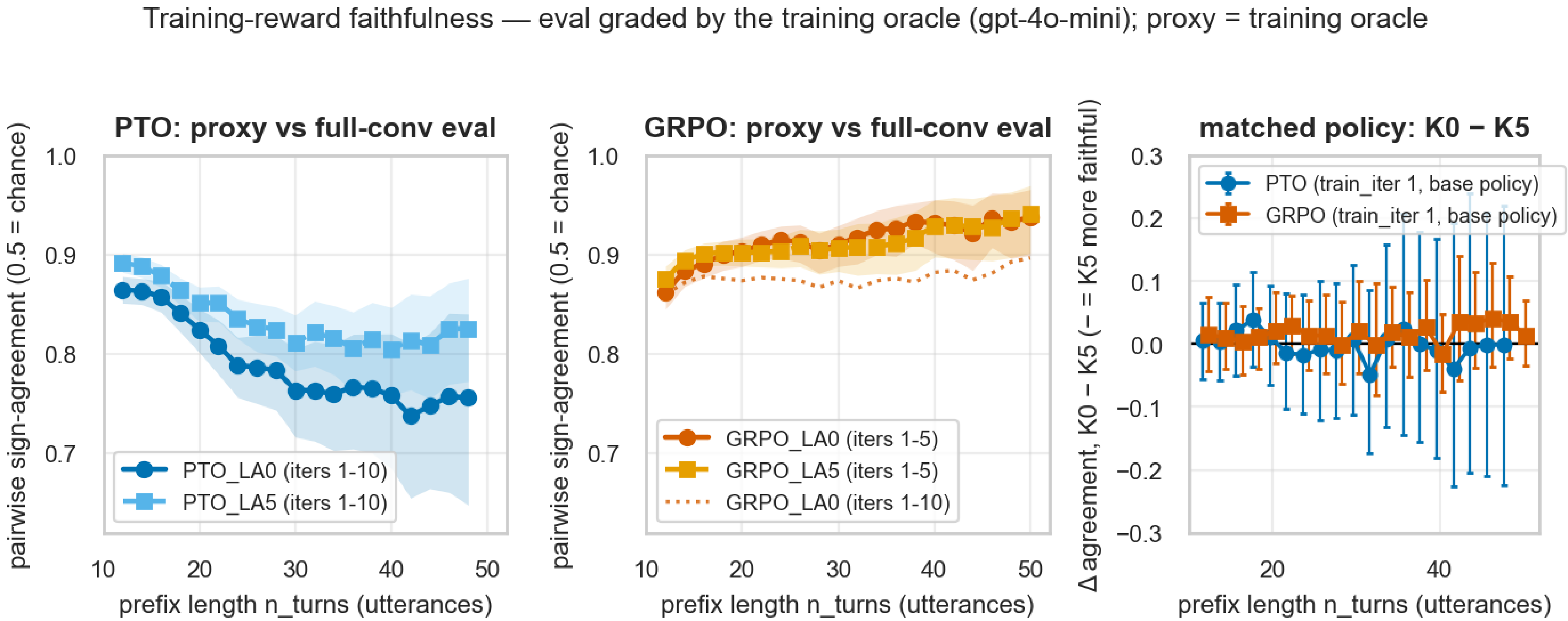
Y AXIS
Share of branch points that yield a pair at that τ.

SERIES
(a) iteration 1 only, where both arms share the same base policy; (b) iterations 1–10 pooled. Solid circle = K=0, dashed square = K=5.

OPEN MARKERS
The same curves after rescaling by the iteration-1 within-group SD ratio (1.53). Vertical dotted line = τ = 0.1, the value used.

source: `lookahead/mechanism/figures/dispersion_tau.png`

Does a short-prefix score predict the finished conversation?



X AXIS
Prefix length in turns — how much of the conversation the training reward saw.

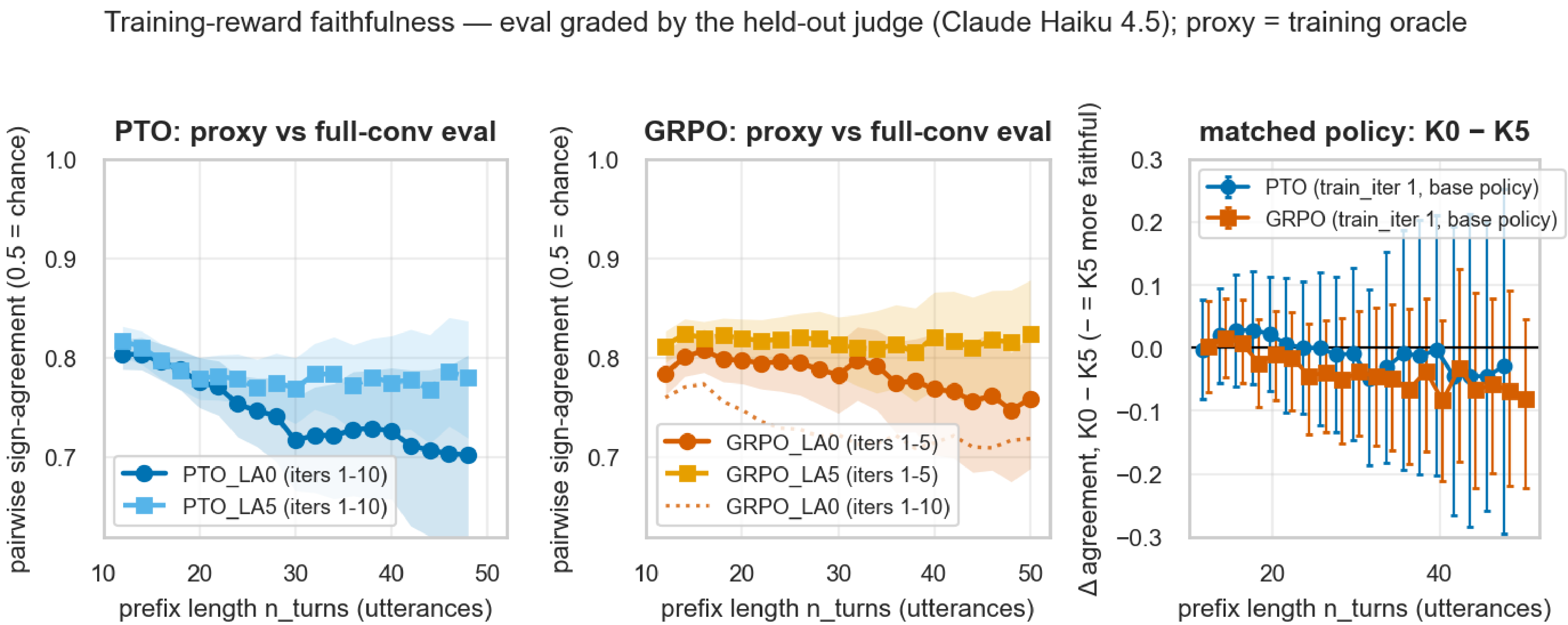
Y AXIS
Pairwise rank agreement between that short-prefix score and the full-conversation Q1+Q2. 0.5 = chance.

SERIES
(a) PTO, (b) GRPO; K=0 solid circle, K=5 dashed square; ribbons = 95% cluster bootstrap over conversations. (c) = the K difference at a matched policy (iteration 1), where negative means K=5 agreed more.

⚠ DEFECT
The GRPO K=5 series pools iterations 1–6 under a column labelled 1–5 (faithfulness.py:110). The GRPO panel is not like-for-like.

source: `lookahead/mechanism/figures/faithfulness.png`

The same question, scored by the held-out judge



X AXIS
Prefix length in turns.

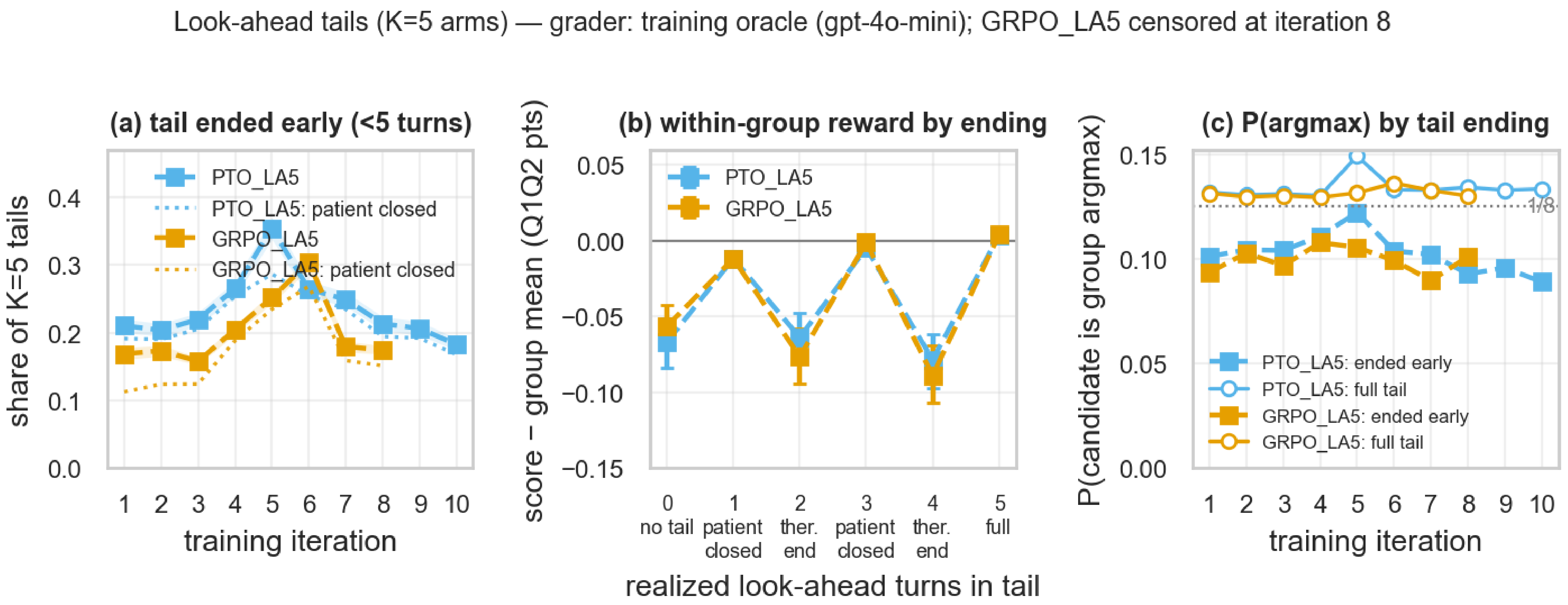
Y AXIS
Rank agreement between the training-oracle prefix score and the full-conversation score AS GRADED BY Claude Haiku 4.5. 0.5 = chance.

SERIES
Same layout as the previous slide.

⚠ DEFECT
Same GRPO K=5 pooling defect as the previous slide.

source: `lookahead/mechanism/figures/faithfulness_heldout.png`

What happens in the 5 simulated look-ahead turns



PANEL A
Share of candidates whose 5-turn tail ended early, by iteration, with the patient-closed share dotted underneath. 95% bootstrap CI.

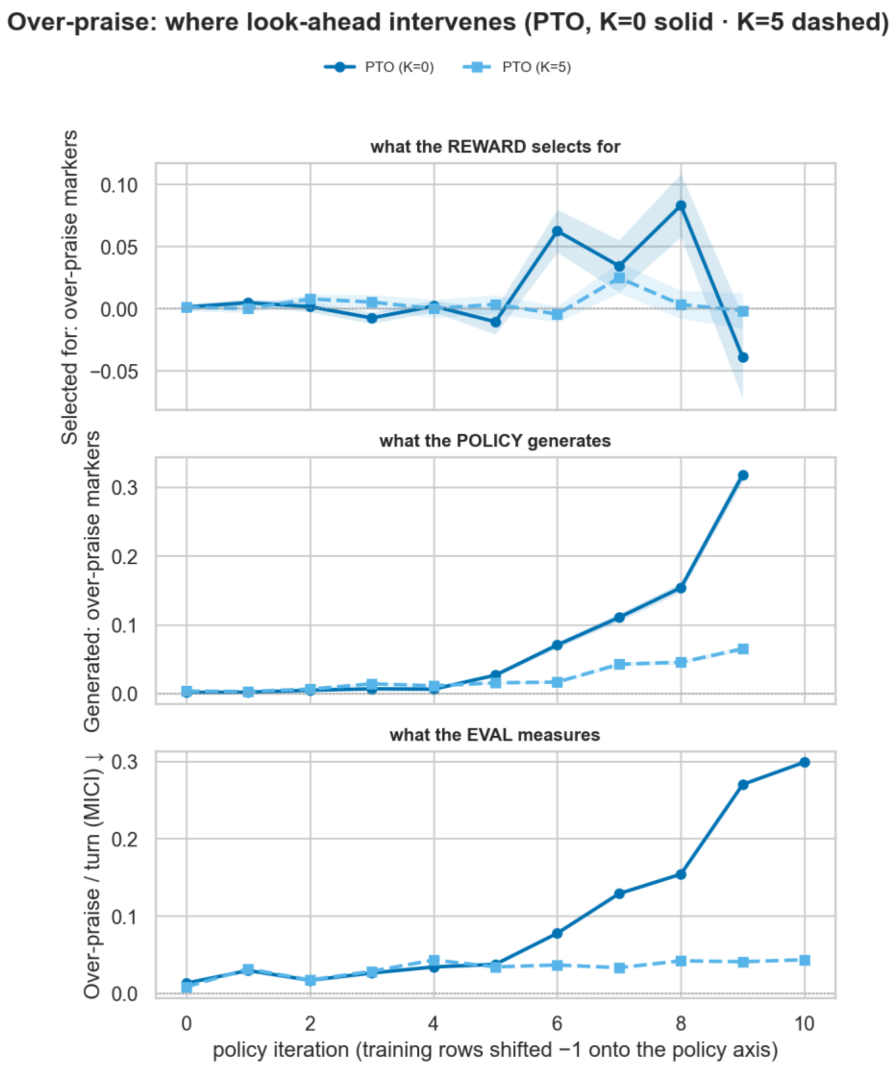
PANEL B
Within-group reward (score minus group mean, in Q1+Q2 points) against the realized tail length. Odd = the patient closed; even = ended on a therapist turn.

PANEL C
Probability that a candidate is its group's argmax, split by whether the tail ended early. 1/8 = chance.

DATA
K=5 arms only — K=0 has no tail. Grader: gpt-4o-mini.

source: `lookahead/mechanism/figures/tail_audit.png`

One channel traced through selection, generation and evaluation



X AXIS
Policy iteration. Training rows are shifted by −1 so all three describe the same policy.

TOP ROW
What the training reward SELECTS for within a group (±1 SE over groups).

MIDDLE ROW
What the policy GENERATES, over every sampled candidate.

BOTTOM ROW
What the full-conversation eval MEASURES per therapist turn.

SERIES
Both PTO look-ahead arms; K=0 solid, K=5 dashed. Top two rows record the training oracle's own choices and are grader-invariant.

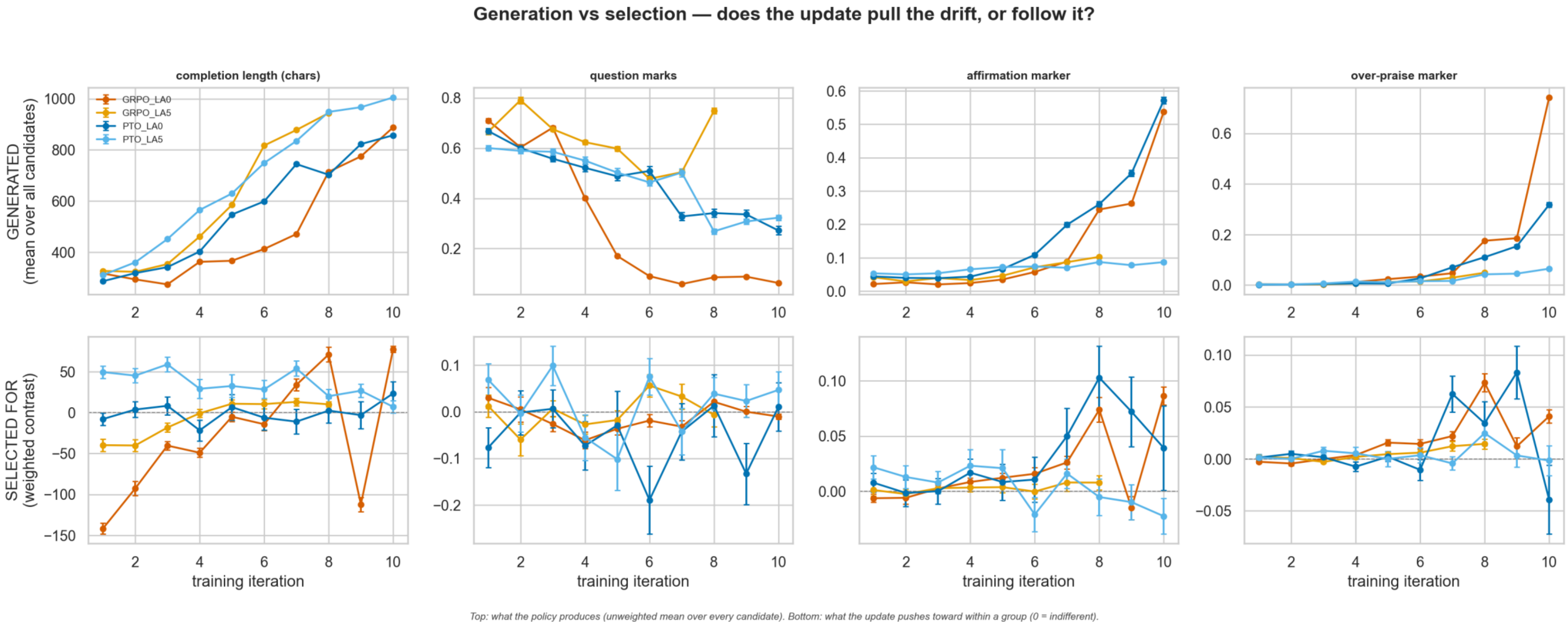
source: `lookahead/mechanism/figures/k_mechanism_overpraise.png`

EXP3 · TRAINING SIGNAL

Exp3 — the preference data itself

Built from the per-generation records each trainer writes: every candidate, its score, and which one was chosen.

What the policy generates vs what the oracle selects



X AXIS
Training iteration.

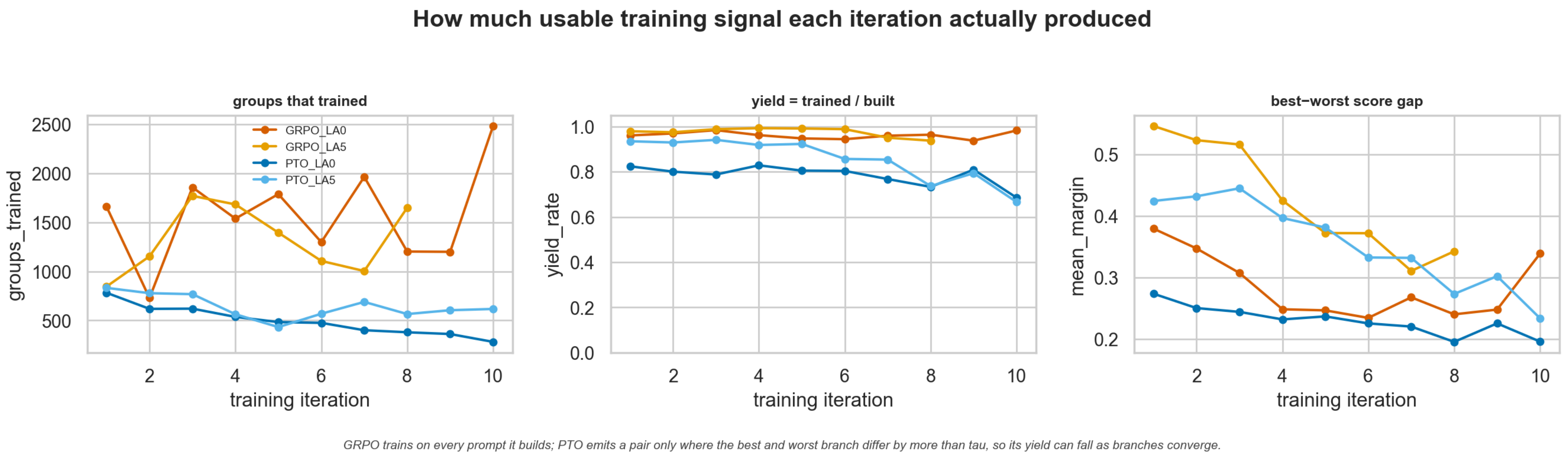
Y AXIS
Two quantities on the same axis: the rate at which a behaviour appears in generated candidates, and the weight the oracle's selection puts on it.

SERIES
One line per arm.

DATA
Read from iteration_N/eda/generations.jsonl — the trainer's own record of every candidate it sampled. Grader: gpt-4o-mini.

source: `arms/preference/figures/gpt-4o-mini/generation_vs_selection.png`

How many usable training rows each iteration produced



X AXIS
Training iteration.

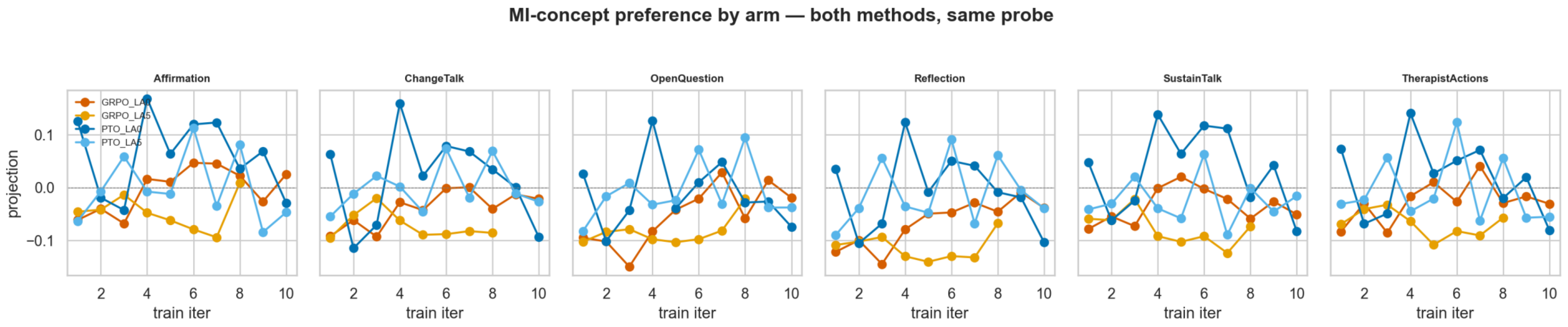
Y AXIS
Number of preference pairs emitted (PTO) — branch points where the best-worst margin cleared τ = 0.1.

SERIES
One line per PTO arm.

NOTE
GRPO has no equivalent: every prompt produces a gradient step, so it cannot yield zero rows.

source: `arms/preference/figures/gpt-4o-mini/training_signal_yield.png`

What the chosen-minus-rejected direction pushes toward



X AXIS
Behaviour category.

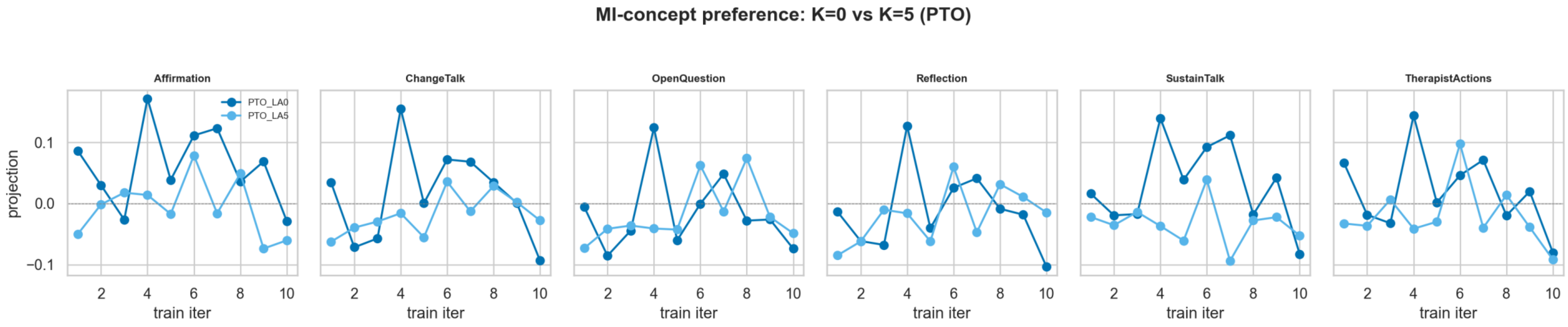
Y AXIS
Net direction of the update — how much more often a category appears in the chosen completion than the rejected one.

SERIES
One group of bars per arm.

DATA
Computed over every emitted preference pair. Grader: gpt-4o-mini.

source: `arms/preference/figures/gpt-4o-mini/update_category_by_arm.png`

The same categories, K=0 against K=5



X AXIS
Behaviour category.

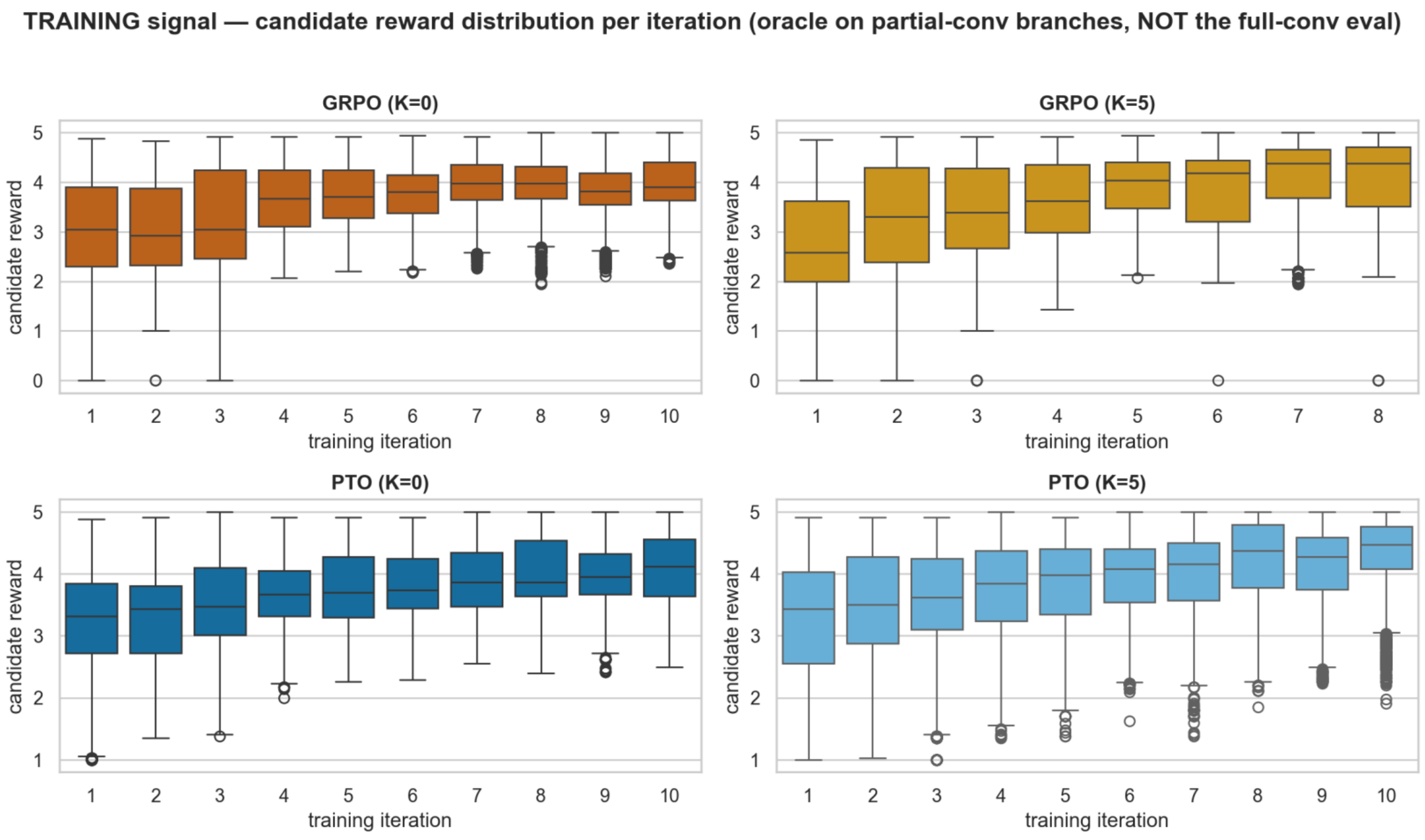
Y AXIS
Preference weight on that category.

SERIES
K=0 against K=5, PTO arms.

DATA
Every emitted preference pair, both PTO arms.

source: `arms/preference/figures/gpt-4o-mini/pref_category_K0_vs_K5.png`

Distribution of the training reward, per arm



X AXIS
Training-reward value (Q1+Q2 of the scored candidate).

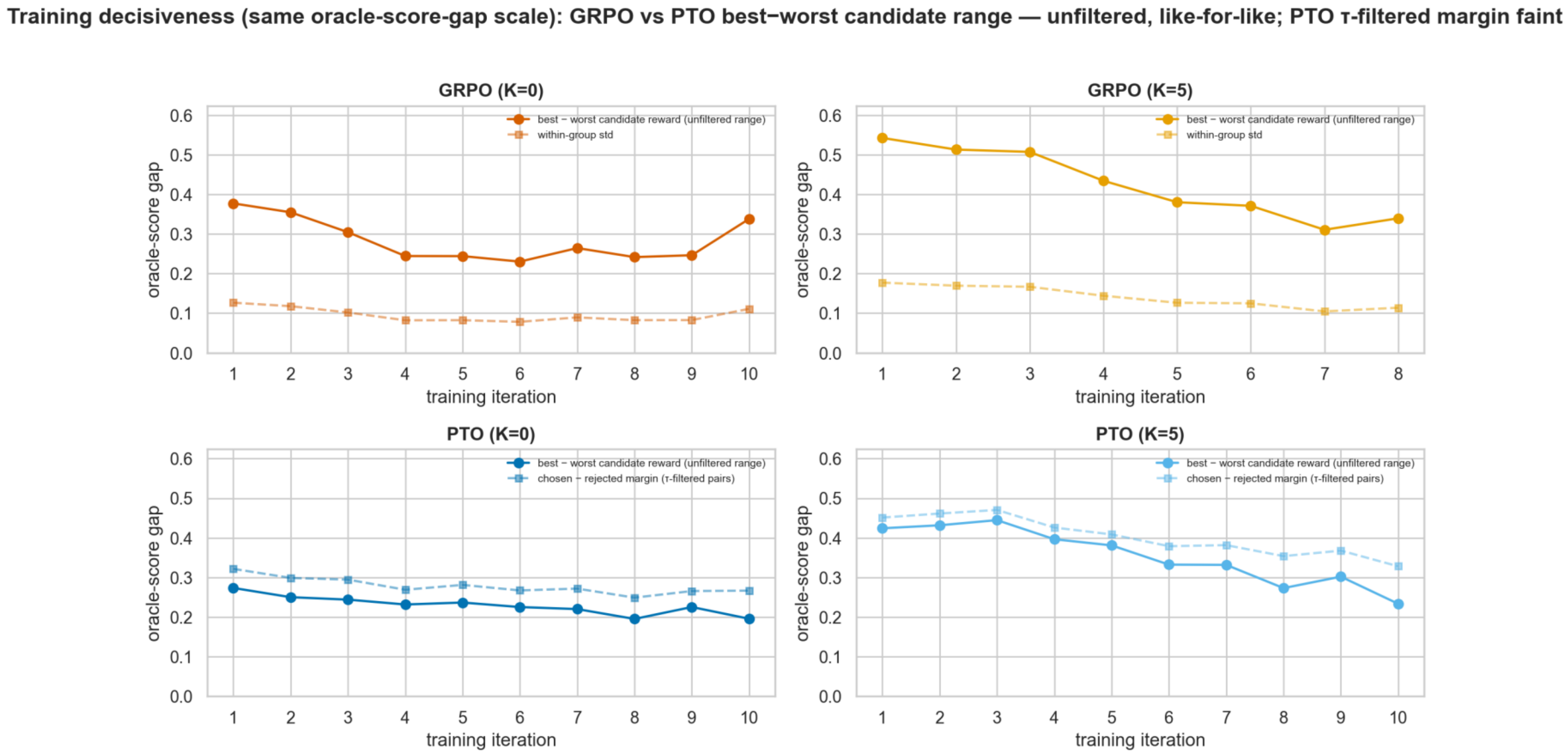
Y AXIS
Density / count of candidates.

SERIES
One panel or colour per arm.

DATA
Every scored candidate across all iterations, from generations.jsonl. Grader: gpt-4o-mini.

source: `arms/training/figures/gpt-4o-mini/reward_distribution_by_arm.png`

The advantage signal the two methods see



X AXIS
Training iteration or advantage value, per panel.

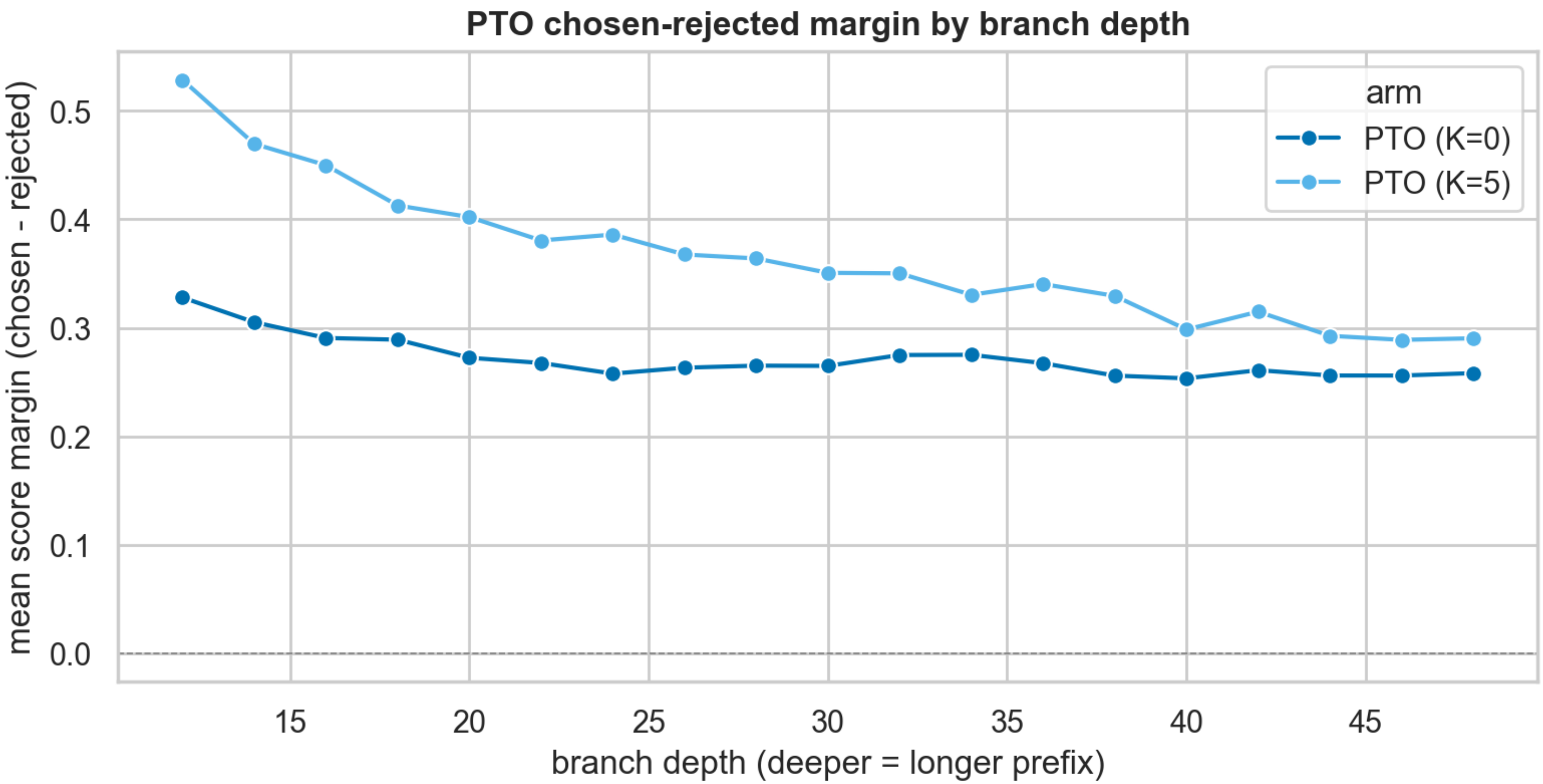
Y AXIS
The group-relative advantage GRPO computes beside the best-worst margin PTO selects on.

SERIES
One panel per method.

NOTE
The two are not the same quantity — this puts them side by side on a common axis rather than claiming equivalence.

source: `arms/training/figures/gpt-4o-mini/advantage_signal_sidebyside.png`

PTO's best-worst margin by depth in the trunk



X AXIS
Depth in the preference trunk — how many therapist turns into the conversation the branch point sits.

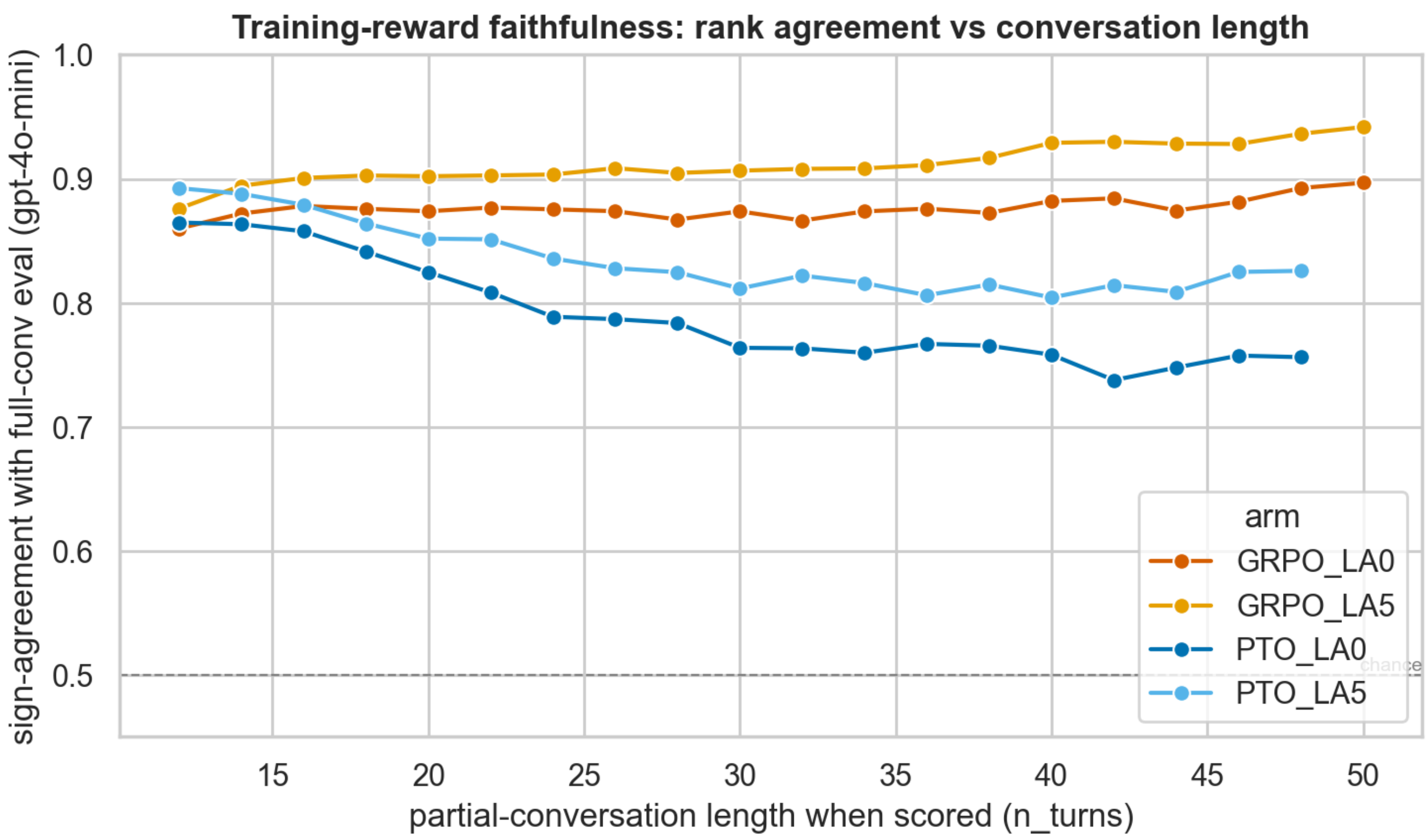
Y AXIS
Best-minus-worst score margin among the 8 candidates.

SERIES
One line per PTO arm.

NOTE
PTO's branch_id IS the trunk depth and repeats across conversations, so aggregation is keyed on (conversation_id, branch_id).

source: `arms/training/figures/gpt-4o-mini/pto_margin_by_depth.png`

Reward reliability against prefix length



X AXIS
Conversation-so-far length in turns.

Y AXIS
Agreement between the score at that prefix and the full-conversation score. 0.5 = chance.

SERIES
Per arm.

WHY IT MATTERS
MIN_CONV_LENGTH = 12 was set from this curve: no training context shorter than that is used.

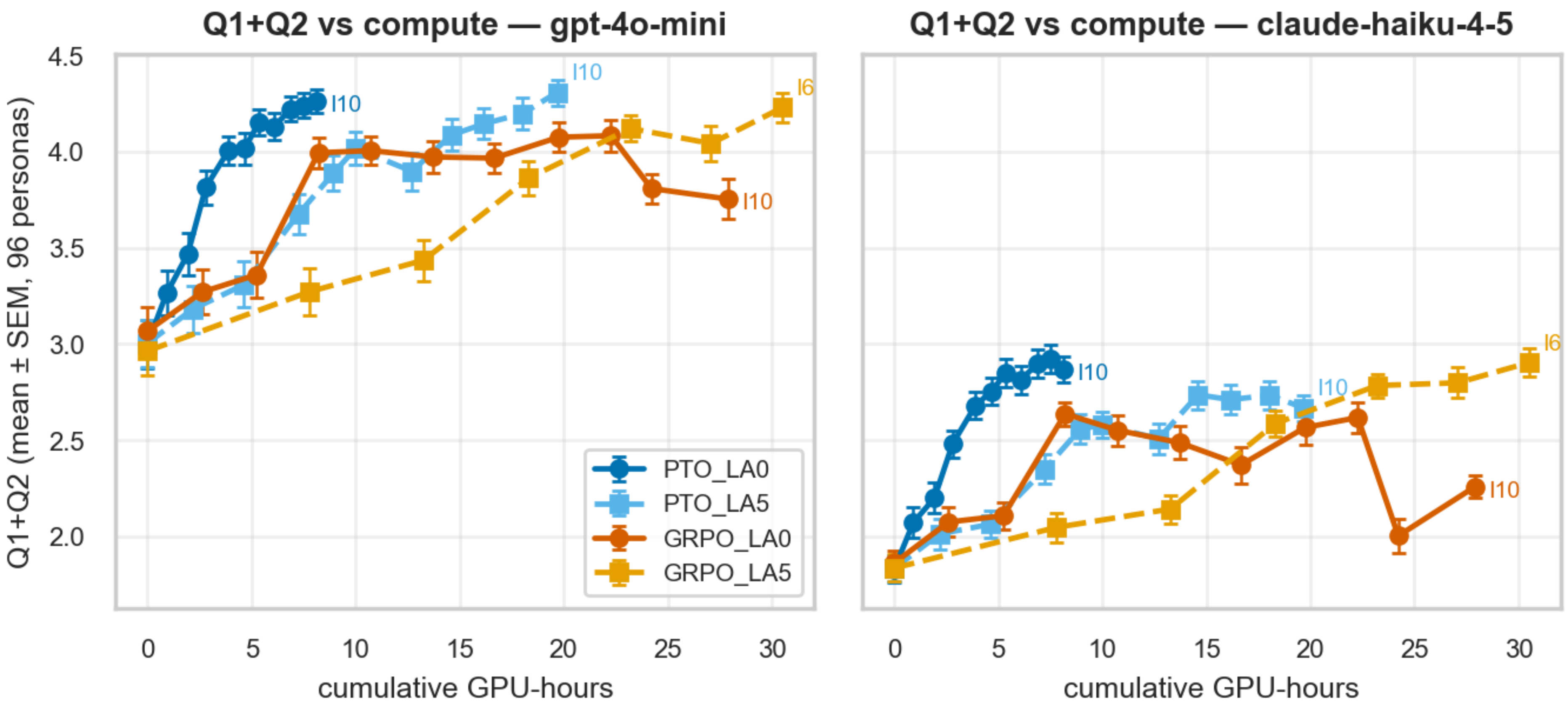
source: `arms/training/figures/gpt-4o-mini/reward_reliability_curve.png`

EXP3 · COMPUTE

Exp3 — what each arm cost

GPU-hours reconstructed from artifact modification times, and API call counts. An iteration is not a fixed unit of spend.

Score against cumulative GPU-hours instead of iteration



X AXIS
Cumulative GPU-hours, reconstructed from artifact mtimes. Iteration 0 sits at 0 h.

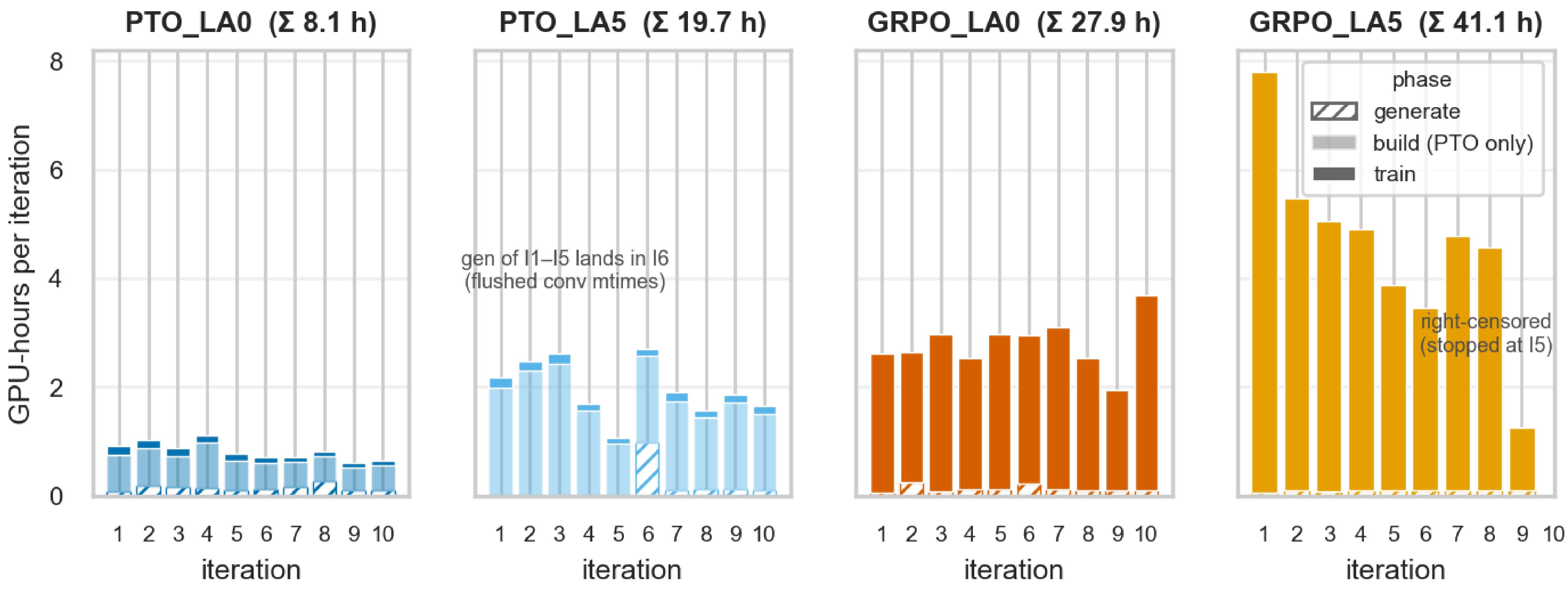
Y AXIS
Q1+Q2, mean ± SEM over the 96 personas.

SERIES
Four arms: PTO K=0, PTO K=5, GRPO K=0, GRPO K=5. K=0 solid + circle, K=5 dashed + square. GRPO K=5's line ends at iteration 6, its last scored state. The last point of each arm is labelled with its iteration.

READ THE SPACING
Marker spacing along x is uneven by construction: an arm with cheap iterations gets many markers close together.

source: `compute/cost/figures/compute_trajectory.png`

Where each arm's GPU-hours went



X AXIS
Training iteration, one panel per arm.

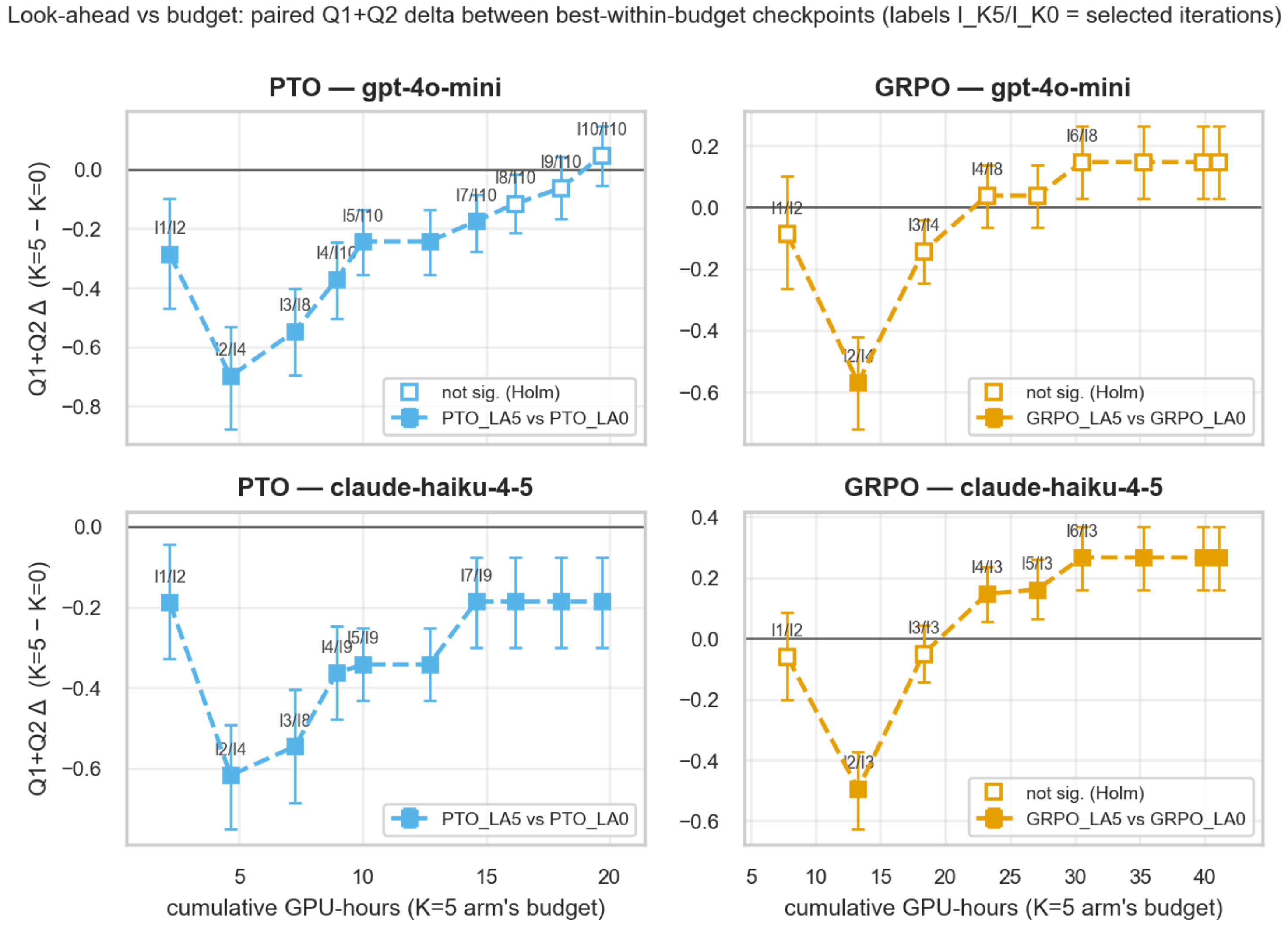
Y AXIS
GPU-hours, stacked by phase: generate (hatched), build (light, PTO only), train (solid).

SERIES
Panel title = that arm's total.

KNOWN ARTEFACT
PTO K=5's generation for iterations 1–5 lands in iteration 6 because conversation mtimes were batch-flushed; the cumulative total is right, the per-iteration split for that arm is not.

source: `compute/cost/figures/cost_breakdown.png`

The K contrast read at matched budget rather than matched iteration



X AXIS
The K=5 arm's cumulative GPU-hours — the budget.

Y AXIS
Paired Q1+Q2 difference between the best checkpoint each arm reached WITHIN that budget. Above zero = the K=5 arm is ahead.

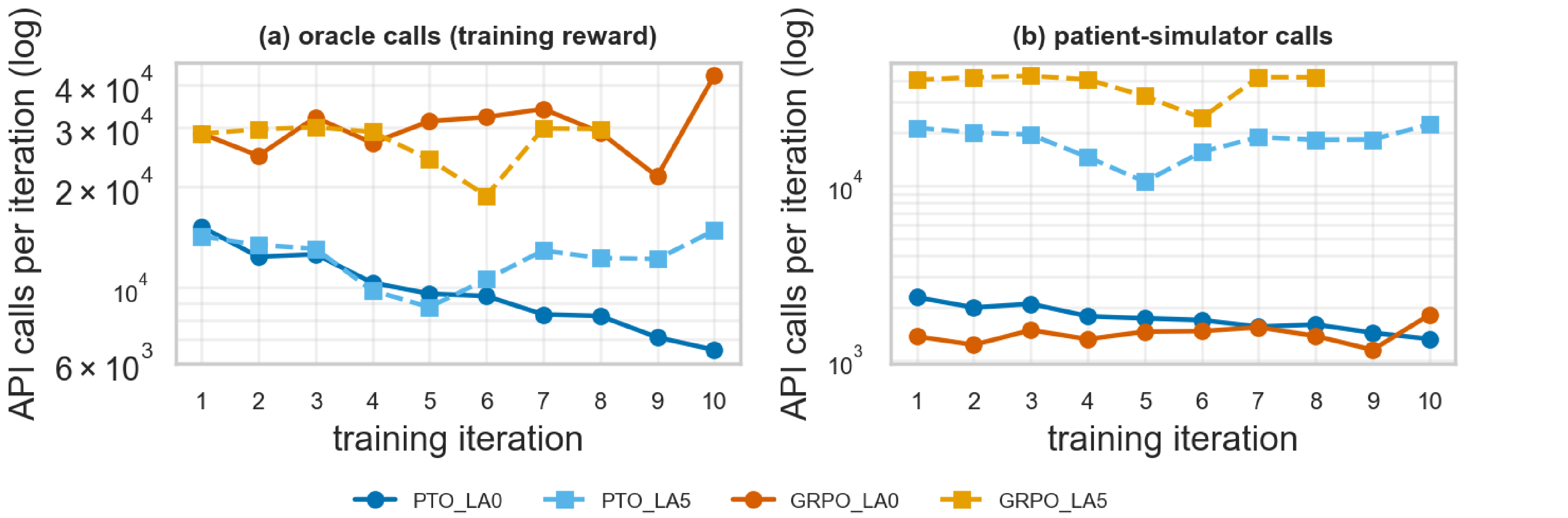
SERIES
Rows = grader, columns = method. Bars = bootstrap 95% CI; hollow = Holm p ≥ .05. Labels show which iterations were selected.

NOTE
Sign here is the opposite of the k_table1 convention — this one is K=5 minus K=0. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

source: `compute/cost/figures/budget_sweep.png`

API calls per training iteration

API calls per training iteration = eval convs of pi_{n-1} + training-time calls (GRPO rescaled to its true step count); GRPO_LA5 censored at iteration 8



X AXIS
Training iteration.

Y AXIS
Number of API calls, LOG scale.

SERIES
(a) oracle calls for the training reward — Q1 and Q2 per scored candidate; (b) patient-simulator calls, including the K=5 tail turns. One line per arm.

DATA
Counts, not scores, so this figure is identical under both graders.

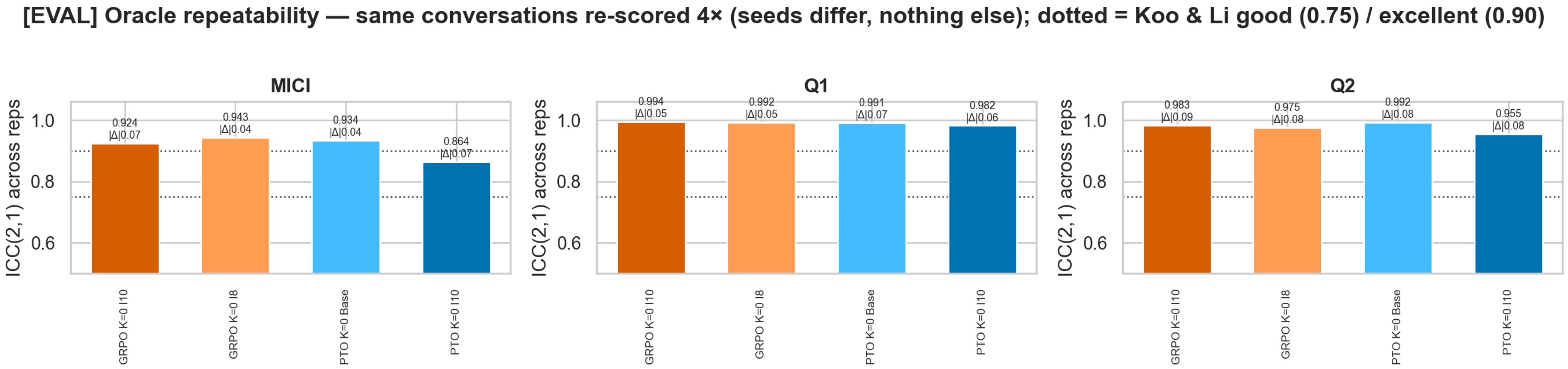
source: `compute/cost/figures/api_calls.png`

EXP3 · MEASUREMENT

Exp3 — how much the grader matters

Repeatability of the primary oracle, and everything the held-out judge does and does not reproduce.

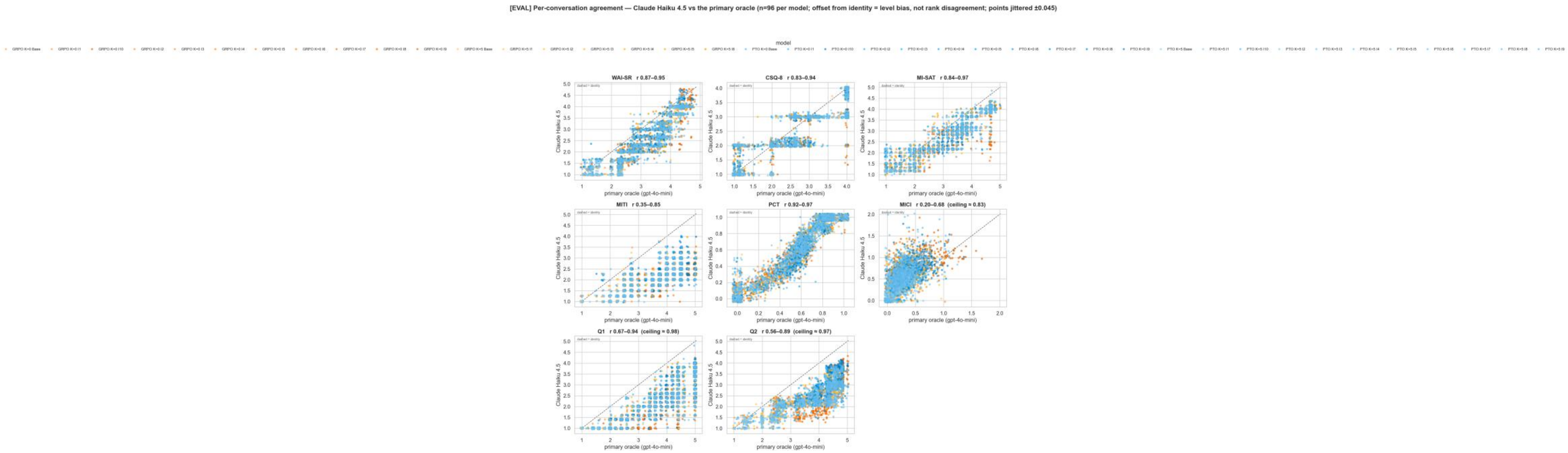
Repeatability of the primary oracle



X AXIS	Y AXIS	SERIES	REFERENCE LINES
Metric.	ICC(2,1) computed from 3 re-scorings of the SAME conversations by gpt-4o-mini.	One point per model × metric.	Dotted at 0.75 and 0.90 — the Koo & Li 'good' and 'excellent' thresholds.

source: `measurement/validity/figures/oracle_repeatability_icc.png`

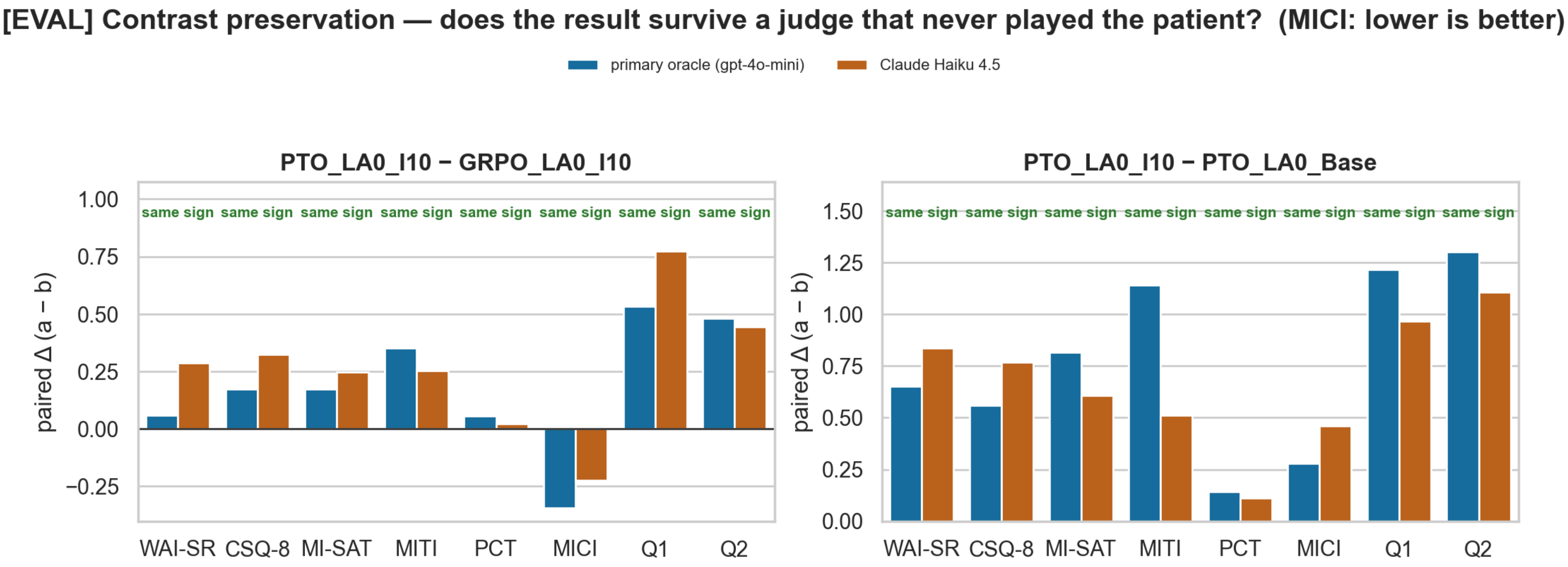
The two graders on the same conversations



X AXIS	Y AXIS	SERIES	REFERENCE
Score given by the primary oracle, gpt-4o-mini.	Score given by Claude Haiku 4.5 on the same conversation.	One panel per metric; all 40 model states pooled; one dot per conversation.	Dashed identity line. Vertical distance from it is level offset; scatter around the trend is rank disagreement.

source: `measurement/validity/figures/judge_agreement_scatter.png`

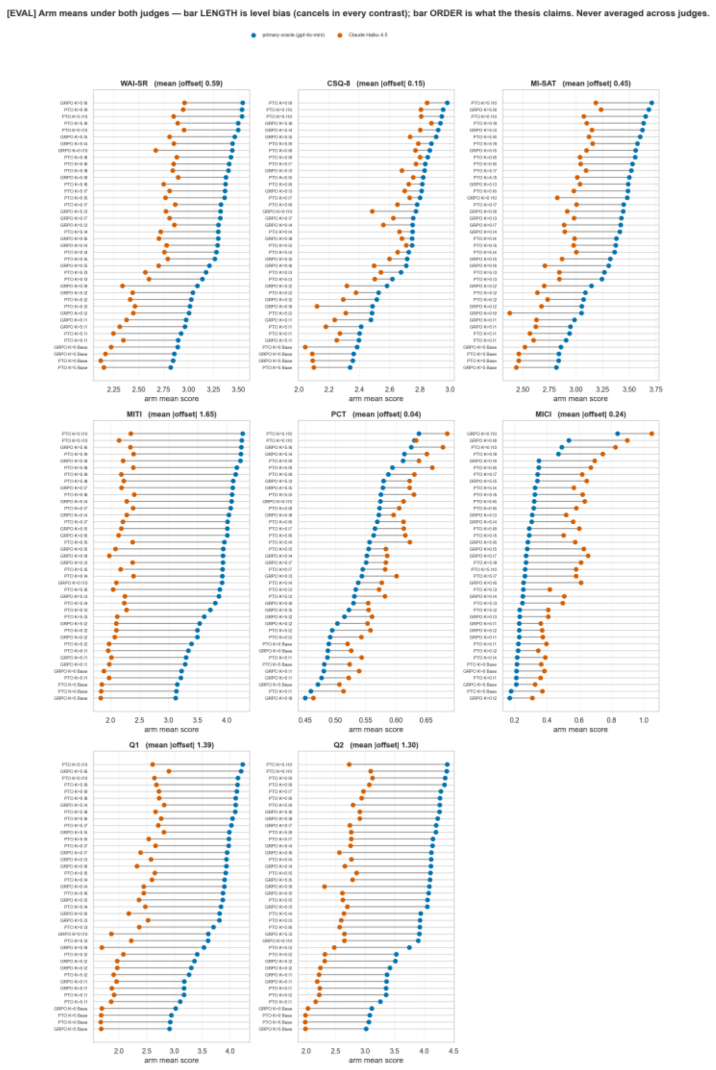
Do the two graders agree on the DIRECTION of each contrast?



X AXIS	Y AXIS	SERIES	SIGN
Contrast.	Effect size under each grader, side by side.	Primary oracle against held-out judge, at the K=0 endpoints.	Same-sign bars mean the contrast does not depend on the grader also having played the patient. MICI is lower-is-better.

source: `measurement/validity/figures/judge_contrast_preservation.png`

Arm means under each grader



X AXIS
Score.

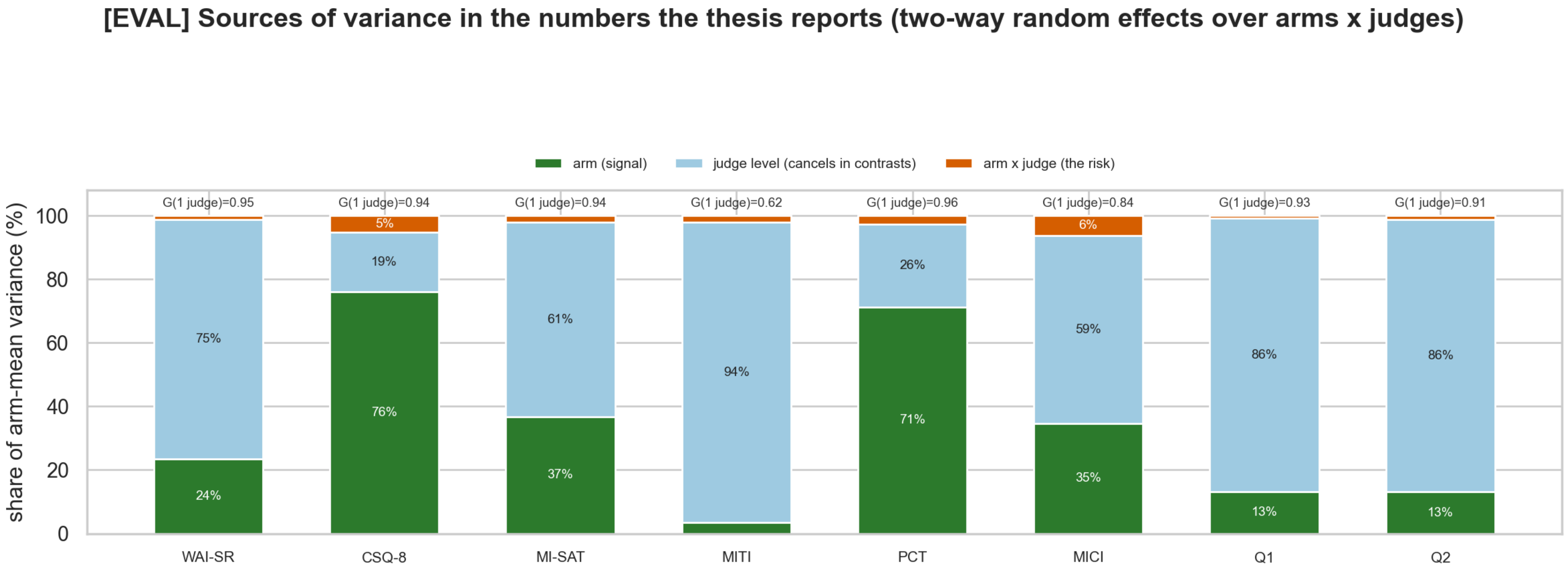
Y AXIS
Model state, all four arms.

SERIES
Each bar joins that state's mean under the primary oracle to its mean under Claude Haiku 4.5.

READ
Bar LENGTH is the level offset; bar ORDER is the ranking. The two are deliberately not averaged.

source: `measurement/validity/figures/multijudge_arm_means_dumbbell.png`

Where the variance in arm means comes from



X AXIS
Metric.

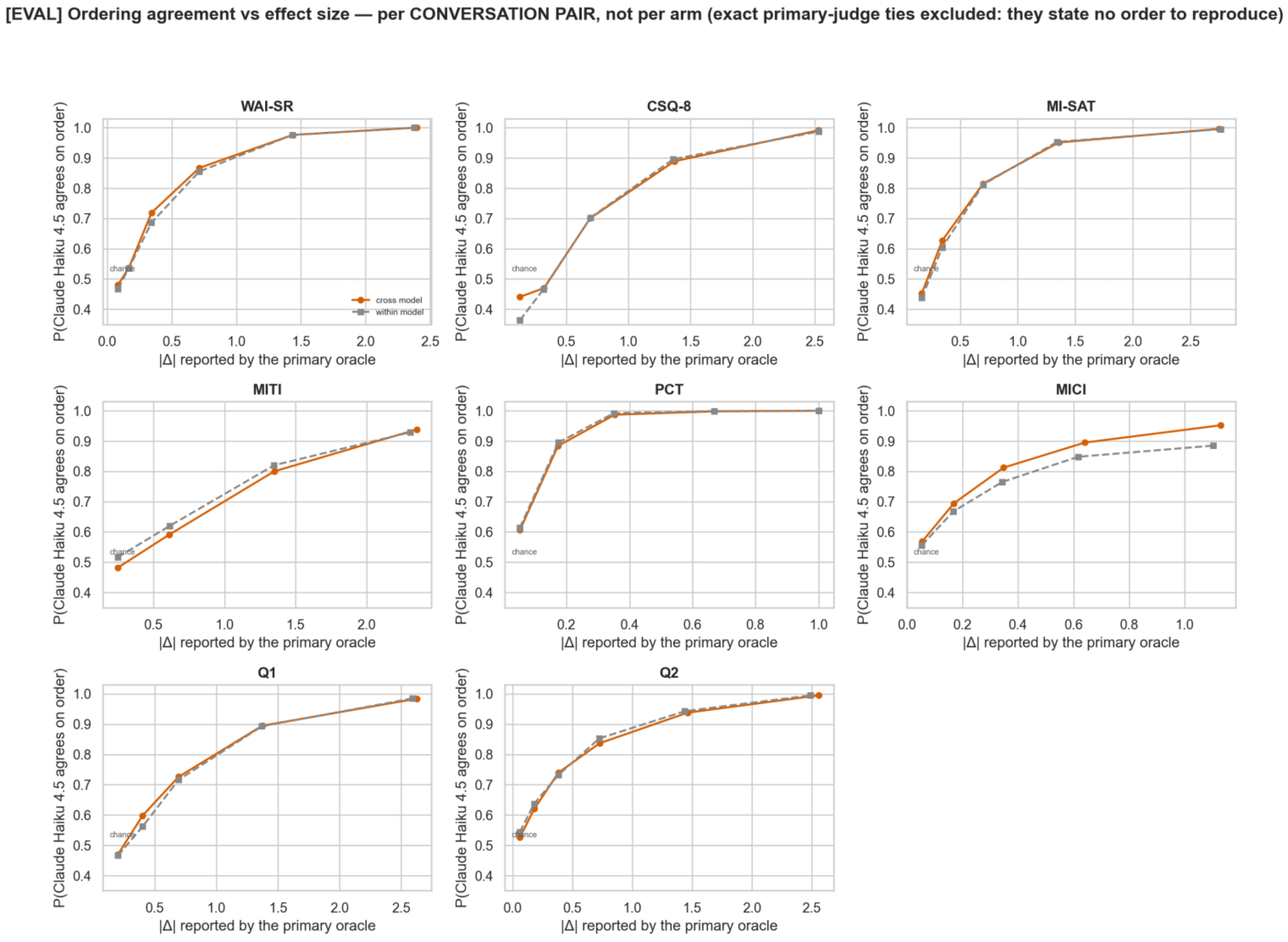
Y AXIS
Share of arm-mean variance attributed to each source: arm, judge level, and arm × judge.

SERIES
Stacked components across all judged model states.

NOTE
A judge-level component cancels in any contrast; the arm × judge component is the one that does not.

source: `measurement/validity/figures/multijudge_variance_decomposition.png`

Cross-grader agreement as a function of effect size



X AXIS
Size of the gap between two conversations, under the primary oracle.

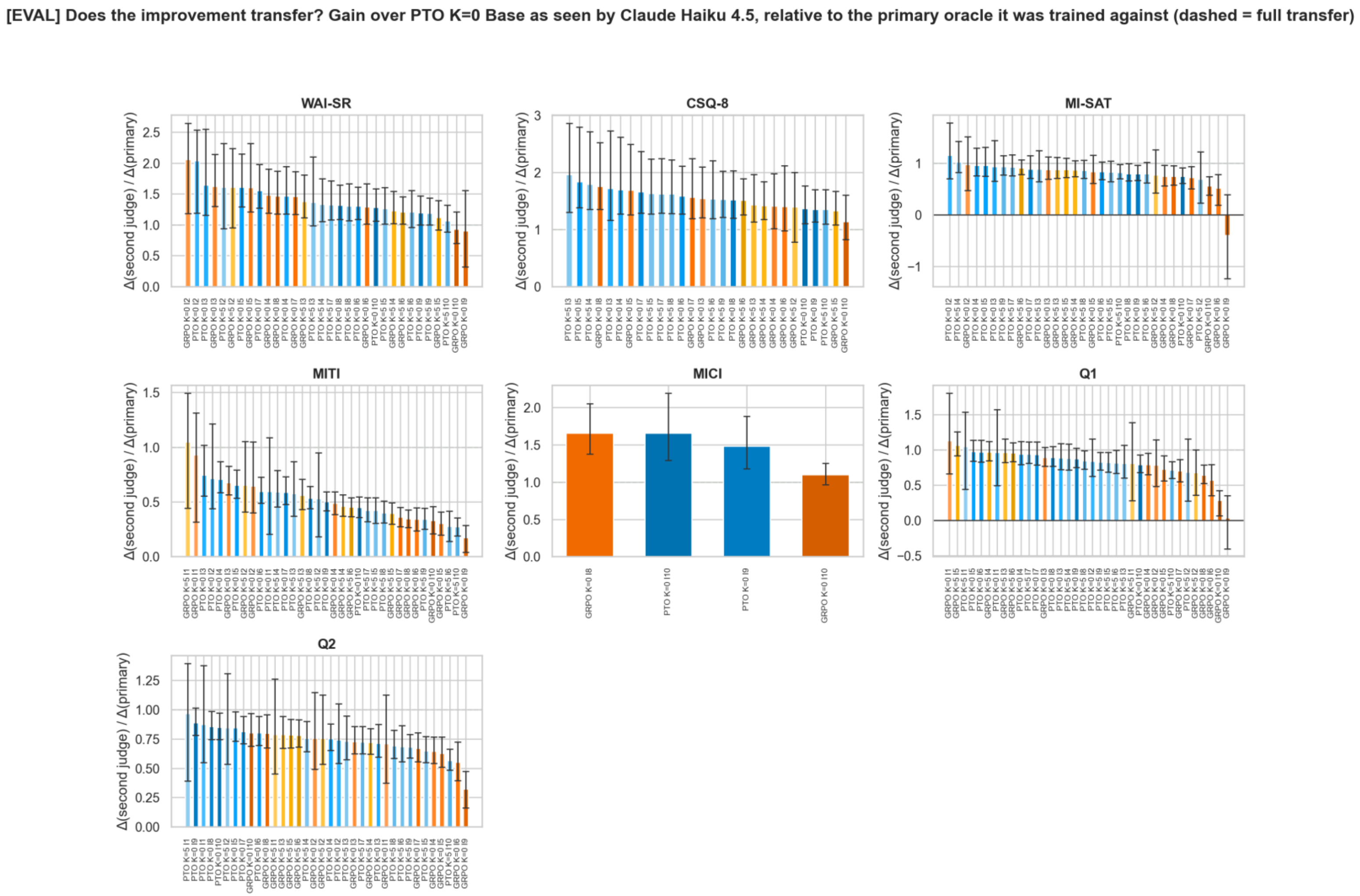
Y AXIS
Probability that Claude Haiku 4.5 orders that pair the same way.

SERIES
One curve, computed over conversation pairs.

USE
Reads how much resolving power a given gap carries per conversation.

source: `measurement/validity/figures/multijudge_concordance_curve.png`

Gain retention, every model state



X AXIS
Model state, all four arms.

Y AXIS
(gain under the held-out judge) ÷ (gain under the primary oracle), with persona-bootstrap CI.

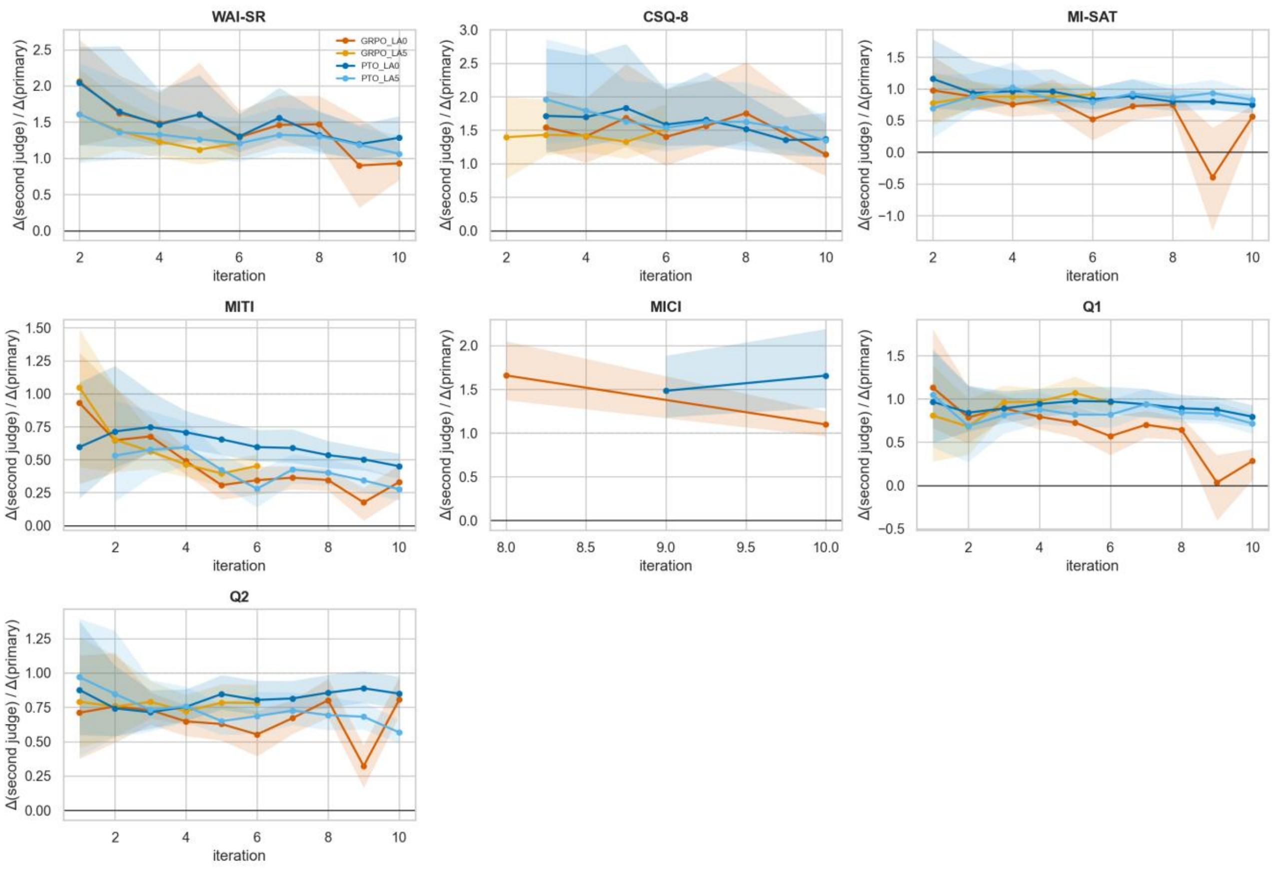
SERIES
One bar per state.

NOTE
Uniform bars across arms and one arm dropping alone are different situations; the figure shows which.

source: `measurement/validity/figures/multijudge_gain_retention.png`

Gain retention against iteration

[EVAL] When do the gains stop transferring? Retention vs iteration under Claude Haiku 4.5 (dashed = full transfer; a declining line = progressively fitting the trained-against grader)



X AXIS
Training iteration.

Y AXIS
Retention under Claude Haiku 4.5, referenced to the PTO K=0 base.

SERIES
One line per arm.

REFERENCE
1.0 means the held-out judge sees the whole of the gain the training oracle credited.

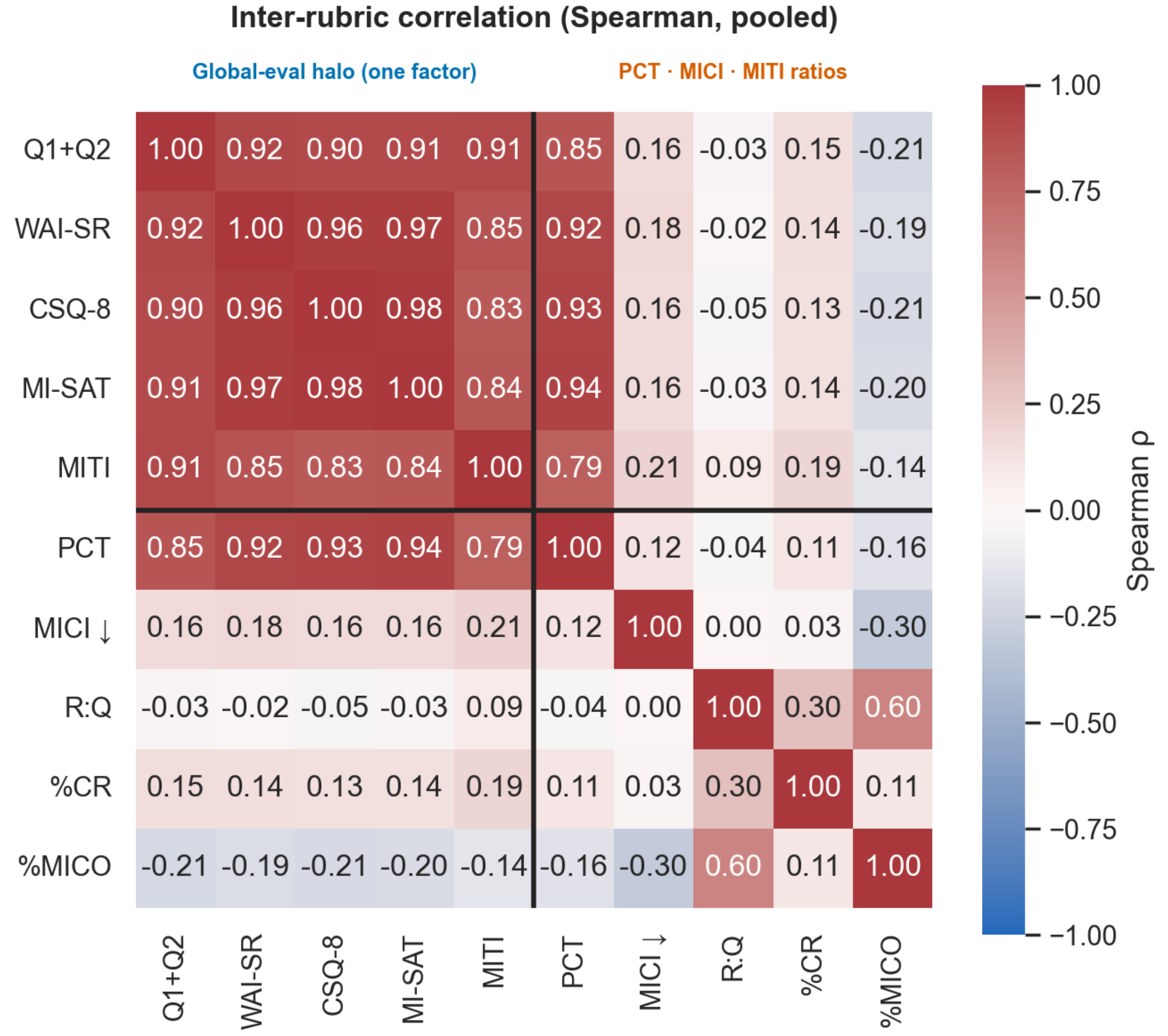
source: `measurement/validity/figures/multijudge_retention_trajectory.png`

EXP3 · RUBRICS

Exp3 — the instruments themselves

How the eight rubrics relate to each other, and what the item-level detail looks like.

How correlated are the eight rubrics?



AXES
Rubric × rubric.

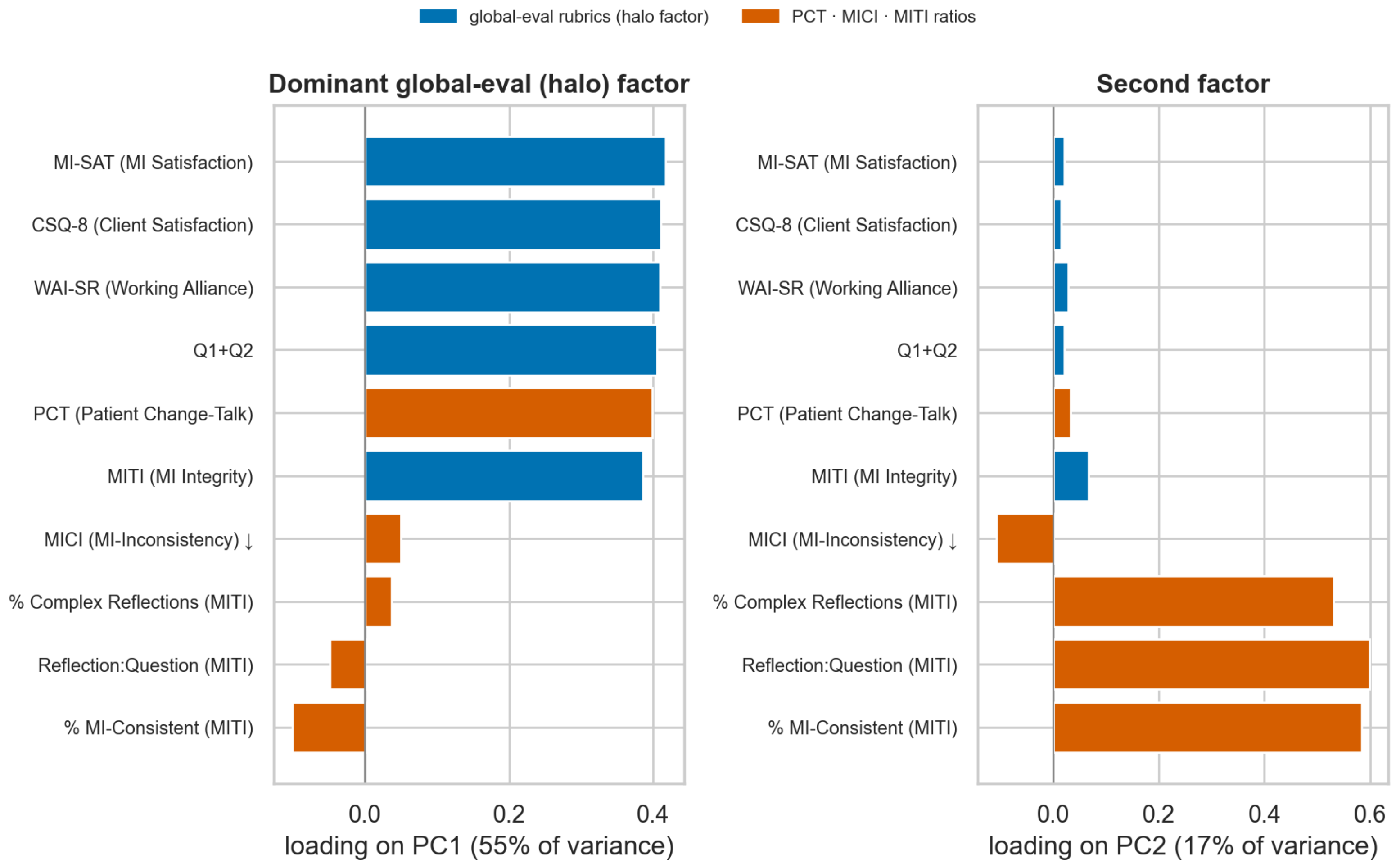
CELL
Correlation between the two rubrics' per-conversation scores, pooled over all model states.

DATA
Grader: gpt-4o-mini. MICI is lower-is-better, so its correlations with the others are expected to carry the opposite sign.

USE
Bears on whether the eight instruments are eight independent readings.

source: `arms/validity/figures/gpt-4o-mini/rubric_correlation.png`

Factor structure across the rubrics



X AXIS
Factor.

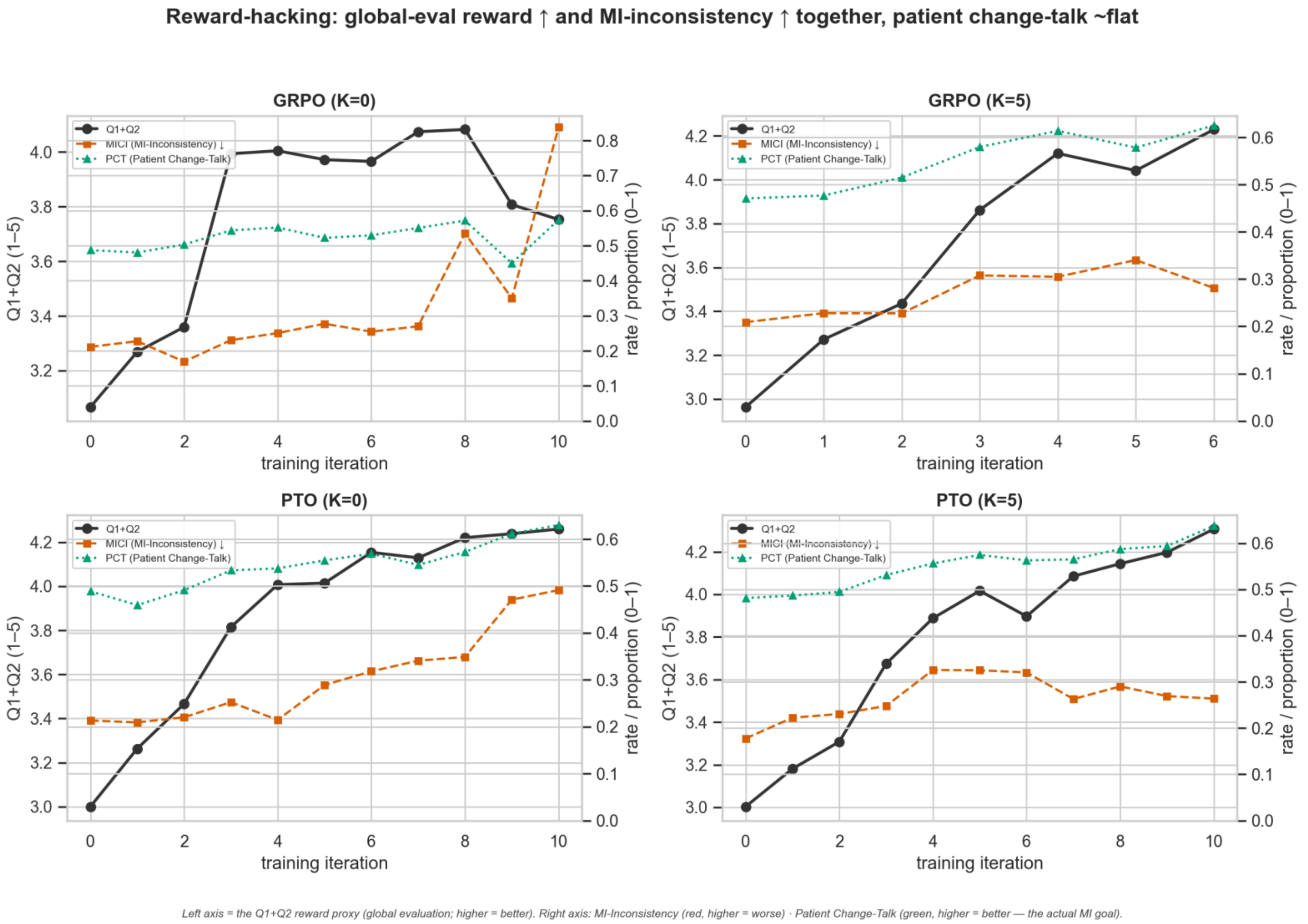
Y AXIS
Loading of each rubric on that factor.

DATA
Computed on the per-conversation score matrix, grader gpt-4o-mini.

NOTE
Descriptive factor extraction — no rotation claim is made on this slide.

source: `arms/validity/figures/gpt-4o-mini/factor_loadings.png`

The reward-hack panel



PANELS
The training reward beside the behaviour channels that move with it, over training iterations.

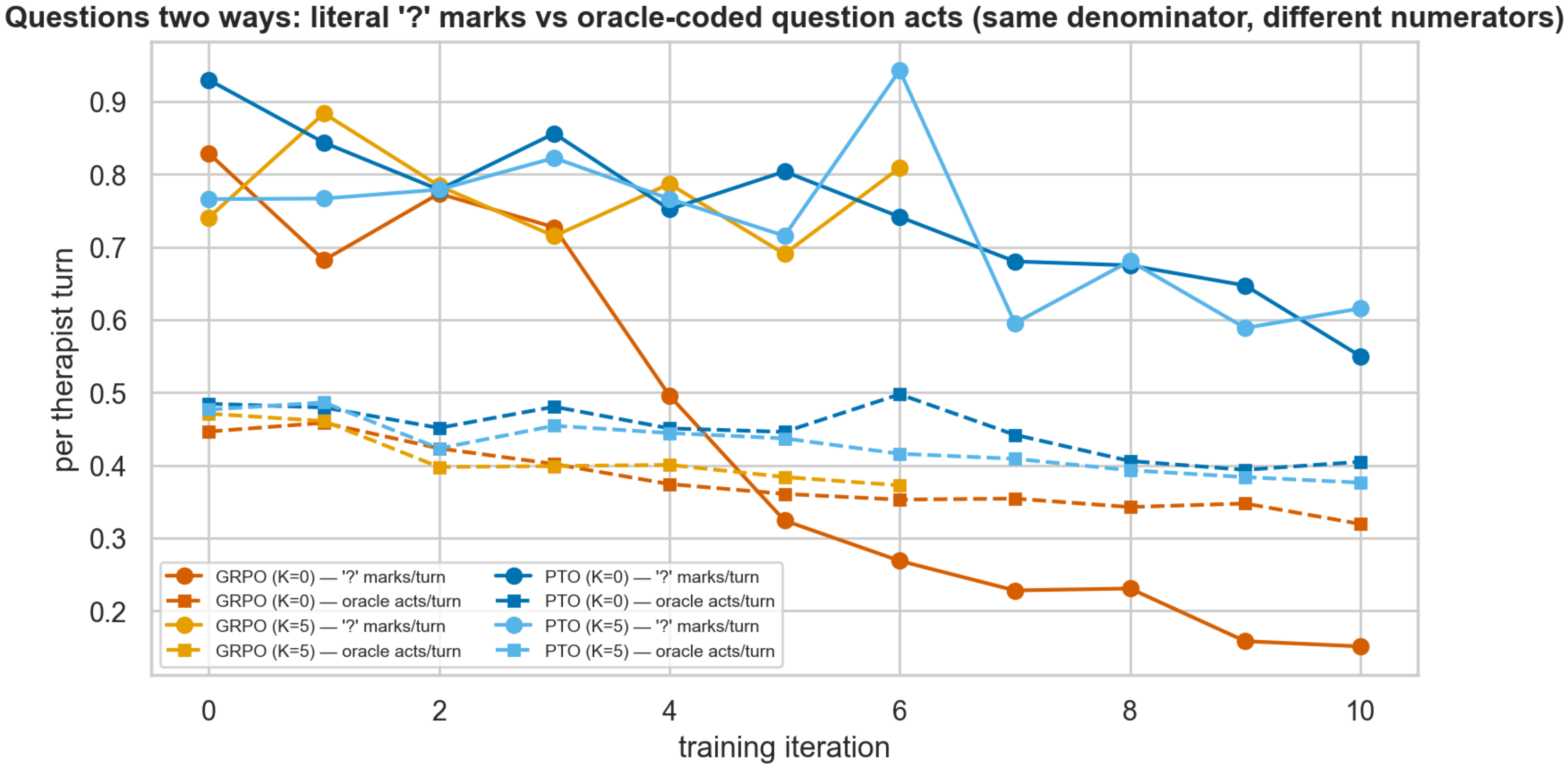
Y AXES
Each panel on its own scale — rubric points for the reward, rates for the channels.

SERIES
One line per arm.

DATA
Grader: gpt-4o-mini. The held-out twin of this figure exists at the same path under claude-haiku-4-5.

source: `arms/validity/figures/gpt-4o-mini/reward_hack_panel.png`

Two crosschecks: questions and over-praise



X AXIS
Training iteration.

Y AXIS
Questions per therapist turn, two counts: literal '?' MARKS (solid) vs oracle-coded question ACTS (dashed).

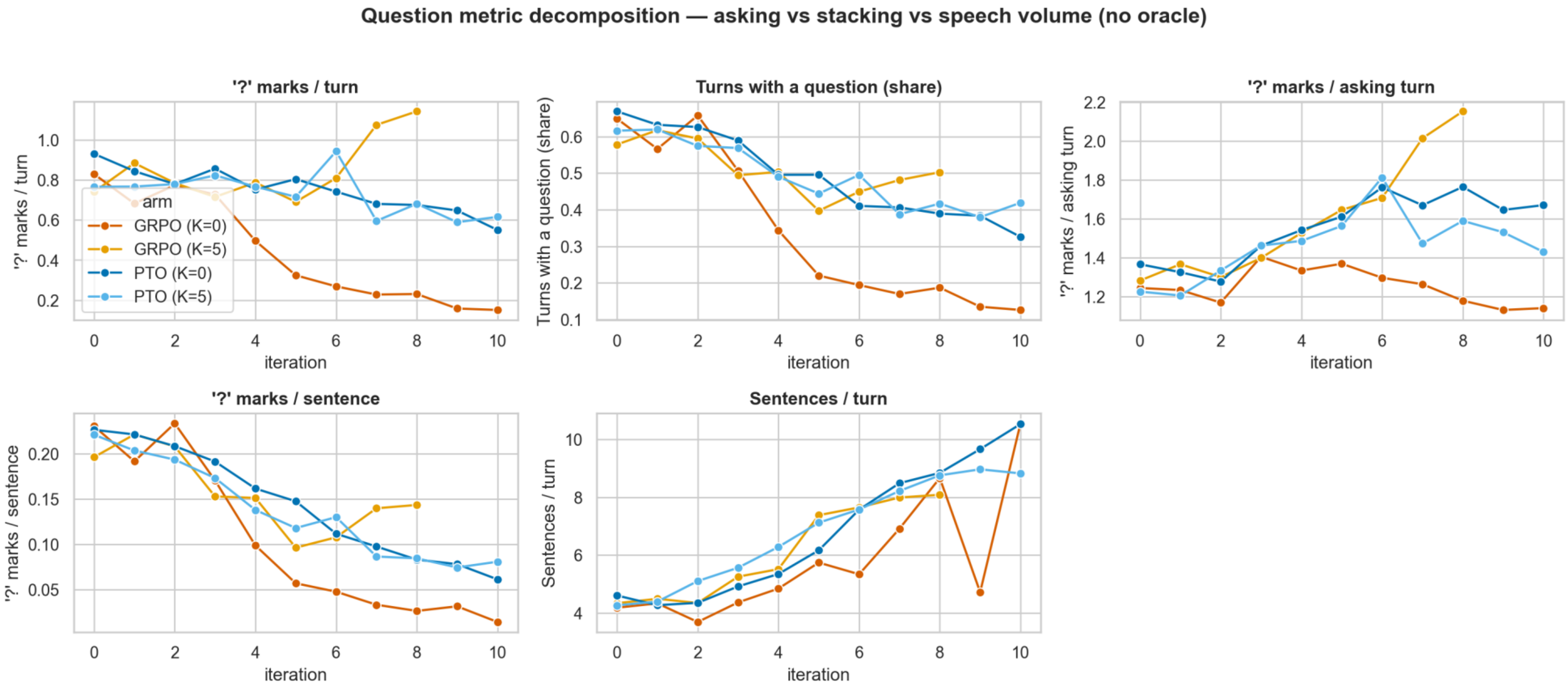
SERIES
One line per counting method, per arm.

LEVELS
Marks run 1.4–2.3× above acts in three of four arms — a construct difference, not disagreement.

USE
Direction agrees everywhere; only GRPO K=0 flips the ordering, from iteration 5. Over-praise twin: overpraise_crosscheck.png.

source: `arms/validity/figures/gpt-4o-mini/question_rate_crosscheck.png`

What the question rate pools



PANELS

Headline '?' marks/turn, then its parts: share of turns asking, marks per asking turn, marks per sentence.

WHY

Headline = panels 2 × 3, and they move oppositely. All four arms ask in FEWER turns; three stack MORE.

THE EXCEPTION

GRPO K=0 collapses on both: asking 0.65 → 0.13, stacking flat 1.24 → 1.14. The one arm that stops asking.

WATCH

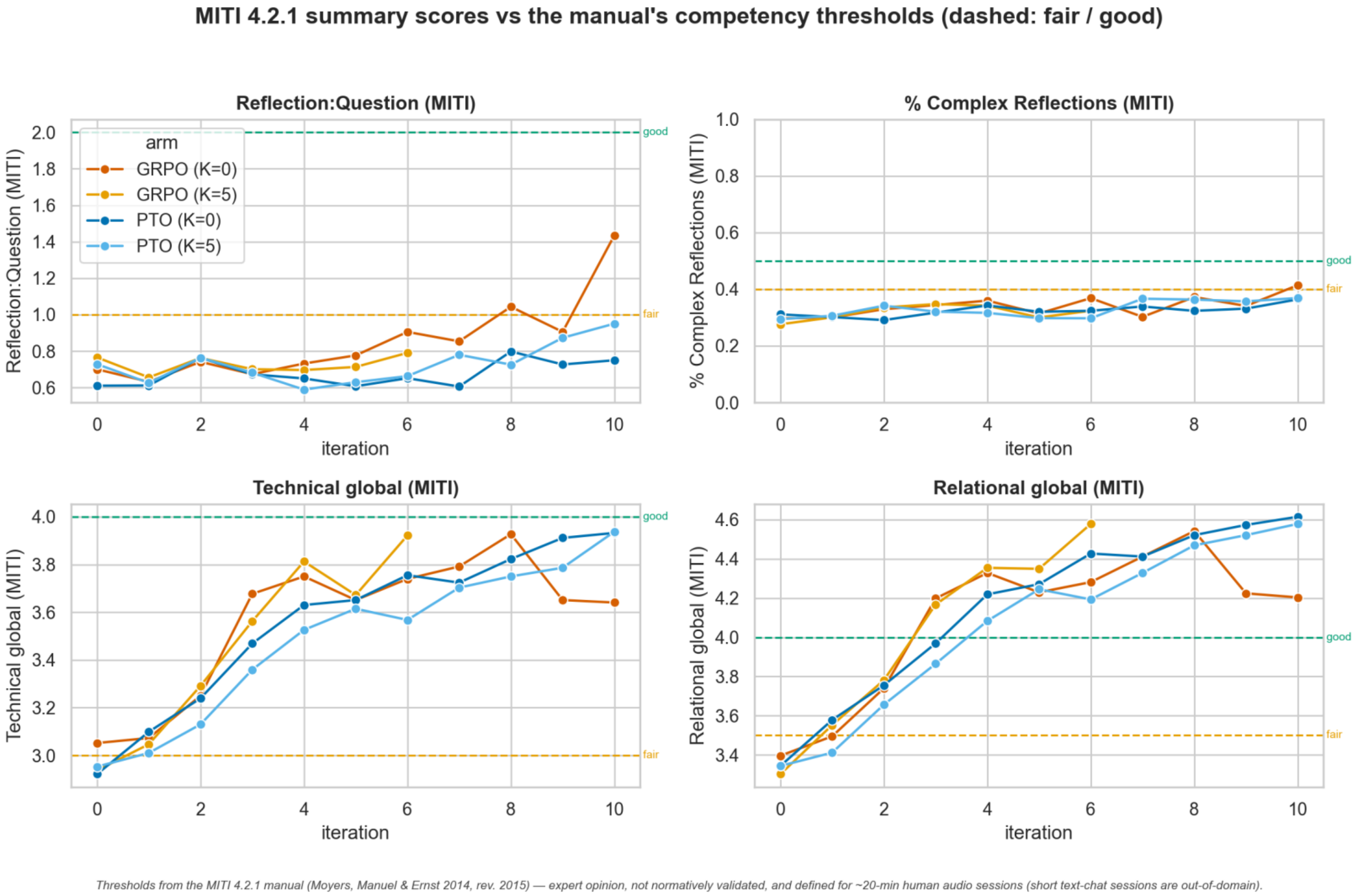
GRPO K=5's headline RISES from iteration 5 (0.69 → 1.14) — stacking, not asking (share 0.40 → 0.50).

DATA

Deterministic, no grader. GRPO K=5 has transcripts to iteration 8, scores only to 5/6.

source: `arms/validity/figures/gpt-4o-mini/question_decomposition.png`

MITI against published proficiency thresholds



X AXIS
Model state.

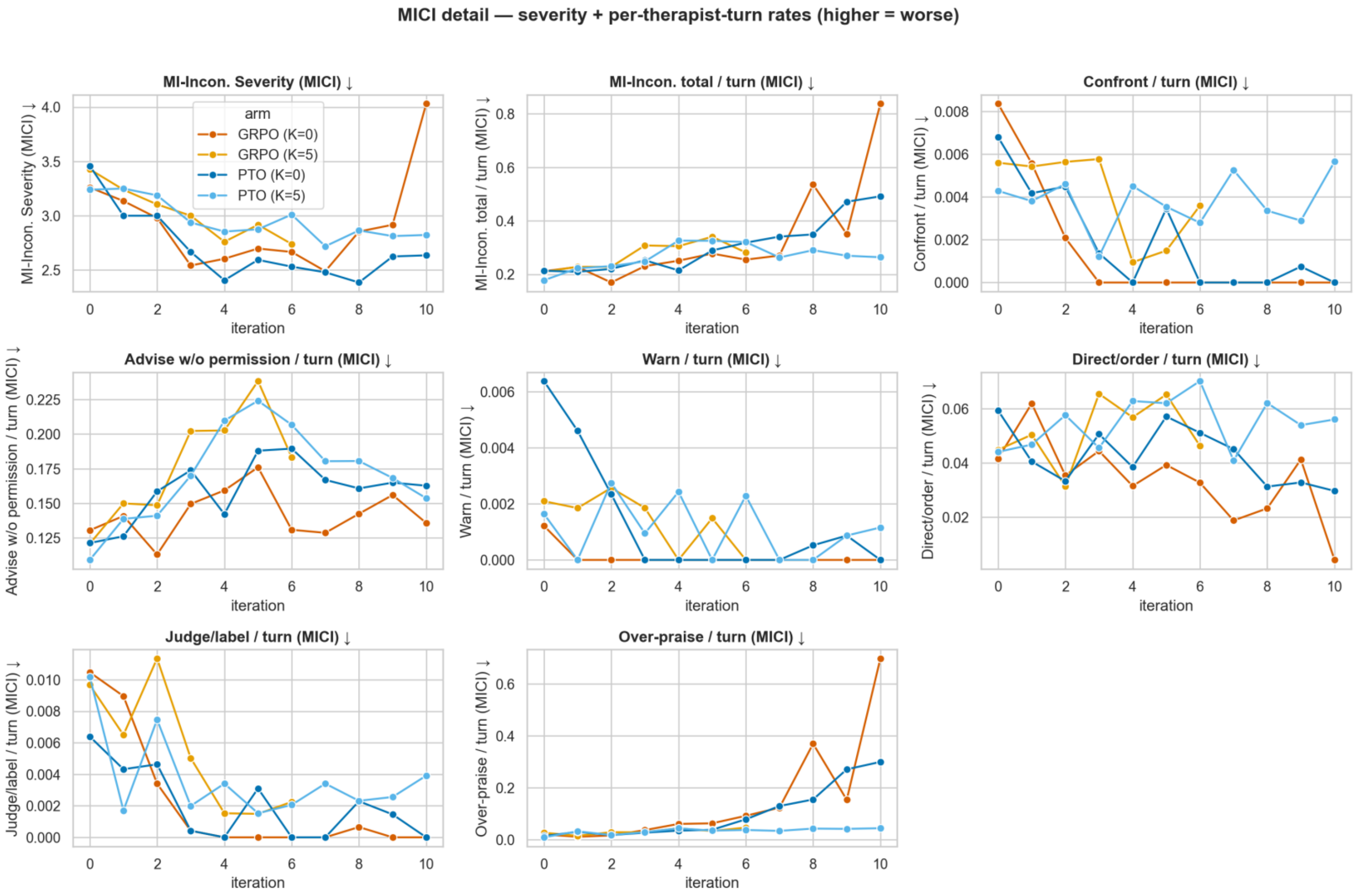
Y AXIS
MITI global and component scores.

REFERENCE LINES
The published MITI 4.2 'fair' and 'good' proficiency thresholds.

DATA
Grader: gpt-4o-mini. MITI is the weakest instrument in this battery on single-judge dependability.

source: `arms/questionnaires/figures/gpt-4o-mini/miti_proficiency_thresholds.png`

MI-inconsistency, item by item



X AXIS
Training iteration, per panel.

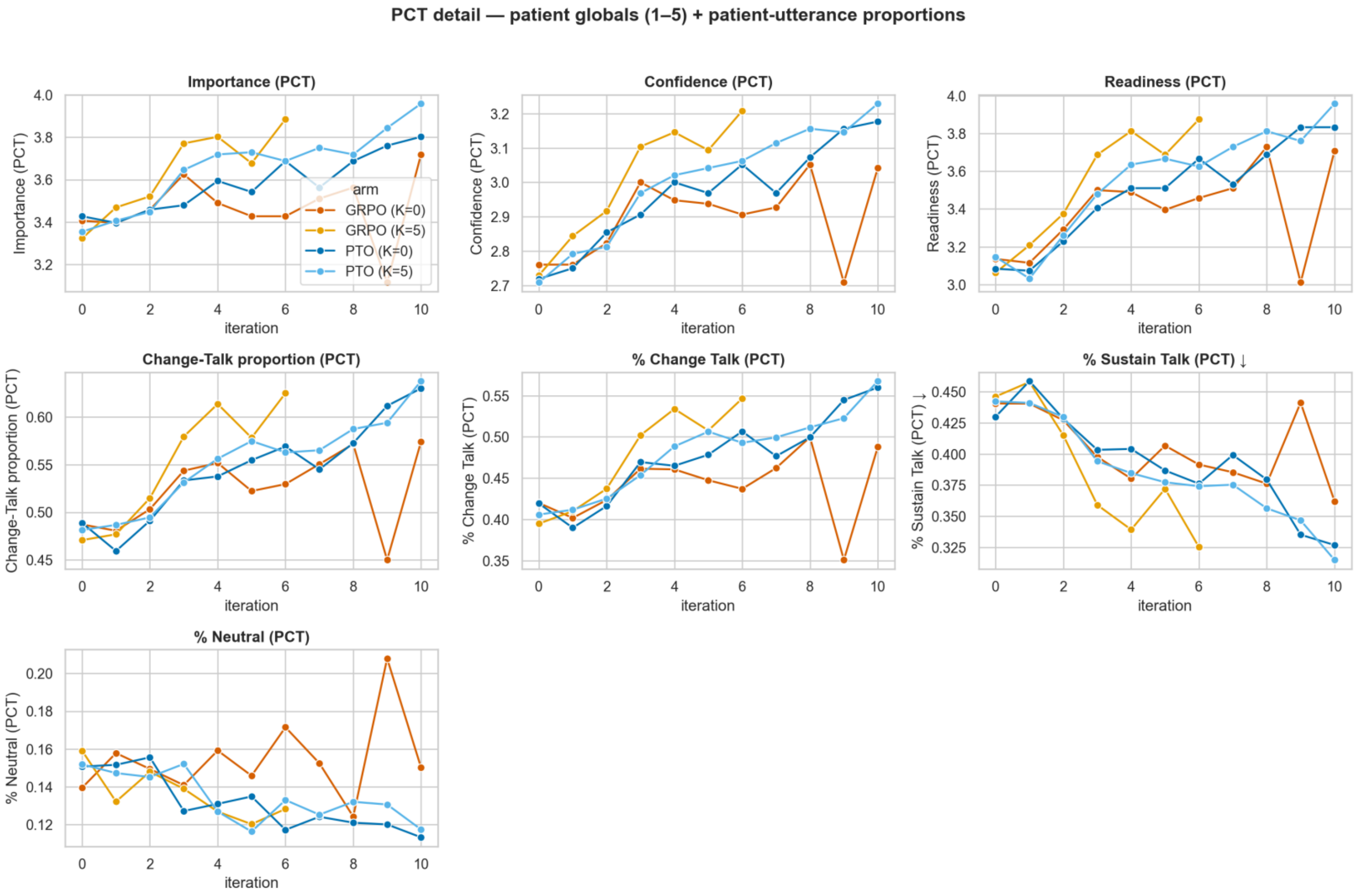
Y AXIS
Rate of that specific MI-inconsistent behaviour.

PANELS
One per coded behaviour: over-praise, confront, direct, judge, warn, advise without permission, plus the severity global.

SERIES
One line per arm. Higher = worse throughout. Grader: gpt-4o-mini.

source: `arms/questionnaires/figures/gpt-4o-mini/mici_detail_grid.png`

Patient change talk, item by item



X AXIS
Training iteration, per panel.

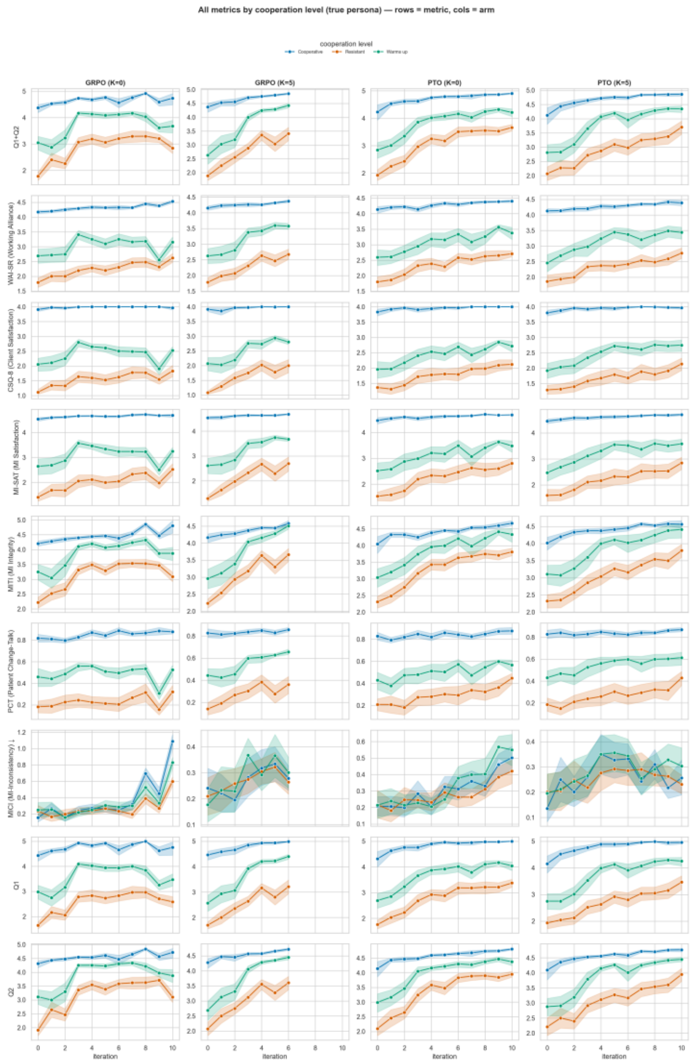
Y AXIS
The panel's PCT component — change-talk proportion, sustain-talk proportion, neutral proportion, importance, confidence, readiness.

SERIES
One line per arm.

NOTE
PCT codes the PATIENT's language, not the therapist's. Grader: gpt-4o-mini.

source: `arms/questionnaires/figures/gpt-4o-mini/pct_detail_grid.png`

Outcomes split by patient cooperation level



X AXIS

Training iteration, per panel.

Y AXIS

The panel's rubric.

SERIES

One line per (arm × cooperation stratum). 32 personas per stratum.

DATA

Grader: gpt-4o-mini. The same grid split by presenting problem is at problem_all_metrics.png.

source: `arms/heterogeneity/figures/gpt-4o-mini/cooperation_level_all_metrics.png`

Rendered but not shown here

— **Per-judge twins.** Almost every figure under arms/* exists once per grader. This deck shows the gpt-4o-mini copy for the detail grids and both copies for the headline outcome figures. The held-out twin sits at the same path with claude-haiku-4-5 substituted.

— **Per-item figures.** arms/questionnaires/figures/<judge>/{mici,miti,pct}/ hold one figure per individual coded item; the detail grids in this deck aggregate them.

— **Per-arm training curves.** arms/training/figures/<judge>/tb_curves/<arm>.png and reward_distribution/<arm>.png hold the single-arm versions.

— **PTO preference drift.** arms/preference/figures/<judge>/PTO_LA{0,5}_pref_*.png — category, direction, word-level and learn/unlearn drift across iterations, per arm.

— **Single-column variants.** compute_trajectory_col.png and crossgen_col.png are the same data laid out for a paper column.

— **Everything is regenerable.** Exp3: python tools/render_results.py from Exp3_PTO_GRPO/eda/. Exp1 and Exp2: python make_exp1_exp2_figs.py from meetings/build/.

source: `Exp3_PTO_GRPO/eda/results/INDEX.md maps every family to its notebook`